

Decoding transcription factor binding to the DNA

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Abstract

TODO: Abstract of the thesis.

Sammendrag

TODO: abstract in Norwegian (mandatory)

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Preface

This thesis is submitted in partial fulfilment of the requirements for the degree of *Philosophiae Doctor* at the University of Oslo. The presented work was conducted at the Centre for Molecular Medicine Norway (NCMM), organized by the University of Oslo, under the supervision of Professor Dr. Anthony Mathelier and co-supervision of Professor Dr. Jonas Paulsen. This research was funded by the XXXXXXXXXXXX.

A collection of three articles is the basis of this thesis. Two of the papers are published and one is in peer-review at the date of the submission of the thesis. The overarching theme of presented articles is gene regulation by TF-DNA interactions. Introductory chapters are provided to give necessary theoretical overview and motivation for this research.

Acknowledgements

Preface

Abbreviations

A	adenine
bp	base pair
C	cytosine
DHS	DNase I hypersensitivity
DNA	deoxyribonucleic acid
G	guanine
IC	information content
kb	(or kbp) kilobase pair
Mb	(or Mbp) megabase pair
mESC	mouse embryonic stem cell
NMF	non-negative matrix factorization
PFM	position frequency matrix
PPI	protein-protein interaction
PSSM	position specific scoring matrix
PWM	position weight matrix
RNA	ribonucleic acid
SMF	single-molecule footprinting
T	thymine
TAD	topologically associated domain
TF	transcription factor
TFBS	transcription factor binding site

Preface

Introduction

Chapter 1

Background

The genome is a blueprint of life. It codes a recipe for growth, development, regulation of all processes in the cell whether it is a multicellular or a single-cell organism. In multicellular organisms, such as humans, the genome dictates intricate intracellular interactions that enables a collection of cells to become one coordinated organism. The genome is composed of deoxyribonucleic acid (DNA) and through the production of proteins and RNA, complex cellular mechanisms are organizing all the biological processes. By studying these mechanisms, we hope to better understand how genome is functioning.

1.1 Organization of the genome

In Eukaryotes hereditary information is stored as DNA in the cell nucleus. Genomic DNA is transcribed and replicated. For example, if stretched into a single thread, human DNA is around 2 meters long and must be compacted into a microscopic nucleus. Therefore, multiple levels of organization are applied, starting from DNA helix and histone proteins and ending with chromosomes and their territories in the nucleus (Figure 1.1) [1]. Specific organization ensures proper functioning of the genome — genes can be expressed in the right place and time.

1.1.1 DNA

The existence of DNA was discovered in 1869 by Friedrich Miescher when he found a new substance inside the nuclei of the human white blood cells that did not have the same properties as proteins or lipids [2, 3]. Interestingly, this is not the discovery related to DNA that is most remembered. In 1953 contributions of Rosalind Franklin, James Watson and Francis Crick led to a publication of resolved DNA molecule structure and is often considered to be the beginning of molecular biology [4]. This is a fundamental moment in science history as it allowed to explain the possible mechanism of DNA replication, thus information multiplication and transfer.

DNA stores genetic information in the cell. Its structure is based on a sugar backbone (deoxyribose) that holds four main types of nucleotides — purines adenine (A) and guanine (G), and pyrimidines thymine (T) and cytosine (C) [5].

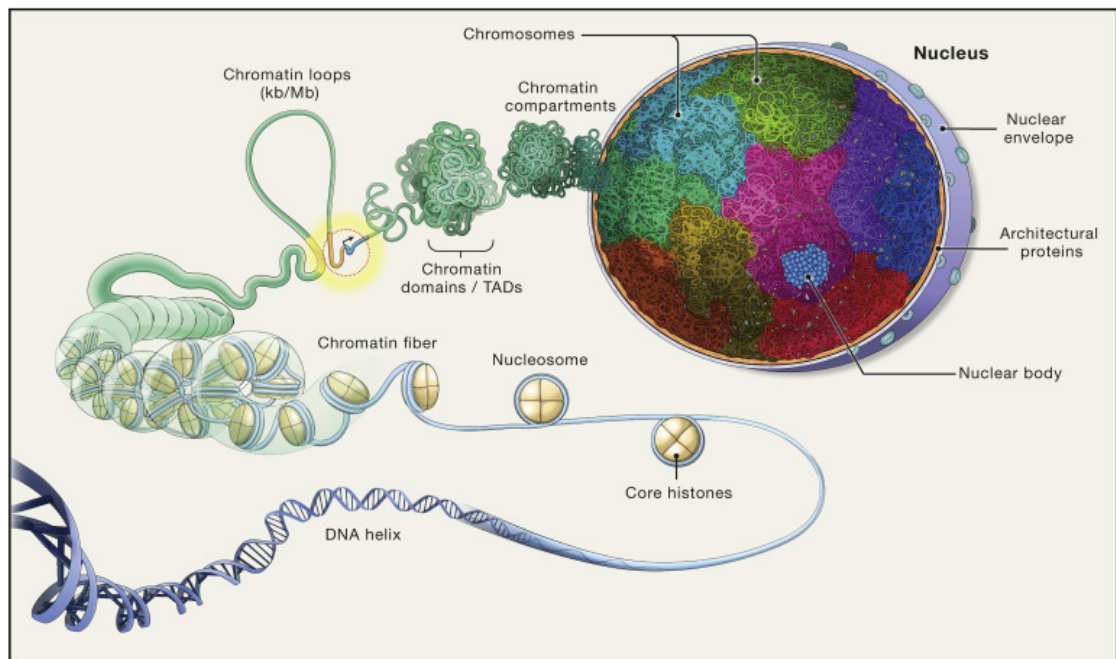


Figure 1.1: Eukaryotic genome organization. Genomic DNA is wrapped up around histones to form nucleosomes - a basic unit of chromatin fibre. Chromatin can form loops that span kilobase pairs (kb) or megabase pairs (Mb) and are essential for gene expression regulation. Bigger stretches of chromatin are organized into domains or topologically associated domains (TADs), which in turn form chromatin compartments to tightly pack everything into the nucleus. Each chromosome usually occupies a territory in the nucleus. The nucleus also contains a nuclear body, architectural proteins and is surrounded by a nuclear envelope. Adapted from [1].

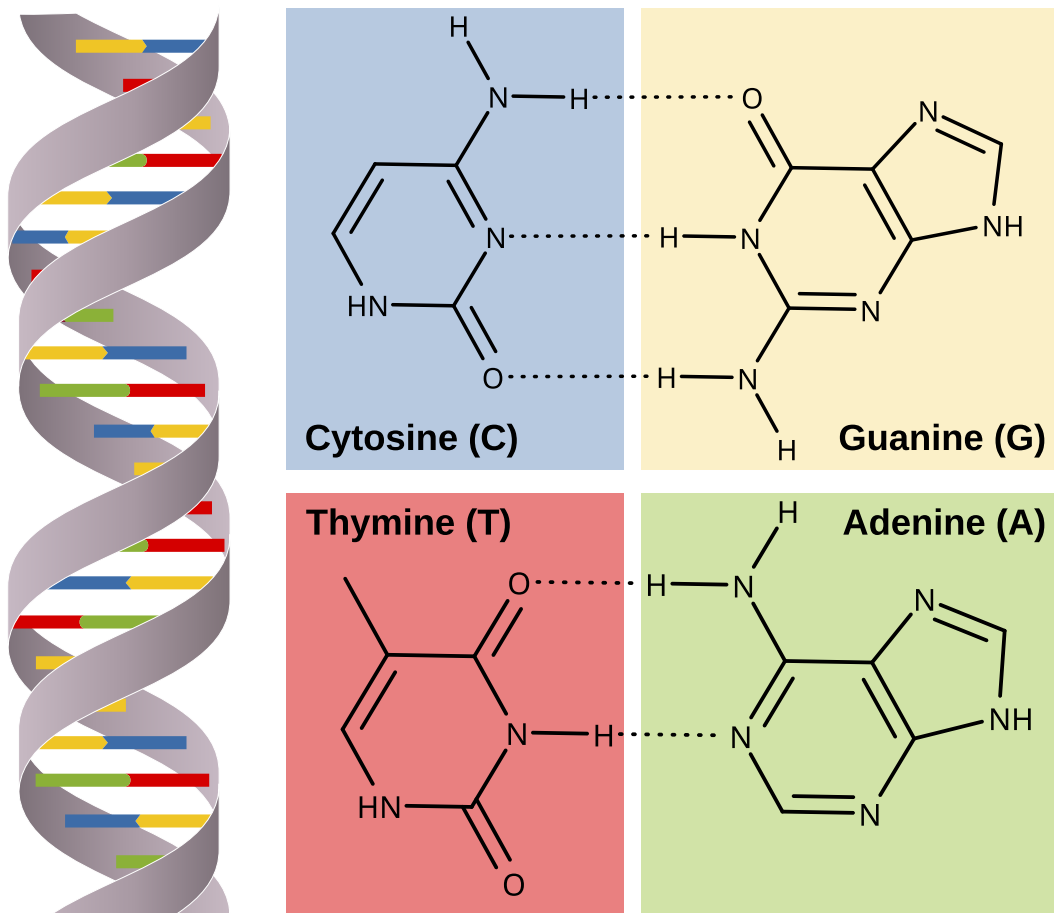


Figure 1.2: DNA structure. The two chains of the DNA are composed of a sugar backbone and four nucleotides: adenine (A), thymine (T), cytosine (C) and guanine (G). Nucleotides in the two chains are connected via hydrogen bonds in a strict complementarity, where A is bound to T and C is bound to G.

Chargaff's rule states that in the DNA of any species, the proportions of A and T nucleotides and C and G are always equal, demonstrating unique base-pairing (complementarity), when forming a double-stranded DNA helix [5, 6]. A-T and C-G nucleotide pairs in two strands of the DNA are held by two and three hydrogen bonds respectively (Figure 1.2).

The most commonly found DNA structure is called a B form DNA. Two helical DNA strands are entwined in a way that forms spaces between the strands — major and minor grooves, which are 22 and 12 Å wide, respectively. The grooves create opportunities for the proteins to interact with the double-stranded DNA, which is a basis for transcription factor (TF) binding [7].

1.1.2 Chromatin

The basic element of chromatin is a 146 bp of DNA wrapped around core histone protein octamer defined as nucleosome [1]. Core histone proteins are the most commonly found in nucleosomes, but can be replaced with variants in special

context, such as promoter regions.

Types of chromatin - euchromatin and heterochromatin.

organization of DNA into chromosome territories

More recently, higher level organized chromatin structures are taken into account. For example, chromosome compartments.

Phase separation maybe mention here?

1.2 Gene transcription

What is gene?

1.2.1 RNA

RNA is produced during transcription, when transcription machinery with the main player RNA polymerase transcribes a portion of the genome. RNA molecule is single-stranded and similarly to the DNA, also consists of the sugar backbone (ribose) and four types nucleotides — uracil **U** (**U**!).

1.2.2 RNA polymerase and transcriptional machinery

Chapter 2

Transcription regulation

Multicellular higher organisms, such as human, are formed by thousands of cell types. However, the genome in every of those cells is identical. If the recipe is the same, how is it possible to have so many cell types that manage to work together and run such a complex organism? The expectation of the Human Genome project was that once we have a completed human genome, all these secrets will be revealed. Unfortunately, it turned out that the genome is much more mysterious, and we are studying its secrets until this day. Even though the sequence of the DNA is the same in every cell and a set of genes does not change, transcription regulation is what allows for the cells to differentiate. Gene expression is tightly regulated by a variety of mechanisms and each gene is expressed only in a particular cell at the particular time. Genomic DNA is constantly being approached by regulatory proteins and other small molecules. And all this must be carefully regulated to ensure appropriate cell growth, division and differentiation.

Transcriptional regulation - at the DNA level: - chromatin state modification (histone tail post-translational modifications - histone marks) Histone methyltransferases, histone acetyltransferases, histone deacetylases, polycomb proteins. - DNA methylation DNA methyltransferases, 5mC-reading proteins (could be a lead to the transcription factors, slightly mention the pioneering function, but more later) - TFs Bind enhancers, making a loop. (mention this very briefly, more details later on)

Post-transcriptional regulation - at the RNA level: - small RNAs - capping, RNA splicing, poly(A) tail, RNA editing, mRNA stability, nuclear export.

(Focus on transcriptional regulation.)

2.1 Cis-regulatory regions

A eukaryotic genome contains different types of repetitive regions or motifs. These include transposable elements, tandem repeats, GC- and AT-rich sequences, splice sites and binding sites of non-coding RNA molecules. Importantly, crucial for gene expression regulation are DNA sequence motifs targeted by transcription factors [8].

Gene expression regulation by transcription factors is one of the most studied fields, yet still contains a number of unanswered questions that keep the scientists

busy. <— as a last sentence?

2.2 Transcription factors

2.2.1 Protein structures and classification

Protein domains

Classification

DNA binding domains are highly conserved, thus are used to classify these proteins into families and classes.

One of the most extensive classification of mammalian transcription factors - TFClass - include superclass, class, family, subfamily and genus [9, 10].

2.3 Transcription factor binding

Transcription factors directly or indirectly bind genomic regions, known as promoters and enhancers. The location where TF resides onto DNA is referred to as a transcription factor binding site (TFBS). A promoter or an enhancer can have multiple TFBSs of the same or different TF. Transcription factors generally have binding preference to a set of DNA sequences - DNA binding motif [8].

2.3.1 Transcription factor DNA binding models

In order to interpret TF binding to the regulatory regions, we need ways to represent the DNA binding. Currently, DNA binding is represented using various models.

The most popular and widely used is positional weight matrix (PWM) or position-specific scoring matrix (PSSM).

PFM, PWM. Other models to represent binding.

There are two main strategies applied to analyse genomic sequences from TF binding perspective. One is to find over-represented sequence motifs using existing collections of DNA binding motifs. The second is to discover over-represented motifs *de novo* [8].

(As a last sentence) There is a common misconception that all TFs are binding DNA very specifically with little variation. In fact, there are different binding mechanisms that TFs can adapt to regulate gene expression in different scenarios.

2.3.2 Mechanisms of gene regulation

Transcription factors can bind DNA in different ways, varying from direct binding to the DNA (most well studied) to indirect and cooperative binding. For example, transcription factor PU.1 can bind to DNA directly and indirectly. Direct binding of PU.1 to the consensus motif AGGAA leads to transcriptional activation, while

indirect binding through a partner GATA1 to the GATA motif may lead to transcriptional repression [8].

It is important to understand that protein binding to DNA mechanisms vary. This is essential to increase the complexity in gene transcription regulation, which is a key to development of living organisms.

Scientists started to understand the variability of gene expression regulation mechanisms by inventing and applying a series of experimental and computational methods. The most famous examples include chromatin immunoprecipitation followed by sequencing (ChIP-seq) and *de novo* motif discovery in genomic regions, respectively. Thus, it is important to understand the advantages and disadvantages of these methods.

In this section I will review these binding mechanisms.

(Go from the simplest to more complicated....)

Direct TF binding

Indirect TF binding

Pioneering

TFs are capable of binding heterochromatin through recruitment of histone acetyltransferases [8].

Cooperative transcription factor binding

When TFs interact or compete with each other binding sites co-localize or overlap [8].

Cooperative binding does not always mean that two TFs will bind the DNA directly.

Notes: talk about mechanisms of binding and methods that investigate these mechanisms can be explained here too. So this needs to be quite specific mechanisms and then specifics on the methods as well (experimental and computational methods that is).

Summary of papers

Chapter 3

Summary of papers

This chapter summarizes the three publications that are the foundation of this thesis. The unifying subject of this thesis is the investigation of transcription factor binding patterns and mechanisms. This subject was explored via two main aims: 1) new motif discovery and addition to the JASPAR database and 2) identification of co-binding patterns.

3.1 Papers I-II. Contribution to the 2022 and 2024 releases of JASPAR database

Castro-Mondragon JA⁰, Riudavets-Puig R⁰, **Rauluseviciute I**⁰, Berhanu Lemma R, Turchi L, Blanc-Mathieu R, Lucas J, Boddie P, Khan A, Manosalva Pérez N, Fornes O, Leung TY, Aguirre A, Hammal F, Schmelter D, Baranasic D, Ballester B, Sandelin A, Lenhard B, Vandepoele K, Wasserman WW, Percy F, Mathelier A. *JASPAR 2022: the 9th release of the open-access database of transcription factor binding profiles*. Nucleic acids research. 2022 Jan 7;50(D1):D165-73; doi: 10.1093/nar/gkab1113.

Rauluseviciute I⁰, Riudavets-Puig R⁰, Blanc-Mathieu R, Castro-Mondragon JA, Ferenc K, Kumar V, Lemma RB, Lucas J, Chèneby J, Baranasic D, Khan A, Fornes O, Gundersen S, Johansen M, Hovig E, Lenhard B, Sandelin A, Wasserman WW, Percy F, Mathelier A. *JASPAR 2024: 20th anniversary of the open-access database of transcription factor binding profiles*. Nucleic Acids Research. 2024 Jan 5;52(D1):D174-82; doi: 10.1093/nar/gkad1059.

Paper I and **Paper II** describe the two latest releases of the JASPAR database. JASPAR is a widely used open-access repository of transcription factor binding motifs created and maintained since 2004. One of the biggest database strengths is that all motifs are manually curated before they are added to the database. Motifs subjected for manual curation are discovered *de novo* from peak data, mined from databases (for example, ReMap [11] and GTRD [12]) and publications, or already generated motifs are mined from such sources. Database

⁰These authors should be regarded as joint first authors.

is updated every 2 years by adding new profiles into two collections - CORE and UNVALIDATED. CORE collection contains profiles with orthogonal support, while UNVALIDATED collection has profiles of same good quality, but not yet confirmed by an independent experiment. In 2022 CORE and UNVALIDATED collections were enhanced or updated by 501 and 182 profiles, respectively, while in 2024 by 404 and 298 profiles, respectively.

Since 2022, JASPAR is not only updated with new information, but existing profiles in both collections are consistently revised. We revise UNVALIDATED collection profiles to check for possible new validation to move the profiles into the CORE collection. In 2022, we upgraded 52 such profiles, while in 2024 validation was found for 72 profiles. We also extensively revised the inconsistencies in the metadata that were accumulated over the years since the release of JASPAR. Metadata fields, such as TF class and family were revised and updated with the newest available information. For example, vertebrate profiles are classified using TFClass [9], while plant TF classification was extensively revised both in 2022 and 2024 and now is based on [13]. 75% (1,869 out of 2,506) JASPAR 2022 motifs and all incoming new motifs were updated by trimming. We applied an algorithm to remove the low IC positions on the sides of motifs to improve the quality of our collections.

In addition to the augmenting and refining the motifs and related metadata, we provide additional analysis and resources. This includes motif clustering, familial binding motifs, TFBSs predictions, genome browser tracks, word clouds. For each taxon we cluster CORE collection motifs individually and CORE and UNVALIDATED motifs together. Clusters are visualized as radial trees and annotated with TF class information. Furthermore, the motif clusters were used to construct familial binding profiles. Such motif represent what DNA binding preference has closely related TFs. Latest versions of CORE collection motifs are used to scan the most popular genomes to predict TFBSs, which are available to users along with the genome browser tracks. Finally, since 2022 word clouds are generated for each TF in JASPAR to illustrate their functional roles.

JASPAR database is user-friendly and can be accessed via a number of channels. Using command-line, RESTful API, R Bioconductor package and pyJASPAR. To accompany the data we also provide a number of computational tools. In the past two releases we added an enrichment analysis tool that allows to study the enrichment of the binding of a specific TF in a set of genomic regions, and a TFBS extraction tool for easier extraction of TFBSs in a genomic region of interest.

JASPAR database update process is complex, but We have refined it into three processes, abbreviated as "3D" - discovery, downstream, deployment. Each process has its own software component in a form of a workflow or a pipeline. In the discovery motifs are processed for manual curation. After manual curation, in the downstream analysis, curated motifs are prepared for the website and all additional analysis performed to produce the data needed for the website and other related sources. Finally, the deployment workflow collects all produced data and updates the website. All software components consistently receive updates to implement new features, address users' needs and increase the reliability and

performance of the entire process of JASPAR update.

3.2 Paper III. Identification of transcription factor co-binding patterns with non-negative matrix factorization

Rauluseviciute I, Launay T, Barzaghi G, Nikumbh S, Lenhard B, Krebs AR, Castro-Mondragon JA, Mathelier A. *Identification of transcription factor co-binding patterns with non-negative matrix factorization*. bioRxiv. 2023:2023-04.

Paper III describes a non-negative matrix factorization-based method to identify co-binding DNA patterns. The tool COBIND can be used to identify DNA/RNA patterns in relation to the provided set of reference regions/sequences (anchors). In the paper, COBIND was applied on a collection of known TFBSs to automatically identify TF co-binding patterns in the regions flanking them. Interpreting such predicted DNA patterns can give insights about TF co-binding and a more complex gene regulation mechanism it creates.

In COBIND, NMF is applied to one-hot encoded sequences and pinpoints the enriched DNA patterns at fixed instances. We used 5,699 TFBS datasets from UniBind database for 401 TFs in seven species [14]. As proof-of-concept, we present a classical example of POU5F1 co-binding with SOX2 or SOX17. For SOX2 and SOX17 TFBS datasets we were able to recover the DNA pattern that represents POU5F1 motif with different spacings, corresponding to canonical (SOX2::POU5F1 pair) and compressed (SOX17::POU5F1 pair) motifs. The binding of POU5F1 was inferred through motif similarity and TF pair presence in the protein-protein interaction (PPI) data.

All discovered co-binding patterns were evaluated with motif similarity and PPI analysis, where 78% (462 out of 591) of the patterns had support in the PPI data. For example, TEAD1 co-binding with TEAD family members. However, a fraction of inferred pairs are not yet reported in the PPI data or did not have a good motif match. In *A. thaliana* genome, the same co-binding pattern was found for HY5, ABF1, ABF3, ABF4, GBF2, and GBF3 anchors. Motif analysis revealed the similarity with the motif recognized by human NF-Y TFs (NF-YC and NF-YA). However, such interactions are not currently supported by the PPI data. Another example, where COBIND prediction was not supported by the PPI data, is a co-binding pattern of CTCF. The same co-binding pattern was found with two different spacings in *H. sapiens*, *M. musculus* and *D. rerio* TFBS datasets. In the literature, we found that the predicted DNA pattern is not bound by another TF, but by CTCF itself. CTCF is a zinc-finger protein and in a fraction of its binding sites 9-11 zinc fingers binds to an extended portion of the motif.

We also demonstrated that the regions with predicted co-binding patterns are likely functionally relevant as they are more evolutionarily conserved compared to the regions that contain only the anchor TFBSs. Across species and all predictions we systematically compared the conservation at the locations of anchor and predicted co-binding sites. We observed that 23–83% of the datasets

from different species with predicted co-binding patterns showed a significantly higher conservation at the co-binding motif locations compared to the alternative locations in regions only with the anchor motifs. For anchor sites, 31–67% of the datasets demonstrated a more conserved anchor site in the regions with predicted co-binding patterns. We observed for 23–64% of datasets that the locations of both the anchor and a co-binding patterns were more conserved in the co-bound regions.

Furthermore, we investigated regions with and without predicted co-binders in the open chromatin data by analysing DNase I hypersensitivity (DHS) footprints from matching human cell-types. By performing a similar analysis to the conservation, we found that in 85% of co-binding pattern - DHS dataset pairs, DHS footprints were significantly deeper at the locations of the co-binding patterns in regions with predicted co-binding motifs compared to the regions without (20% was expected by chance). DHS footprints in anchor locations were significantly deeper in co-bound regions for 78% of the pairs (21% expected by chance). When considering deeper footprints for both anchor and co-binding pattern sites, significance was never observed in a random expectation, but with real data it was observed for 73% of the pattern-dataset pairs.

Finally, single-molecule footprinting (SMF) data from mouse embryonic stem cells (mESCs) confirmed that the predicted co-binding events associated for a set of TFs are likely occurring on the same DNA molecules. We analysed predictions for 17 anchor TFs, where co-binding patterns were found in a total of 1,694 regions. For each of these genomic regions and regions with only anchor motifs, we determined proportions of molecules in five states: accessible, occupied by nucleosomes, only occupied at the anchor motif site, only occupied at the co-binding motif site or occupied at both sites. Overall, we found that a significantly larger fraction of molecules was co-occupied at the both sites in the regions with COBIND-predicted co-binding motifs compared to the regions only with the anchor motifs. When looking at the individual anchors, the significance was observed for 8 out of 17 analysed anchor TFs.

We introduced a computational framework COBIND (https://bitbucket.org/CBGR/cobind_tool/src/main/) for DNA motif pattern discovery *de novo* in the vicinity of predetermined regions. During extensive analysis we produced a collection of predicted co-binding patterns in 7 species, which are provided to the community at https://cbgr.bitbucket.io/COBIND_results_page/. DNA patterns discovered in the fixed locations from known TFBSs reveal the possible cooperative events and can be studied to investigate more complex TF-DNA interactions.

Discussion

Discuss motif trimming in JASPAR 2024. Apparently Taipale's selex paper or whatever says that low information positions are the ones that matter when it comes to the binding of a specific TF family member (why one member and not another will bind even though the motifs are the same). I could give an argument for downstream data analysis, that those positions don't really provide any information. Also, maybe cooperation matters more and those low information positions are actually masking the possible grammar, because of technical limitations. You chip only one protein and expect short, so surrounding are kind of averaged out.

Bibliography

Bibliography

- [1] Tom Misteli. “The self-organizing genome: principles of genome architecture and function.” In: *Cell* 183.1 (2020), pp. 28–45.
- [2] Leslie Pray. “Discovery of DNA structure and function: Watson and Crick.” In: *Nature Education* 1.1 (2008), p. 100.
- [3] Ralf Dahm. “Discovering DNA: Friedrich Miescher and the early years of nucleic acid research.” In: *Human genetics* 122 (2008), pp. 565–581.
- [4] James D Watson and Francis HC Crick. “Molecular structure of nucleic acids: a structure for deoxyribose nucleic acid.” In: *Nature* 171.4356 (1953), pp. 737–738.
- [5] Andrew Travers and Georgi Muskhelishvili. “DNA structure and function.” In: *The FEBS Journal* 282.12 (2015), pp. 2279–2295. DOI: <https://doi.org/10.1111/febs.13307>. eprint: <https://febs.onlinelibrary.wiley.com/doi/pdf/10.1111/febs.13307>. URL: <https://febs.onlinelibrary.wiley.com/doi/abs/10.1111/febs.13307>.
- [6] Erwin Chargaff et al. “The composition of the desoxyribonucleic acid of salmon sperm.” In: *Journal of Biological Chemistry* 192.1 (1951), pp. 223–230.
- [7] Carl O. Pabo and Robert T. Sauer. “PROTEIN-DNA RECOGNITION.” In: *Annual Review of Biochemistry* 53.1 (1984). PMID: 6236744, pp. 293–321. DOI: 10.1146/annurev.bi.53.070184.001453. eprint: <https://doi.org/10.1146/annurev.bi.53.070184.001453>. URL: <https://doi.org/10.1146/annurev.bi.53.070184.001453>.
- [8] Valentina Boeva. “Analysis of genomic sequence motifs for deciphering transcription factor binding and transcriptional regulation in eukaryotic cells.” In: *Frontiers in genetics* 7 (2016), p. 24.
- [9] Edgar Wingender et al. “TFClass: expanding the classification of human transcription factors to their mammalian orthologs.” In: *Nucleic acids research* 46.D1 (2018), pp. D343–D347.
- [10] Edgar Wingender. “Criteria for an updated classification of human transcription factor DNA-binding domains.” In: *Journal of Bioinformatics and Computational Biology* 11.01 (2013), p. 1340007.

Bibliography

- [11] Fayrouz Hammal et al. “ReMap 2022: a database of Human, Mouse, Drosophila and Arabidopsis regulatory regions from an integrative analysis of DNA-binding sequencing experiments.” In: *Nucleic acids research* 50.D1 (2022), pp. D316–D325.
- [12] Semyon Kolmykov et al. “GTRD: an integrated view of transcription regulation.” In: *Nucleic acids research* 49.D1 (2021), pp. D104–D111.
- [13] Romain Blanc-Mathieu et al. “Plant-TFClass: a structural classification for plant transcription factors.” In: *Trends in Plant Science* 29.1 (2024), pp. 40–51.
- [14] Rafael Riudavets Puig et al. “UniBind: maps of high-confidence direct TF-DNA interactions across nine species.” In: *BMC genomics* 22.1 (2021), pp. 1–17.

Papers

Paper I

**JASPAR 2022: the 9th release of the
open-access database of
transcription factor binding profiles**

JASPAR 2022: the 9th release of the open-access database of transcription factor binding profiles

Jaime A. Castro-Mondragon^{1,†}, Rafael Riudavets-Puig^{1,†}, Ieva Rauluseviciute^{1,†}, Roza Berhanu Lemma¹, Laura Turchi², Romain Blanc-Mathieu², Jeremy Lucas², Paul Boddie¹, Aziz Khan³, Nicolás Manosalva Pérez^{4,5}, Oriol Fornes⁶, Tiffany Y. Leung⁶, Alejandro Aguirre⁶, Fayrouz Hammal⁷, Daniel Schmelter⁸, Damir Baranasic^{9,10}, Benoit Ballester⁷, Albin Sandelin^{11,*}, Boris Lenhard^{9,10,*}, Klaas Vandepoele^{4,5,12}, Wyeth W. Wasserman^{10,6,*}, François Parcy^{2,*} and Anthony Mathelier^{1,13,*}

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Received September 15, 2021; Revised October 20, 2021; Editorial Decision October 20, 2021; Accepted October 22, 2021

ABSTRACT

JASPAR (<http://jaspar.genereg.net/>) is an open-access database containing manually curated, non-redundant transcription factor (TF) binding profiles for TFs across six taxonomic groups. In this 9th release, we expanded the CORE collection with 341 new profiles (148 for plants, 101 for vertebrates, 85 for urochordates, and 7 for insects), which corresponds to a 19% expansion over the previous release. We added 298 new profiles to the Unvalidated collection when no orthogonal evidence was found in the literature. All the profiles were clustered to provide familial binding profiles for each taxonomic group. Moreover, we revised the structural classification of DNA bind-

ing domains to consider plant-specific TFs. This release introduces word clouds to represent the scientific knowledge associated with each TF. We updated the genome tracks of TFBSs predicted with JASPAR profiles in eight organisms; the human and mouse TFBS predictions can be visualized as native tracks in the UCSC Genome Browser. Finally, we provide a new tool to perform JASPAR TFBS enrichment analysis in user-provided genomic regions. All the data is accessible through the JASPAR website, its associated RESTful API, the R/Bioconductor data package, and a new Python package, pyJASPAR, that facilitates serverless access to the data.

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INTRODUCTION

Transcription factors are proteins that interact with the DNA in a sequence-specific manner through recognition of their TF binding sites (TFBSs) located at cis-regulatory regions (promoters, enhancers) to regulate transcription (1). TF binding to these regions occurs through direct interactions between the DNA-binding domains (DBDs) of TFs and the DNA. DBDs are classified into structural classes and families, and TFs with related DBDs typically have similar DNA binding preferences (2). The binding of TFs to cis-regulatory regions promotes or inhibits the assembly of the transcription machinery, thereby controlling gene expression regulation (1,3–5).

Sequence-specific TF-DNA interactions at TFBSs can be experimentally determined either *in vitro* or *in vivo*. High-throughput *in vitro* methods include systematic evolution of ligands by exponential enrichment (SELEX) (6) and protein binding-microarrays (PBM) (7) where TFs are exposed to synthesized DNA sequences. High-throughput *in vivo* assays include chromatin immunoprecipitation-based methods such as ChIP-seq (8), ChIP-exo (9) and ChIP-nexus (10), and cleavage-based methods such as cleavage under targets and tagmentation (11) or cleavage under targets and release using nuclease (12). These high-throughput assays (reviewed in (1)) provide unprecedented means to characterize the binding properties of individual TFs. Nevertheless, a challenge lies in our understanding of how TFs interact cooperatively at regulatory elements, for instance by forming dimers (13). Recently, CAP-SELEX revealed that TF pairs can bind in a DNA-dependent manner and that the combined binding of TFs can alter their individual binding specificities (14).

Despite the establishment of a wide variety of experimental techniques that delineate TF-DNA binding interactions and TF binding specificities, experimentally identifying all TFBSs for all TFs in various systems and biological conditions is intractable. To address this challenge, researchers rely on computational modeling to predict and investigate TF-DNA interactions. Such methods are helpful for investigating results of experimental methods with low resolution. For instance, ChIP-seq peaks are typically an order of magnitude larger than the actual binding sites of a targeted TF, and therefore computational methods can be used to pinpoint the binding sites within the peaks (15,16).

Given the importance of understanding TF-DNA interactions in studying gene expression regulation, various computational methods have been devised to model and predict TFBSs. The methods utilize experimentally identified TFBSs to build models and computationally predict TFBSs in a given genomic sequence (5). These computational methods range from basic representations such as sequence consensus-based models and position frequency matrices (PFMs) to more complex representations such as Markov and deep learning-based models (reviewed in (13,17–18)). PFMs, which summarize occurrences of each nucleotide at each position in a set of observed TF-DNA interactions, are largely and most commonly used to capture TF binding specificities. Unlike the simple consensus-based models, PFMs can be transformed to probabilistic or energy-based models to obtain position weight matrices (PWMs) (or position-specific scoring matrices (PSSMs))

that can be used to scan any DNA sequence and predict TFBSs with sum weights above a defined threshold (reviewed in (17)). Hence, TF binding preferences can be represented as PFMs, which can be interpreted as TF binding profiles or motifs. In this manuscript, we will use the term PFM, motif, and TF binding profile interchangeably.

JASPAR is a popular and regularly maintained open-access and manually curated database storing TF binding preferences as PFMs. The JASPAR CORE collection provides non-redundant binding preferences for TFs (one versioned profile per TF per taxon, except when a TF has multiple DNA-binding preferences) across 6 taxa: urochordata, vertebrates, plants, insects, nematodes, and fungi. Inclusion of new profiles requires orthogonal evidence for the binding preferences of the TFs, which is rigorously evaluated by our expert curators. To complement the CORE collection, we previously introduced the Unvalidated collection to store high-quality TF-binding profiles that are lacking orthogonal supporting evidence in the literature (19). Beyond the high-quality TF binding profiles and metadata stored in JASPAR, the popularity of the database originates from its simplicity, the tools embedded in its web-interface, and the multitude of popular resources and tools directly integrating JASPAR profiles. Some of these tools include: (i) the MEME suite, allowing various motif enrichment and discovery analysis (20), (ii) TFBSshape allowing investigation of DNA shape features for TFBSs to provide insight on the mechanism of protein–DNA interaction (21,22), (iii) CiiDER (23) for TFBS prediction and analysis such as enrichment assessment in DNA sequences, (iv) RSAT, allowing motif discovery, TFBS motif analyses (24) and (v) i-cisTarget, which allows the prediction of *cis*-regulatory modules and regulatory features (25,26).

In this paper, we present the 9th release of the JASPAR database, which provides a substantial update and expansion of TF binding profiles in the six taxonomic groups. The update includes not only binding profiles (as PFMs) but also revisited metadata. Additionally, we added word clouds to display enriched terms associated with TFs in the scientific literature. Furthermore, a rigorous structural classification of plant TF DBDs is provided to adequately consider the numerous plant-specific TFs. Finally, the update comes with a range of new or updated functionalities and resources such as a TFBS enrichment tool, the pyJASPAR package, new familial binding profiles, and native UCSC human and mouse genome tracks with TFBSs predicted from JASPAR TF binding profiles.

RESULTS

Expansion and update of the JASPAR database

TF binding profiles. In the 9th release of JASPAR, we discarded unused collections introduced in early releases of the database (27–29) that either did not correspond to TF-specific binding profiles or were data-type specific; we maintained the CORE and Unvalidated collections. We computed and compiled TF binding profiles obtained from CAP-SELEX (14), NCAP-SELEX (30), SELEX-seq (31), PBMs (32), ChIP-seq (33–36) and DAP-seq experiments from ReMap 2022 (36) and GEO (37), and ChIP-exo (38)

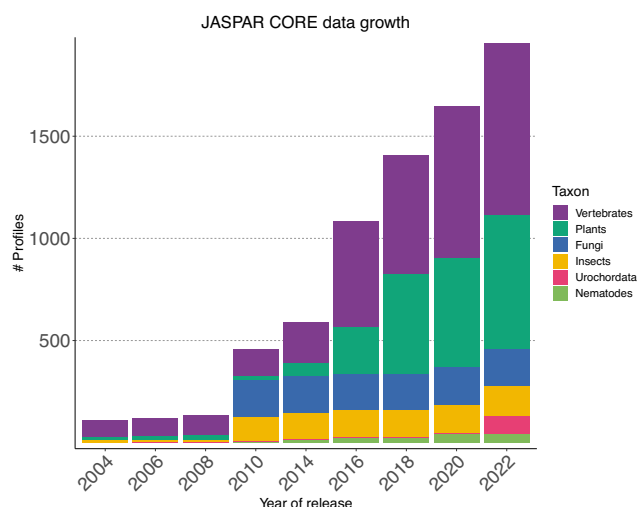


Figure 1. JASPAR CORE collection growth. The number of non-redundant profiles in each taxon (see legend) and overall through all JASPAR releases.

data (Supplementary Data 1 - Text for detailed list of datasets and method details). After manual curation of these profiles to confirm orthogonal supports in the literature, we augmented the CORE collection with 341 new binding profiles for TFs in four taxa (Table 1; Figure 1): 148 profiles in plants (a 24% expansion for this taxon), 101 profiles in vertebrates (a 13% expansion), 85 profiles in urochordates (only one motif was present since the second release of JASPAR in 2006 (27)), and seven profiles in insects (a 5% expansion). Out of these added profiles, 52 were upgraded from the Unvalidated to the CORE collection (27 and 25 for plants and vertebrates, respectively). Moreover, out of the newly introduced PFMs, 31 are associated with TF dimers. The literature that provides orthogonal evidence for the newly introduced TF binding profiles is provided in the metadata. Additionally, we updated 160 TF binding profiles across the six taxa with new PFMs (Table 1).

High-quality PFMs lacking orthogonal support were included in the Unvalidated collection (298 new profiles; Supplementary Data 1—Supplementary Figure S1, Supplementary Data 2—Supplementary Table S1). Specifically, 115 TF binding profiles are associated with zinc-finger TFs and 95 associated with TFs binding DNA as dimers. We provide the Unvalidated collection of TF binding profiles to the community to use with due caution since they are not yet supported with orthogonal evidence. We extend our invitation to the user community to be involved in the motif curation process by providing either new unvalidated profiles to consider or support to existing profiles in the collection.

We exhaustively revised the metadata to update information about the TF names, the structural class and family of the TF DBDs (following TFClass (39)), and links to external databases such as UniProt (40), ReMap (36), UniBind (15,16) and DNA Readout Viewer (41), whenever possible. Finally, we removed 32 profiles from the CORE collection (22 plant, 6 vertebrate and 4 fungi profiles) as they corresponded to synonyms of already present TF profiles,

had low information content, or were derived from consensus strings (Table 1). In addition, we removed 85 profiles from the Unvalidated collection (44 vertebrate, 40 plant and 1 fungi profiles) because: (i) the corresponding profile or a new profile for the same TF was added to the CORE collection; (ii) the profile was of insufficient quality or (iii) the profile was misannotated (Supplementary Data 2—Supplementary Table S1; detailed list of all removed profiles at <https://jaspar.genereg.net/changelog/>).

The JASPAR 2022 CORE collection now stores 1955 non-redundant PFMs (841 for vertebrates, 656 for plants, 179 for fungi, 150 for insects, 43 for nematodes, and 86 for urochordates) (Table 1; Figure 1). Additionally, we maintained the associated collection of transcription factor flexible models (TFFMs; hidden Markov-based models capturing dinucleotide dependencies in TF–DNA interactions (42)) that were initialized using JASPAR CORE PFMs and trained on ChIP-seq data (Supplementary Data 1—Text). This process resulted in 303 new TFFMs (207 for vertebrates and 96 for plants).

Improved structural classification of plant TF DNA-binding domains. In JASPAR, TFs are classified based on TF-Class (39), which provides a hierarchical structural classification (including superclass, class, and family) originally designed for human TFs and later extended to mammals. Since plant genomes contain many classes of TFs absent from TFClass, we expanded the TF structural classification using TFClass guidelines (39) and published structural evidence (Supplementary Data 2—Supplementary Table S2). In some rare cases (e.g. GARP and NF-Y TFs), we slightly diverged from TFClass so that the TF common name expected by users is provided in the structural class or family name. We arbitrarily decided to classify plant specific RAV TFs that contain two types of DBD (B3 and AP2) in the B3 Class. WRKY TFs that have a Zinc finger and a DBD derived from a GCM fold have been classified under the GCM domain factors class and WRKY family, and not in the Zinc-coordinating DNA-binding domains superclass. This homogenised classification introduced 27 novel entries in the TF DBD structural classification (Supplementary Data 2—Supplementary Table S2) and led to numerous corrections in the class and family fields compared to previous JASPAR releases.

Word clouds of terms associated with TFs in the scientific literature

Biological information about TFs, or genes in general, is scattered across many different resources, with PubMed possibly being the most extensive one. In an attempt to provide rich annotations for the TFs in JASPAR, we mined the corpus of article abstracts available in the PubMed database (43). We compiled sets of abstracts associated with each TF and weighted each word present by its relative importance when compared to all abstracts associated with other TFs in the same taxon (Supplementary Data 1—Text for method details). For each TF, the 200 highest weighted words were used to create a word cloud summarizing the annotations associated with that TF. As an example, Figure 2 illustrates the word cloud of terms associated with the PAX6

Table 1. Growth overview of the CORE collection of JASPAR 2022 compared to the previous release

Taxonomic Group	Non-redundant PFM in JASPAR 2020	New non-redundant PFM in JASPAR 2022	Removed profiles	Upgraded profiles (from Unvalidated to CORE)	Updated PFMs in JASPAR 2022	Total PFMs (non-redundant) in JASPAR 2022
Plants	530	121	22	27	44	656
Vertebrates	746	76	6	25	102	841
Urochordata	1	85	-	-	-	86
Insects	143	7	-	-	-	150
Nematodes	43	-	-	-	-	43
Fungi	183	-	4	-	14	179
CORE total	1646	289	32	52	160	1955

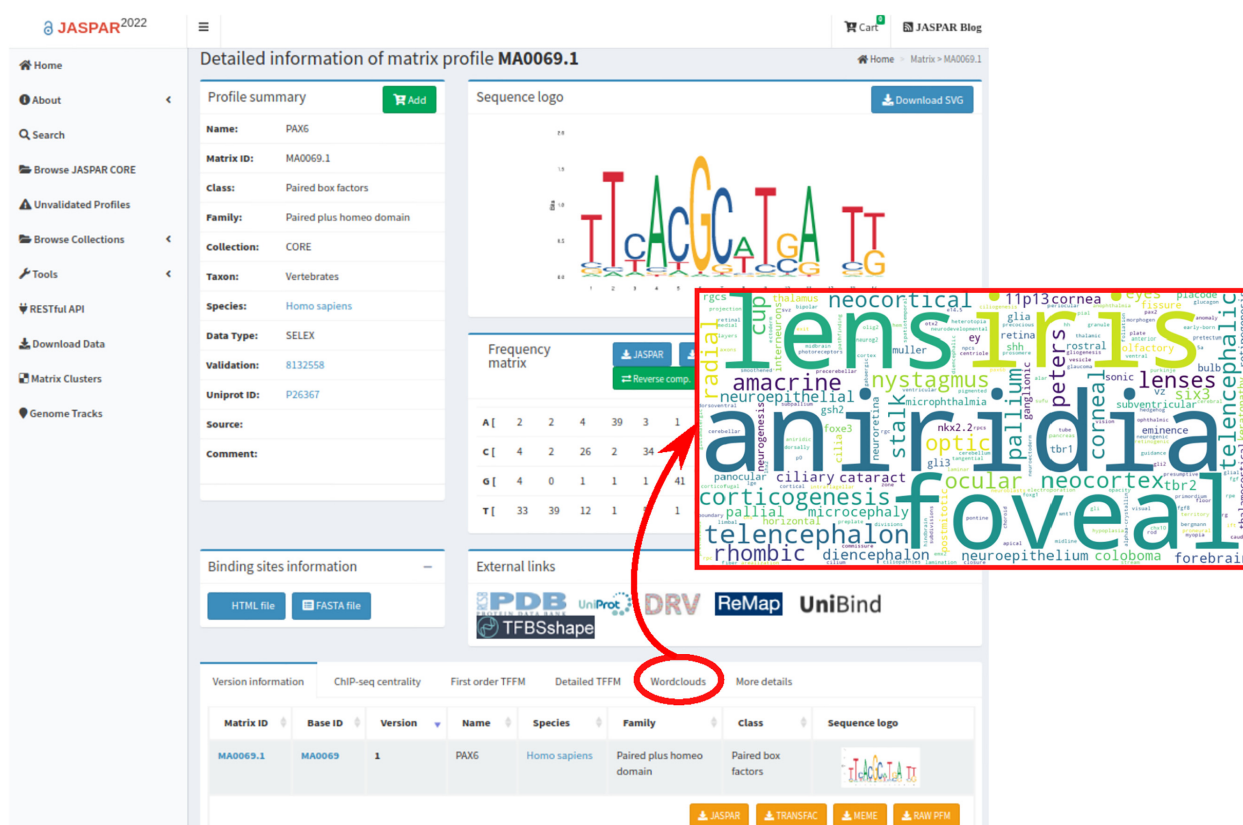


Figure 2. JASPAR TF word clouds. Webpage providing information about the binding profile associated with PAX6. The word cloud of terms obtained for PAX6 is highlighted in red, which supports the role of this TF in eye development and its implication in causing the genetic disorder aniridia.

TF in the scientific literature. Among the most significant terms, we find ‘lens’, ‘iris’, and ‘foveal’ that are representative of the importance of PAX6 in the development of the eye, while the term ‘aniridia’ reflects the link between some PAX6 mutations and the genetic disorder aniridia (44,45).

TF binding profile clusters, familial binding profiles, and genomic tracks

We updated the hierarchical clustering of the JASPAR TF binding profiles for each taxon with the RSAT matrix-clustering tool (46). Users can explore the CORE and Unvalidated collections through radial trees, which highlight the TF DBD structural classes, and directly access the un-

derlying profiles by clicking on the TF name (<https://jaspar.genereg.net/matrix-clusters>).

The hierarchical clustering of JASPAR PFMs was used to generate a collection of familial binding profiles (5,47), following previously published methodologies (16,48). Such familial motifs are useful in applications where motif redundancy (many TFs have similar binding preferences) is not desired. In brief, we defined clusters based on the DBD structural classes along the hierarchical clustering of PFMs. Next, we computed a familial binding profile for each cluster, summarizing the profiles within the clusters following (48) (Supplementary Data 1—Text for method details; Supplementary Data 1—Supplementary Figure S2). The familial binding profiles, also referred to as archetypes in (48), can be explored and downloaded at <https://jaspar>.

genereg.net/matrix-clusters and <https://jaspar.genereg.net/downloads/>, respectively.

One of the primary uses of PFMs is to predict binding sites. To facilitate this, we created ready-made prediction tracks for genome visualization and interpretation. Specifically, we scanned the genomes of eight organisms (*Arabidopsis thaliana*, *Caenorhabditis elegans*, *Ciona intestinalis*, *Danio rerio*, *Drosophila melanogaster*, *Homo sapiens*, *Mus musculus*, and *Saccharomyces cerevisiae*) with the JASPAR CORE PFMs associated with the same taxon to predict TFBSs and update the JASPAR TFBS genomic tracks. Moreover, we created a collection of familial TFBSs by merging overlapping TFBSs that were predicted from PFMs associated with the same familial binding profile (Supplementary Data 1—Text for method details). The TFBS predictions associated with all PFMs are available at http://expdata.cmmmt.ubc.ca/JASPAR/downloads/UCSC_tracks/2022/. The familial binding TFBSs are available at <https://jaspar.genereg.net/downloads/>. Finally, we provide JASPAR TFBS predictions as genomic tracks, which can be visualized in genome browsers. Notably, the UCSC Genome Browser (49) now presents predicted human (for the hg19 and hg38 genome assemblies) and mouse (for the mm10 and mm39 genome assemblies) JASPAR TFBS data as a native tracks for the human and mouse genomes with information such as TF names, TFBS prediction scores, and PFM logos (Supplementary Data 1 - Supplementary Figure S3).

A command-line tool to evaluate JASPAR TFBS enrichment in genomic regions

A common challenge in the field of transcriptional regulation is to predict the TF(s) most likely to control a set of cis-regulatory regions. This challenge is classically addressed by evaluating the enrichment for potential TFBSs associated with candidate TFs in the genomic regions of interest compared to background regions (16,26,50–53). We previously introduced an enrichment tool that evaluates the enrichment for sets of direct TF–DNA interactions from UniBind in user-provided DNA regions compared to background regions (16). Following the same strategy, we introduce a TFBS enrichment tool to predict TFs with an enrichment of JASPAR TFBSs using the Locus Overlap Analysis (LOLA) tool (54). The enrichment tool is available as a command-line tool (<https://jaspar.genereg.net/enrichment/>, https://bitbucket.org/CBGR/jaspar_enrichment/).

As a use case, we studied the differential enrichment of predicted TFBSs at DNase-seq peaks observed in A549 cells before and after 2 h treatment with 100 nM dexamethasone. DNase-seq is an assay capturing open chromatin regions (55). Dexamethasone is a known agonist of the glucocorticoid receptor (NR3C1), a nuclear receptor that binds the DNA upon ligand-based activation. Figure 3 provides a visual representation of the differential TFBS enrichment analysis results when considering DNase-seq peaks in treated versus untreated cells. As expected, NR3C1 (a member of the Steroid hormone receptors (NR3) family) was the top enriched TF ($-\log_{10}(P) = 58.77$). Among other TFs showing a high enrichment of TFBSs, we observed many members of the Three-zinc finger Kruppel-

related family (e.g. KLF factors, SP3, and SP9) (Supplementary Data 2—Supplementary Table S3). In another example, we observed the enrichment of TFBSs for the TFs FOXA1 and GATA3 in regions surrounding CpGs that are hypomethylated in estrogen receptor positive (ER+) breast cancers (56) (Supplementary Data 1—Supplementary Figure S4, Supplementary Data 2—Supplementary Table S4). These TFs are well established drivers of ER+ breast cancers binding to hypomethylated enhancers in ER+ breast cancers (56).

pyJASPAR—serverless pythonic interface to JASPAR data

All data is accessible through the JASPAR website (<https://jaspar.genereg.net/>), its associated RESTful API (<https://jaspar.genereg.net/api/>) (57), and the JASPAR2022 R/Bioconductor data package (source code at <https://github.com/da-bar/JASPAR2022>). The JASPAR database can also be accessed using Biopython (58) but it requires a local MySQL server to query the underlying database, which limits its access and use. To make access to JASPAR data easier, we introduce a new Python package, pyJASPAR (59), which allows users to query and access all JASPAR data without setting up the underlying MySQL database.

pyJASPAR is implemented in Python 3 using the Biopython *motifs* module and SQLite3 to provide a serverless Pythonic interface to the JASPAR database. The package allows users to query and access TF binding profiles across various releases of JASPAR. The releases currently available are: JASPAR2014, JASPAR2016, JASPAR2018, JASPAR2020, and JASPAR2022. The pyJASPAR package will be updated when future JASPAR releases become available. TF binding profiles can be retrieved using JASPAR matrix IDs, TF names, or other metadata information (Supplementary Data 1—Text for more details).

pyJASPAR is open source and the code is available at <https://github.com/asntech/pyjaspar/> under the GPL-3.0 License. The module can easily be installed with Conda from the bioconda channel (<https://anaconda.org/bioconda/pyjaspar>) (60) or from the Python Package Index with the *pip* command. Detailed documentation with usage examples is available at <https://pyjaspar.rtfd.io/>.

CONCLUSIONS AND PERSPECTIVES

For the 9th release of the JASPAR database, we substantially expanded the JASPAR CORE collection by 19% (341 added motifs). The newly introduced TF binding profiles were obtained after manual curation of PFMs predicted de novo from >3500 ChIP-seq/-exo datasets (from ReMap 2022 (36) and GEO (61)) or retrieved from publicly available repositories. While we continued our commitment to provide non-redundant, high-quality TF binding profiles for TFs across six taxa, this release comes with an important increase in the number of profiles for urochordata, with 86 PFMs available when JASPAR has contained a single one since 2006 (27). We now also provide TFBS predictions in *Ciona intestinalis* using the 86 JASPAR binding profiles. This increase exemplifies how the investigation of transcriptional regulation is expanding across more model organisms.

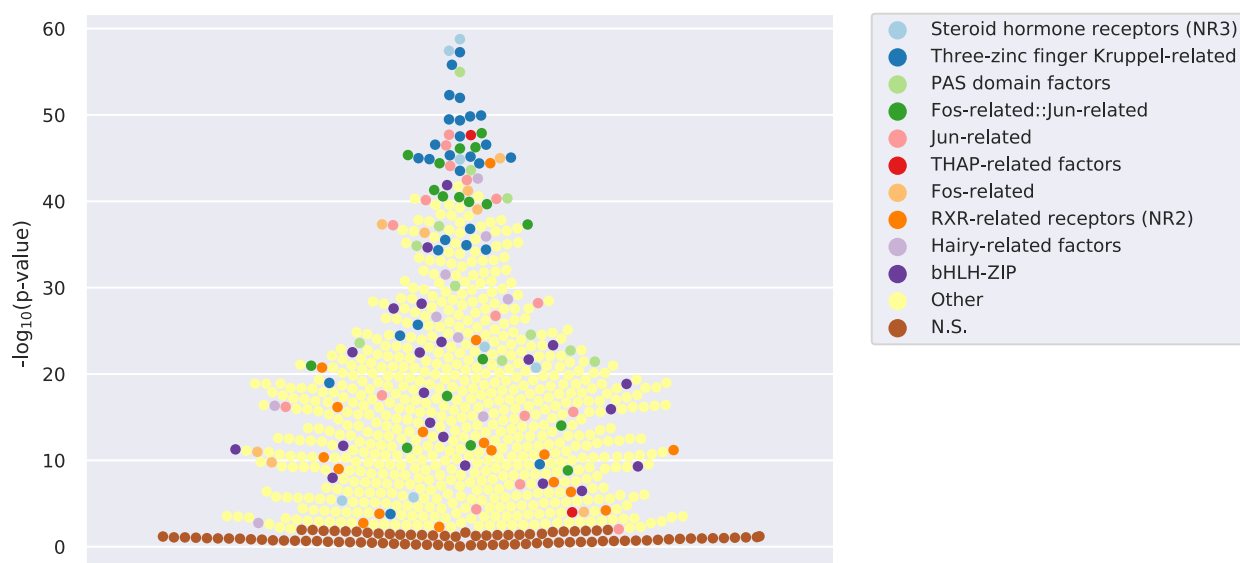


Figure 3. TFBS differential enrichment analysis on DNase-seq data for A549 cells before and after 2 h of dexamethasone treatment. Enrichment significance for each JASPAR profile from the vertebrate CORE collection is shown in the y-axis as $-\log_{10}(P)$ in this beeswarm plot. Each point depicts the Fisher exact test P -value (P) corresponding to a TF. The Points are colored based on the TF DBD structural family annotation, with a distinct color for each of the top 10 enriched families (see legend). Light yellow represents TF families outside of the top 10 enriched and with $-\log_{10}(P) > 3$ (Other) and brown represents TF families for which $-\log_{10}(P) \leq 3$ (non significant, N.S.).

An important question is what fraction of TFs have a binding profile in JASPAR. For humans, the JASPAR vertebrates CORE collection contains a binding profile for 43% of the 1639 human TFs (1), 56% when including the Unvalidated collection. If we consider the 1717 reported TFs for *A. thaliana* (62), 21% of these TFs have a profile in the JASPAR plants CORE collection, 22% when including the Unvalidated collection.

From the previous version of the Unvalidated collection (19), we found literature support for 81 profiles. Unfortunately, our team of curators did not succeed in identifying orthogonal validation in the literature for several high-quality motifs found enriched at ChIP-seq/-exo peak summits. As a result, 298 of such profiles were added to the previously introduced Unvalidated collection (19). The lack of experimental support for these profiles indicates an opportunity for the research field to explore these understudied TFs (63). Notably, 61% of the profiles in the vertebrates Unvalidated collection is associated with C2H2 zinc finger factors. A potential contributing challenge to obtaining orthogonal evidence may be the fact that many zinc-fingers, which represent the largest class of TFs, have been reported to regulate a limited number or even a single gene (e.g. Zfp568 (64), ZNF558 (65), ZNF410 (66) and ZFP64 (67)).

This JASPAR update comes with a new tool to compute TFBS enrichment given user-provided input and background sequences, mimicking a similar tool available with the UniBind database (16). The tool relies on the genome-wide TFBSs predicted using PFMs from the JASPAR CORE collection. Even though JASPAR predicted TFBSs will contain a high number of false positives, the enrichment

tool could be useful to suggest roles for TFs for which no direct TF-DNA interactions are available in UniBind (16).

Consistent with Weidemüller *et al.* (63), we noticed that limited scientific literature (i.e. at most a single manuscript in PubMed) exists for many TFs, which clearly impacts the utility of the JASPAR word clouds. This constraint varies between taxa. For example, while the average number of PubMed manuscripts per vertebrate TF was ~ 500 , urochordata TFs were associated with an average of only four manuscripts. Furthermore, a large number of TFs associated with individual PubMed manuscripts was observed. The average number of vertebrate TFs associated with PubMed IDs was ~ 19 with some associated with hundreds of TFs. An example is PubMed ID 21873635 that describes methods development of the Gene Ontology database (822 TFs), PubMed ID 12477932 that describes the Mammalian Gene Collection (MGC) Program (805 TFs), and PubMed ID 15618518 that analyzes the expression of TFs in the mouse brain (722 TFs). These manuscripts include general information about TFs. Therefore, we see opportunities to further improve the literature annotation engine, by decreasing the influence of outlier manuscripts and incorporating emerging natural language processing methods.

PFMs are still the most widely used models to represent TF binding preferences to DNA, despite their well-established caveats such as fixed-length and the failure to account for nucleotide interdependencies. A novel generation of computational models based on machine learning approaches such as deep learning are arising (68,69). Nevertheless, how to best share these models in a unified manner is still unclear despite some recent efforts (70) and will require discussion in the community. As the field moves to-

wards a unified framework to share such models, we expect their inclusion in future JASPAR releases.

AUTHORS' NOTE

The authors wish it to be known that, in their opinion, the first three authors should be regarded as joint first authors. The order of co-first authors provided here was decided through a mushroom picking competition around the Sognsvann lake, Oslo, Norway. Co-first authors can prioritise their names when adding this paper's reference to their résumés.

SUPPLEMENTARY DATA

[Supplementary Data](#) are available at NAR Online.

ACKNOWLEDGEMENTS

We thank the user community for useful input and the scientific community for performing experimental assays of TF-DNA interactions and for publicly releasing the data. We thank Shaun Mahony and Franklin Pugh for providing early access to ChIP-exo data from (38) and Emma Farley for pointers to urochordata data sets. We thank Vipin Kumar for contribution in the early curation sessions. We thank Walter Santana for his technical assistance to generate the motif radial trees, the UCSC Genome Browser project team for their assistance with the genome tracks, Harold Gutch and the NCMM IT team for their IT support, and Ingrid Kjelsvik for administrative support.

FUNDING

Norwegian Research Council [187615]; Helse Sør-Øst; University of Oslo through the Centre for Molecular Medicine Norway (NCMM) (to Mathelier group); Norwegian Research Council [288404 to R.R.P., J.A.C.M., Mathelier group]; Norwegian Cancer Society [197884 to R.B.L., Mathelier group]; GRAL program [ANR-10-LABX-49-01] with the frame of the CBH-EUR-GS [ANR-17-EURE-0003 to Parcy group]; PhD fellowship from CNRS Prime80 (to L.T.); NHGRI [5U41HG002371-20 to D.S.]; BOF grant from Ghent University [BOF24Y2019001901 to N.M.P.]; PhD Fellowship from the Provence-Alpes-Côte d'Azur Regional Council (Région SUD); Institut National de la Santé et de la Recherche Médicale (INSERM) (to F.H.); Novo Nordisk Foundation [NNF20OC0059951, NNF19OC0058262]; Danish Cancer Foundation [R204-A12359]; Danish Independent Research Fund [6110-00207B, 7014-00120B]; Carlsberg Foundation; ERC the European Union's Horizon 2020 research and innovation programme (MSCA ITN pHionIC) (to Sandelin group and collaborators); Canadian Institutes of Health Research [PJT-162120]; Natural Sciences and Engineering Research Council of Canada (NSERC) Discovery Grant [RGPIN-2017-06824]; BC Children's Hospital Foundation and Research Institute (to Wasserman group). The open access publication charge for this paper has been waived by Oxford University Press – NAR Editorial Board members are entitled to one free paper per year in recognition of their work on behalf of the journal.

Conflict of interest statement. None declared.

REFERENCES

- Lambert, S.A., Jolma, A., Campitelli, L.F., Das, P.K., Yin, Y., Albu, M., Chen, X., Taipale, J., Hughes, T.R. and Weirauch, M.T. (2018) The human transcription factors. *Cell*, **172**, 650–665.
- Weirauch, M.T., Yang, A., Albu, M., Cote, A.G., Montenegro-Montero, A., Drewe, P., Najafabadi, H.S., Lambert, S.A., Mann, I., Cook, K. *et al.* (2014) Determination and inference of eukaryotic transcription factor sequence specificity. *Cell*, **158**, 1431–1443.
- Reiter, F., Wienerroither, S. and Stark, A. (2017) Combinatorial function of transcription factors and cofactors. *Curr. Opin. Genet. Dev.*, **43**, 73–81.
- Venters, B.J. and Pugh, B.F. (2009) How eukaryotic genes are transcribed. *Crit. Rev. Biochem. Mol. Biol.*, **44**, 117–141.
- Wasserman, W.W. and Sandelin, A. (2004) Applied bioinformatics for the identification of regulatory elements. *Nat. Rev. Genet.*, **5**, 276–287.
- Jolma, A., Kivioja, T., Toivonen, J., Cheng, L., Wei, G., Enge, M., Taipale, M., Vaquerizas, J.M., Yan, J., Sillanpää, M.J. *et al.* (2010) Multiplexed massively parallel SELEX for characterization of human transcription factor binding specificities. *Genome Res.*, **20**, 861–873.
- Berger, M.F., Philippakis, A.A., Qureshi, A.M., He, F.S., Estep, P.W. 3rd and Bulky, M.L. (2006) Compact, universal DNA microarrays to comprehensively determine transcription-factor binding site specificities. *Nat. Biotechnol.*, **24**, 1429–1435.
- Johnson, D.S., Mortazavi, A., Myers, R.M. and Wold, B. (2007) Genome-wide mapping of in vivo protein-DNA interactions. *Science*, **316**, 1497–1502.
- Franklin Pugh, B. (2012) Ultra-high resolution mapping of protein-genome interactions using ChIP-exo. *BMC Proc.*, **6**, O27.
- He, Q., Johnston, J. and Zeitlinger, J. (2015) ChIP-nexus enables improved detection of in vivo transcription factor binding footprints. *Nat. Biotechnol.*, **33**, 395–401.
- Kaya-Okur, H.S., Wu, S.J., Codomo, C.A., Pledger, E.S., Bryson, T.D., Henikoff, J.G., Ahmad, K. and Henikoff, S. (2019) CUT&Tag for efficient epigenomic profiling of small samples and single cells. *Nat. Commun.*, **10**, 1930.
- Skene, P.J. and Henikoff, S. (2017) An efficient targeted nuclease strategy for high-resolution mapping of DNA binding sites. *Elife*, **6**, e21856.
- Slatery, M., Zhou, T., Yang, L., Dantas Machado, A.C., Gordân, R. and Rohs, R. (2014) Absence of a simple code: how transcription factors read the genome. *Trends Biochem. Sci.*, **39**, 381–399.
- Jolma, A., Yin, Y., Nitta, K.R., Dave, K., Popov, A., Taipale, M., Enge, M., Kivioja, T., Morgunova, E. and Taipale, J. (2015) DNA-dependent formation of transcription factor pairs alters their binding specificity. *Nature*, **527**, 384–388.
- Gheorghe, M., Sandve, G.K., Khan, A., Chêneby, J., Ballester, B. and Mathelier, A. (2019) A map of direct TF-DNA interactions in the human genome. *Nucleic Acids Res.*, **47**, e21.
- Puig, R.R., Boddie, P., Khan, A., Castro-Mondragon, J.A. and Mathelier, A. (2021) UniBind: maps of high-confidence direct TF-DNA interactions across nine species. *BMC Genomics*, **22**, 482.
- Stormo, G.D. (2013) Modeling the specificity of protein-DNA interactions. *Quant Biol*, **1**, 115–130.
- Koo, P.K. and Ploenzke, M. (2020) Deep learning for inferring transcription factor binding sites. *Curr Opin Syst Biol.*, **19**, 16–23.
- Fornes, O., Castro-Mondragon, J.A., Khan, A., van der Lee, R., Zhang, X., Richmond, P.A., Modi, B.P., Correard, S., Gheorghe, M., Baranašić, D. *et al.* (2020) JASPAR 2020: update of the open-access database of transcription factor binding profiles. *Nucleic Acids Res.*, **48**, D87–D92.
- Bailey, T.L., Johnson, J., Grant, C.E. and Noble, W.S. (2015) The MEME Suite. *Nucleic Acids Res.*, **43**, W39–W49.
- Yang, L., Zhou, T., Dror, I., Mathelier, A., Wasserman, W.W., Gordân, R. and Rohs, R. (2014) TFBSshape: a motif database for DNA shape features of transcription factor binding sites. *Nucleic Acids Res.*, **42**, D148–D155.
- Chiu, T.-P., Xin, B., Markarian, N., Wang, Y. and Rohs, R. (2020) TFBSshape: an expanded motif database for DNA shape features of transcription factor binding sites. *Nucleic Acids Res.*, **48**, D246–D255.

23. Gearing, L.J., Cumming, H.E., Chapman, R., Finkel, A.M., Woodhouse, I.B., Luu, K., Gould, J.A., Forster, S.C. and Hertzog, P.J. (2019) CiiDER: a tool for predicting and analysing transcription factor binding sites. *PLoS One*, **14**, e0215495.
24. Nguyen, N.T.T., Contreras-Moreira, B., Castro-Mondragon, J.A., Santana-Garcia, W., Ossio, R., Robles-Espinoza, C.D., Bahin, M., Collombet, S., Vincens, P., Thieffry, D. *et al.* (2018) RSAT 2018: regulatory sequence analysis tools 20th anniversary. *Nucleic Acids Res.*, **46**, W209–W214.
25. Herrmann, C., Van de Sande, B., Potier, D. and Aerts, S. (2012) i-cisTarget: an integrative genomics method for the prediction of regulatory features and cis-regulatory modules. *Nucleic Acids Res.*, **40**, e114.
26. Imrichová, H., Hulselmans, G., Atak, Z.K., Potier, D. and Aerts, S. (2015) i-cisTarget 2015 update: generalized cis-regulatory enrichment analysis in human, mouse and fly. *Nucleic Acids Res.*, **43**, W57–W64.
27. Vlieghe, D., Sandelin, A., De Bleser, P.J., Vleminckx, K., Wasserman, W.W., van Roy, F. and Lenhard, B. (2006) A new generation of JASPAR, the open-access repository for transcription factor binding site profiles. *Nucleic Acids Res.*, **34**, D95–D97.
28. Bryne, J.C., Valen, E., Tang, M.-H.E., Marstrand, T., Winther, O., da Piedade, I., Krogh, A., Lenhard, B. and Sandelin, A. (2008) JASPAR, the open access database of transcription factor-binding profiles: new content and tools in the 2008 update. *Nucleic Acids Res.*, **36**, D102–D106.
29. Portales-Casamar, E., Thongjuea, S., Kwon, A.T., Arenillas, D., Zhao, X., Valen, E., Yusuf, D., Lenhard, B., Wasserman, W.W. and Sandelin, A. (2010) JASPAR 2010: the greatly expanded open-access database of transcription factor binding profiles. *Nucleic Acids Res.*, **38**, D105–D110.
30. Zhu, F., Farnung, L., Kaasinen, E., Sahu, B., Yin, Y., Wei, B., Dodonova, S.O., Nitta, K.R., Morgunova, E., Taipale, M. *et al.* (2018) The interaction landscape between transcription factors and the nucleosome. *Nature*, **562**, 76–81.
31. Brozovic, M., Dantec, C., Dardailon, J., Dauga, D., Faure, E., Gineste, M., Louis, A., Naville, M., Nitta, K.R., Piette, J. *et al.* (2018) ANISEED 2017: extending the integrated ascidian database to the exploration and evolutionary comparison of genome-scale datasets. *Nucleic Acids Res.*, **46**, D718–D725.
32. Lambert, S.A., Yang, A.W.H., Sasse, A., Cowley, G., Albu, M., Caddick, M.X., Morris, Q.D., Weirauch, M.T. and Hughes, T.R. (2019) Similarity regression predicts evolution of transcription factor sequence specificity. *Nat. Genet.*, **51**, 981–989.
33. Ricardi, M.M., González, R.M., Zhong, S., Domínguez, P.G., Duffy, T., Turjanski, P.G., Salgado Salter, J.D., Alleva, K., Carrari, F., Giovannoni, J.J. *et al.* (2014) Genome-wide data (ChIP-seq) enabled identification of cell wall-related and aquaporin genes as targets of tomato ASR1, a drought stress-responsive transcription factor. *BMC Plant Biol.*, **14**, 29.
34. Du, M., Zhao, J., Tzeng, D.T.W., Liu, Y., Deng, L., Yang, T., Zhai, Q., Wu, F., Huang, Z., Zhou, M. *et al.* (2017) MYC2 orchestrates a hierarchical transcriptional cascade that regulates jasmonate-mediated plant immunity in tomato. *Plant Cell*, **29**, 1883–1906.
35. Liu, Y., Shi, Y., Zhu, N., Zhong, S., Bouzayen, M. and Li, Z. (2020) SGRAS4 mediates a novel regulatory pathway promoting chilling tolerance in tomato. *Plant Biotechnol. J.*, **18**, 1620–1633.
36. Hammal, F., de Langen, P., Bergon, A., Lopez, F. and Ballester, B. (2021) ReMap 2022: a database of Human, Mouse, Drosophila and Arabidopsis regulatory regions from an integrative analysis of DNA-binding sequencing experiments. *Nucleic Acids Res.*, <https://doi.org/10.1093/nar/gkab996>.
37. Barrett, T., Wilhite, S.E., Ledoux, P., Evangelista, C., Kim, I.F., Tomashevsky, M., Marshall, K.A., Phillippy, K.H., Sherman, P.M., Holko, M. *et al.* (2013) NCBI GEO: archive for functional genomics data sets—update. *Nucleic Acids Res.*, **41**, D991–D995.
38. Rossi, M.J., Kuntala, P.K., Lai, W.K.M., Yamada, N., Badjatia, N., Mittal, C., Kuzu, G., Bocklund, K., Farrell, N.P., Blanda, T.R. *et al.* (2021) A high-resolution protein architecture of the budding yeast genome. *Nature*, **592**, 309–314.
39. Wingender, E., Schoeps, T., Haubrock, M., Krull, M. and Dönitz, J. (2018) TFCClass: expanding the classification of human transcription factors to their mammalian orthologs. *Nucleic Acids Res.*, **46**, D343–D347.
40. UniProt, Consortium (2021) UniProt: the universal protein knowledgebase in 2021. *Nucleic Acids Res.*, **49**, D480–D489.
41. Adam, K., Gyorgypal, Z. and Hegedus, Z. (2020) DNA Readout Viewer (DRV): visualization of specificity determining patterns of protein-binding DNA segments. *Bioinformatics*, **36**, 2286–2287.
42. Mathelier, A. and Wasserman, W.W. (2013) The next generation of transcription factor binding site prediction. *PLoS Comput. Biol.*, **9**, e1003214.
43. Sayers, E.W., Beck, J., Bolton, E.E., Bourexis, D., Brister, J.R., Canese, K., Comeau, D.C., Funk, K., Kim, S., Klimke, W. *et al.* (2021) Database resources of the National Center for Biotechnology Information. *Nucleic Acids Res.*, **49**, D10–D17.
44. Jordan, T., Hanson, I., Zaletayev, D., Hodgson, S., Prosser, J., Seawright, A., Hastie, N. and van Heyningen, V. (1992) The human PAX6 gene is mutated in two patients with aniridia. *Nat. Genet.*, **1**, 328–332.
45. Gehring, W.J. and Ikeo, K. (1999) Pax 6: mastering eye morphogenesis and eye evolution. *Trends Genet.*, **15**, 371–377.
46. Castro-Mondragon, J.A., Jaeger, S., Thieffry, D., Thomas-Chollier, M. and van Helden, J. (2017) RSAT matrix-clustering: dynamic exploration and redundancy reduction of transcription factor binding motif collections. *Nucleic Acids Res.*, **45**, e119.
47. Mahony, S., Auron, P.E. and Benos, P.V. (2007) DNA familial binding profiles made easy: comparison of various motif alignment and clustering strategies. *PLoS Comput. Biol.*, **3**, e61.
48. Vierstra, J., Lazar, J., Sandstrom, R., Halow, J., Lee, K., Bates, D., Diegel, M., Dunn, D., Neri, F., Haugen, E. *et al.* (2020) Global reference mapping of human transcription factor footprints. *Nature*, **583**, 729–736.
49. Navarro Gonzalez, J., Zweig, A.S., Speir, M.L., Schmelter, D., Rosenbloom, K.R., Raney, B.J., Powell, C.C., Nassar, L.R., Maulding, N.D., Lee, C.M. *et al.* (2021) The UCSC Genome Browser database: 2021 update. *Nucleic Acids Res.*, **49**, D1046–D1057.
50. Kwon, A.T., Arenillas, D.J., Worsley Hunt, R. and Wasserman, W.W. (2012) oPOSSUM-3: advanced analysis of regulatory motif over-representation across genes or ChIP-Seq datasets. *G3*, **2**, 987–1002.
51. Puente-Santamaria, L., Wasserman, W.W. and Del Peso, L. (2019) TFEA.ChIP: a tool kit for transcription factor binding site enrichment analysis capitalizing on ChIP-seq datasets. *Bioinformatics*, **35**, 5339–5340.
52. Roopra, A. (2020) MAGIC: A tool for predicting transcription factors and cofactors driving gene sets using ENCODE data. *PLoS Comput. Biol.*, **16**, e1007800.
53. Arenillas, D.J., Forrest, A.R.R., Kawaji, H., Lassmann, T., Wasserman, W.W., Mathelier, A. and Consortium FANTOM (2016) CAGED-oPOSSUM: motif enrichment analysis from CAGE-derived TSSs. *Bioinformatics*, **32**, 2858–2860.
54. Sheffield, N.C. and Bock, C. (2015) LOLA: enrichment analysis for genomic region sets and regulatory elements in R and Bioconductor. *Bioinformatics*, **32**, 587–589.
55. Song, L. and Crawford, G.E. (2010) DNase-seq: a high-resolution technique for mapping active gene regulatory elements across the genome from mammalian cells. *Cold Spring Harb. Protoc.*, **2010**, db.prot5384.
56. Fleischer, T., Tekpli, X., Mathelier, A., Wang, S., Nebdal, D., Dhakal, H.P., Sahlberg, K.K., Schlichting, E., Børresen-Dale, A.-L., Oslo Breast Cancer Research Consortium (OSBREAC) *et al.* (2017) DNA methylation at enhancers identifies distinct breast cancer lineages. *Nat. Commun.*, **8**, 1379.
57. Khan, A. and Mathelier, A. (2017) JASPAR RESTful API: accessing JASPAR data from any programming language. *Bioinformatics*, **34**, 1612–1614.
58. Cock, P.J.A., Antao, T., Chang, J.T., Chapman, B.A., Cox, C.J., Dalke, A., Friedberg, I., Hamelryck, T., Kauff, F., Wilczynski, B. *et al.* (2009) Biopython: freely available Python tools for computational molecular biology and bioinformatics. *Bioinformatics*, **25**, 1422–1423.
59. Khan, A. (2021) pyJASPAR: a Pythonic interface to JASPAR transcription factor motifs. <https://doi.org/10.5281/zenodo.5062370>.
60. Grüning, B., Dale, R., Sjödin, A., Chapman, B.A., Rowe, J., Tomkins-Tinch, C.H., Valieris, R. and Köster, J. (2018) Bioconda: sustainable and comprehensive software distribution for the life sciences. *Nat. Methods*, **15**, 475–476.

61. Edgar, R. (2002) Gene Expression Omnibus: NCBI gene expression and hybridization array data repository. *Nucleic Acids Res.*, **30**, 207–210.
62. Jin, J., Tian, F., Yang, D.-C., Meng, Y.-Q., Kong, L., Luo, J. and Gao, G. (2017) PlantTFDB 4.0: toward a central hub for transcription factors and regulatory interactions in plants. *Nucleic. Acids. Res.*, **45**, D1040–D1045.
63. Weidemüller, P., Kholmatov, M., Petsalaki, E. and Zaugg, J.B. (2021) Transcription factors: Bridge between cell signaling and gene regulation. *Proteomics*, e2000034.
64. Yang, P., Wang, Y., Hoang, D., Tinkham, M., Patel, A., Sun, M.-A., Wolf, G., Baker, M., Chien, H.-C., Lai, K.-Y.N. *et al.* (2017) A placental growth factor is silenced in mouse embryos by the zinc finger protein ZFP568. *Science*, **356**, 757–759.
65. Johansson, P.A., Brattås, P.L., Douse, C.H., Hsieh, P., Pontis, J., Grassi, D., Garza, R., Jönsson, M.E., Atacho, D.A.M., Pirce, K. *et al.* (2020) A human-specific structural variation at the ZNF558 locus controls a gene regulatory network during forebrain development. bioRxiv doi: <https://www.biorxiv.org/content/10.1101/2020.08.18.255562>, 18 August 2020, preprint: not peer reviewed.
66. Lan, X., Ren, R., Feng, R., Ly, L.C., Lan, Y., Zhang, Z., Aboreden, N., Qin, K., Horton, J.R., Grevet, J.D. *et al.* (2021) ZNF410 uniquely activates the NuRD component CHD4 to silence fetal hemoglobin expression. *Mol. Cell*, **81**, 239–254.
67. Lu, B., Klingbeil, O., Tarumoto, Y., Somerville, T.D.D., Huang, Y.-H., Wei, Y., Wai, D.C., Low, J.K.K., Milazzo, J.P., Wu, X.S. *et al.* (2018) A transcription factor addiction in leukemia imposed by the MLL promoter sequence. *Cancer Cell*, **34**, 970–981.
68. Avsec, Ž., Weilert, M., Shrikumar, A., Krueger, S., Alexandari, A., Dalal, K., Fropp, R., McAnany, C., Gagneur, J., Kundaje, A. *et al.* (2021) Base-resolution models of transcription-factor binding reveal soft motif syntax. *Nat. Genet.*, **53**, 354–366.
69. Minnoye, L., Taskiran, I.I., Mauduit, D., Fazio, M., Van Aerschot, L., Hulselmans, G., Christiaens, V., Makhzami, S., Seltenhammer, M., Karras, P. *et al.* (2020) Cross-species analysis of enhancer logic using deep learning. *Genome Res.*, **30**, 1815–1834.
70. Avsec, Ž., Kreuzhuber, R., Israeli, J., Xu, N., Cheng, J., Shrikumar, A., Banerjee, A., Kim, D.S., Beier, T., Urban, L. *et al.* (2019) The Kipoi repository accelerates community exchange and reuse of predictive models for genomics. *Nat. Biotechnol.*, **37**, 592–600.

Paper II

**JASPAR 2024: 20th anniversary of
the open-access database of
transcription factor binding profiles**

JASPAR 2024: 20th anniversary of the open-access database of transcription factor binding profiles

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Abstract

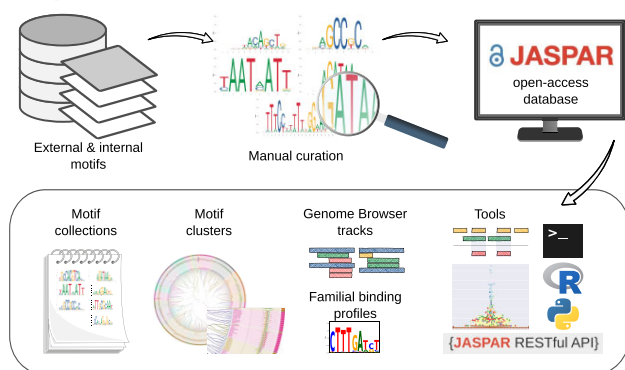
JASPAR (<https://jaspar.elixir.no/>) is a widely-used open-access database presenting manually curated high-quality and non-redundant DNA-binding profiles for transcription factors (TFs) across taxa. In this 10th release and 20th-anniversary update, the CORE collection has expanded with 329 new profiles. We updated three existing profiles and provided orthogonal support for 72 profiles from the previous release's UNVALIDATED collection. Altogether, the JASPAR 2024 update provides a 20% increase in CORE profiles from the previous release. A trimming algorithm enhanced profiles by removing low information content flanking base pairs, which were likely uninformative (within the capacity of the PFM models) for TFBS predictions and modelling TF-DNA interactions. This release includes enhanced metadata, featuring a refined classification for plant TFs' structural DNA-binding domains. The new JASPAR collections prompt updates to the genomic tracks of predicted TF binding sites (TFBSs) in 8 organisms, with human and mouse tracks available as native tracks in the UCSC Genome browser. All data are available through the JASPAR web interface and programmatically through its API and the updated Bioconductor and pyJASPAR packages. Finally, a new TFBS extraction tool enables users to retrieve predicted JASPAR TFBSs intersecting their genomic regions of interest.

Received: September 15, 2023. Revised: October 20, 2023. Editorial Decision: October 20, 2023. Accepted: October 31, 2023

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Graphical abstract



Introduction

Transcriptional gene regulation is mediated through the interactions of specific regulatory proteins, notably transcription factors (TFs), with *cis*-regulatory genomic elements, including promoters and enhancers (1). TFs are a broad class of proteins that regulate and mediate transcription; they can be classified as general TFs, sequence-specific DNA binding TFs, or transcriptional co-regulators (we refer the readers to (2) for more details). Within this report, we limit the application of the TF term to the subset that engages with DNA in a sequence-specific manner via DNA binding domains (DBDs) (1,2). The sequence-specific binding of TFs at *cis*-regulatory elements occurs at TF binding sites (TFBSs), delineated genomic regions that are typically 6–20 bp in length (3). TFs can be classified into structurally related families based on their DBDs. TFs with DBDs from the same structural family tend to recognise similar DNA sequence motifs, except for zinc finger proteins (4,5). Although several biochemical and genome-wide assays exist to assess TF-DNA affinities and detect TFBSs, these assays cannot be performed for all TFs in all cell types and biological conditions. Thus, computational approaches and models for TF binding remain critical. Aside from prediction of TFBS, such models can also be used as part of other analyses, such as enrichment of TFBSs in sets of promoters or enhancers, prediction of impacts of mutations in non-coding regions and guided *in vitro* mutagenesis (6,7).

Position frequency matrices (PFMs) remain the most common computational representation of TF-DNA interactions. PFMs are quantitative summaries of the DNA-binding preferences of a given TF, tallying the frequency of each nucleotide at each position of an aligned set of TFBSs and storing this information in a matrix form. These matrices can be converted into probabilistic matrices termed position weight matrices (PWMs) (8). The primary function of PWMs is to model the binding affinity or probability of interaction between a TF and a DNA sequence (8). As such, PWMs are used to predict TFBSs within any DNA sequence. To systematically explore the functional effects stemming from the binding process these models describe, it is necessary to have an extensive compilation that mirrors their diversity. To address this issue, multiple database solutions have been developed to collect and store PFMs, such as JASPAR (9), CIS-BP (10), and HOCOMOCO (11).

JASPAR is a regularly maintained, open-access database that stores manually curated high-quality DNA binding pro-

files of TFs as PFMs. For the past two decades, JASPAR has consistently upheld its core principles of (i) providing high-quality TF binding profiles, (ii) fostering open access, and (iii) ensuring ease of use. These core principles drove its evolution and growth and contributed to JASPAR's usefulness in the scientific community studying gene transcription regulation. JASPAR is now a standard resource in computational regulatory genomics (Supplementary Figure S1).

Central to JASPAR's mission is to provide the community with an extensively curated, non-redundant collection of profiles from published resources and literature. This effort produces a CORE collection of profiles where at least two orthogonal experimental supports validate each entry. The quality-control process grew along with JASPAR's expansion, introducing tools to aid curation. For example, we rely on the inference tool introduced in 2016 to support TF binding based on the similarity of DBDs between TFs (12). In 2020, we complemented the JASPAR CORE collection with the UNVALIDATED collection to reflect the increase in profiles derived from the broader use of high-throughput sequencing methods for which independent validation is yet to be produced (13). In order to have credible profiles within the UNVALIDATED collection, we kept the notion of high quality by putting the profiles under rigorous curation, where we look at the enrichment for TFBSs close to ChIP-seq peak summits (14), among other criteria. Although the profiles in the UNVALIDATED collection are computationally sound, we explicitly inform the users that these profiles should be used cautiously due to a lack of orthogonal support.

JASPAR offered the first open-access TF-binding profiles database with a web interface enabling direct download of the PFMs collected. The ease of data access and download manifests another core principle of JASPAR: the active effort to promote open science. As open science initiatives emerged in this field over the years, we witnessed a growing tendency to integrate those various resources into an ecosystem where tools and repositories build upon one another. These efforts eventually produced a synergistic ecosystem aligned with FAIR principles (15). The mutual integration of these various resources benefited from their early interoperability with an active effort to share standards for data formats and the possibility to leverage their respective source data. This latter point relies on the open accessibility of the data, which JASPAR adopted as a design choice from its beginning, with all profiles made available as simple flat text files. An additional

dimension to this ecosystem is the engagement with our user community in our effort to strengthen the growth and quality of JASPAR's content. For instance, we manifested our engagement using Google Groups for Q&As and by introducing an online form enabling users to notify the JASPAR team directly about profiles to add, validate, or update (13).

From its inception, JASPAR provided a web interface catering to the needs of both wet and dry researchers, illustrating JASPAR's emphasis on 'ease of use.' This principle translates into design choices for the JASPAR database, starting with the data organised in a simple schema trying to make one profile correspond to one TF or one dimeric complex (e.g. MYC::MAX) in one taxon, corresponding to the non-redundant aspect of JASPAR. However, this condition was later relaxed in 2020 to introduce binding variants (13), reflecting the possibility for some TFs to bind two or more distinct DNA motifs. This simple architecture makes JASPAR easy to engage for any user. We further provide different interfaces ranging from straightforward web-interface to programmatic access through various packages in Perl (16), Python (17), R/Bioconductor (18), and Ruby (12). Further catering to the community's needs, access to JASPAR data has been expanded to incorporate a platform- and language-independent interface through the recent introduction of the JASPAR RESTful API (19). JASPAR's ease of use was further facilitated by introducing the new web-interface and including the 'JASPAR dynamic tour' in 2018, which guides users through the typical tasks and novel features of the JASPAR website (20).

The team behind JASPAR has continuously tried to encompass the current scope of the data produced in the field. Of note in this effort is the rapid expansion witnessed following the introduction of high-throughput sequencing assays such as ChIP-seq (21), DAP-seq (22), PBM (23), SMILE-seq (24), and HT-SELEX (25), which accelerated the generation of datasets suitable for modelling TF-DNA interactions (26,27). This process, which started with vertebrates, eventually reached all taxa present in JASPAR. Faced with this expansion, we adapted the procedures and pipelines at the source of JASPAR, moving from the original manual survey of journals and subsequent construction of profiles directly from article tables or images to a systematic motif processing pipeline from online resources. This process was also fueled by the integration of data from ReMap (28), GTRD (29), and CIS-BP (4) directly into our pipelines, illustrating the merit of such open science efforts in consolidating the field as a whole again. Furthermore, the increasing number of profiles inferred across numerous taxa allowed new functionalities, such as interactive profile clustering trees (20) or archetypes (9), to assist users in interpreting individual profiles within a broader context.

Here, we present the 10th release of the JASPAR database, providing a substantial update and expansion of TF binding profiles in seven taxonomic groups. This update includes the addition of 329 profiles as PFMs, orthogonal support for 72 profiles stored in the previous release's UNVALIDATED collection (i.e. they are now part of the CORE collection), an update of three profiles and an update of the metadata for 241 profiles. Moreover, 182 new PFMs were added to the UNVALIDATED collection. This release further includes updates to the word clouds displaying enriched terms associated with TFs in the literature, further improvement of the struc-

tural classification of plant TF DBDs, updated native UCSC human and mouse genome tracks with TFBSs predicted from JASPAR TF binding profiles, and updates on the various JASPAR tools such as the TFBS enrichment tool, pyJASPAR and R/Bioconductor packages. We introduce a new motif trimming algorithm to remove flanking positions from PFMs with low information content. Finally, we provide a new TFBS extraction tool to perform extraction of predicted JASPAR TFBSs intersecting with an input set of genomic regions provided by users.

Results

Expansion and update of the TF binding profiles

We retrieved TF binding profiles as PFMs from Lai *et al.* (30) for the PBM experiments, from CIS-BP (4) for TFs in insects, nematodes, and plants, from Bass *et al.* for worms (31), and from the UNVALIDATED collection of the JASPAR 2022 release (9) (Supplementary Table S1). We processed ChIP-seq, ChIP-exo, and DAP-seq datasets from GTRD (29) and ChIP-exo data from Lai *et al.* (30) using the RSAT *peak-motifs* tool (32) to identify enriched motifs (as PFMs) in the corresponding peak sets (Supplementary Table S1 and Supplementary Text for dataset and method details). Our expert curators manually selected the PFMs supported by orthogonal evidence from the literature to either add them to or update former TF binding profiles in the JASPAR CORE collection. The PFMs deemed high quality, but for which our curators did not find any orthogonal support in the literature were added to the JASPAR UNVALIDATED collection. We complemented the JASPAR CORE collection with 329 TF binding profiles and updated three existing profiles with new PFMs (Table 1 and Figure 1). We identified orthogonal support in the literature for 72 profiles previously stored in the JASPAR UNVALIDATED collection, promoting them to the CORE collection. Overall, the new JASPAR 2024 CORE collection represents a 20% increase in the number of profiles compared to the previous release (Supplementary Table S2). We augmented the JASPAR UNVALIDATED collection with 182 profiles (Supplementary Table S3). Finally, we updated the metadata associated with the profiles wherever possible (for 241 and 55 CORE and UNVALIDATED profiles, respectively) and removed 28 profiles (11 from the CORE collection and 17 from the UNVALIDATED collection) as these profiles were either redundant with other profiles, incorrectly supported by the literature, or associated with a protein not considered as a DNA-binding specific TF.

The JASPAR 2024 release culminates with 2346 TF binding profiles in the CORE collection and 643 in the UNVALIDATED collection (Figure 1 and Supplementary Figure S2). In addition, we generated transcription factor flexible models (TFFMs; hidden Markov-based models capturing dinucleotide dependencies in TF-DNA interactions (33)) using all new CORE PFMs for which ChIP-seq data was available, resulting in 75 new TFFMs (45 for plants, 12 for vertebrates, 11 for insects, and 7 for nematodes). The JASPAR 2024 release compiles 1135 TFFMs (Supplementary Table S4). The web interface to access and visualise all profiles and metadata is accessible at <https://jaspar.elixir.no>, now hosted by ELIXIR Norway and recognised as a Norwegian bioinformatics service.

Table 1. Summary of the JASPAR 2024 update compared to the previous release

Taxonomic group in CORE collection	Non-redundant PFMs in JASPAR 2022	New non-redundant PFMs	Removed PFMs	Promoted PFMs (from UNVALIDATED to CORE)	Updated PFMs	Total non-redundant PFMs in JASPAR 2024
<i>Plants</i>	656	114	7	42	2	805
<i>Vertebrates</i>	841	19	1	20	1	879
<i>Urochordata</i>	86	-	-	8	-	94
<i>Insects</i>	150	135	1	2	-	286
<i>Nematodes</i>	43	61	1	-	-	103
<i>Fungi</i>	179	-	1	-	-	178
<i>Diatoms</i>	1	-	-	-	-	1
CORE total	1956	329	11	72	3	2346

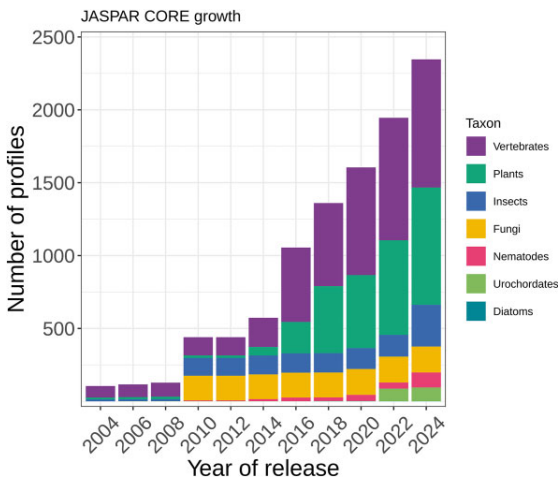


Figure 1. Overview of the growth of the number of profiles in JASPAR CORE collection across releases and taxons.

Trimming of TF binding profiles

Most TF binding profiles stored in JASPAR derive from computational *de novo* motif discovery tools applied to *in vitro* and *in vivo* data. The underlying algorithms sometimes report PFMs with low information content (IC) at the flanks (Figure 2A–B, top logos). The corresponding positions with low information content are likely uninformative (within the capacity of the PFM models) for predicting TFBSs and modelling TF–DNA interactions. We designed an algorithm to remove these uninformative flanking positions in the latest version of all the TF binding profiles available in the JASPAR CORE and UNVALIDATED collections (Supplementary Text for the detailed method). The bottom logos in Figures 2A–B illustrate case examples of the TF binding profile trimming algorithm results. The algorithm trimmed up to 19 positions (Figure 2C) in 1869 (1457 from the JASPAR 2022 CORE collection and 412 from the JASPAR 2022 UNVALIDATED collection) out of 2506 profiles. All newly curated profiles were trimmed by default. After trimming, the PFMs stored in JASPAR 2024 are 4–33 bp long (Supplementary Figure S3). As expected, the trimmed profiles concentrate on informative positions, as determined by Gini coefficients, which measure the inequality of values in a distribution (Figure 2D).

Improved structural classification of plant TFs

Previous JASPAR versions have used TFClass as the reference structural classification for TF DBDs (34,35). However,

this classification was built for mammal TFs and lacks many DBD types in plant genomes. This new JASPAR release benefited from the recent creation of a plant classification (Plant-TFClass) that includes 8 TF classes and 37 families absent from TFClass (36). We curated all entries in the JASPAR plant collection with this new classification.

TF binding profile clusters, familial binding profiles, word clouds and genomic tracks

Beyond TF binding profiles, JASPAR provides several complementary features to the community to compare, analyse, and interpret genomic data in the context of transcriptional regulation of gene expression. Users can visualise the TF binding profiles’ similarity in the CORE and UNVALIDATED collections through a radial tree. We updated the RSAT *matrix-clustering* tool (37) into a faster and expanded stand-alone version (https://github.com/jaimicore/matrix-clustering_stand-alone). We applied the tool to the PFMs stored in JASPAR to provide hierarchical clustering of the profiles in every taxon. To remove redundancy due to similar profiles, we computed familial binding profiles, which summarise similar profiles with a single PFM, by relying on hierarchical clustering (38). We followed the same methodology as introduced in the previous JASPAR release (9) to construct 408 familial binding profiles (155 for vertebrates, 52 for plants, 62 for fungi, 44 for nematodes, 76 for insects and 19 for urochordates). We provide all hierarchical clusters and familial binding profile summaries at <https://jaspar.elixir.no/matrix-clusters>.

In the JASPAR 2022 release, we introduced word clouds to summarise biological information associated with each TF. Specifically, the word clouds illustrate the significance of each word found in the abstracts associated with each TF by comparing their occurrences to those found in the abstracts of other TFs within the same taxon. For JASPAR 2024, we have created word clouds for newly added profiles and updated existing ones with up-to-date literature queries from PubMed.

We scanned the latest genome assemblies of eight species (*Arabidopsis thaliana*, *Caenorhabditis elegans*, *Ciona intestinalis*, *Danio rerio*, *Drosophila melanogaster*, *Homo sapiens*, *Mus musculus*, and *Saccharomyces cerevisiae*) with the latest version of all TF binding profiles from the corresponding taxon in the JASPAR CORE collection. This release includes both TF binding profiles for new TFs and trimmed profiles from the previous release to focus on informative positions. Upon comparing genome-wide TFBS predictions using JASPAR 2022 profiles with the corresponding trimmed profiles in JASPAR 2024, we found that most profiles yielded a

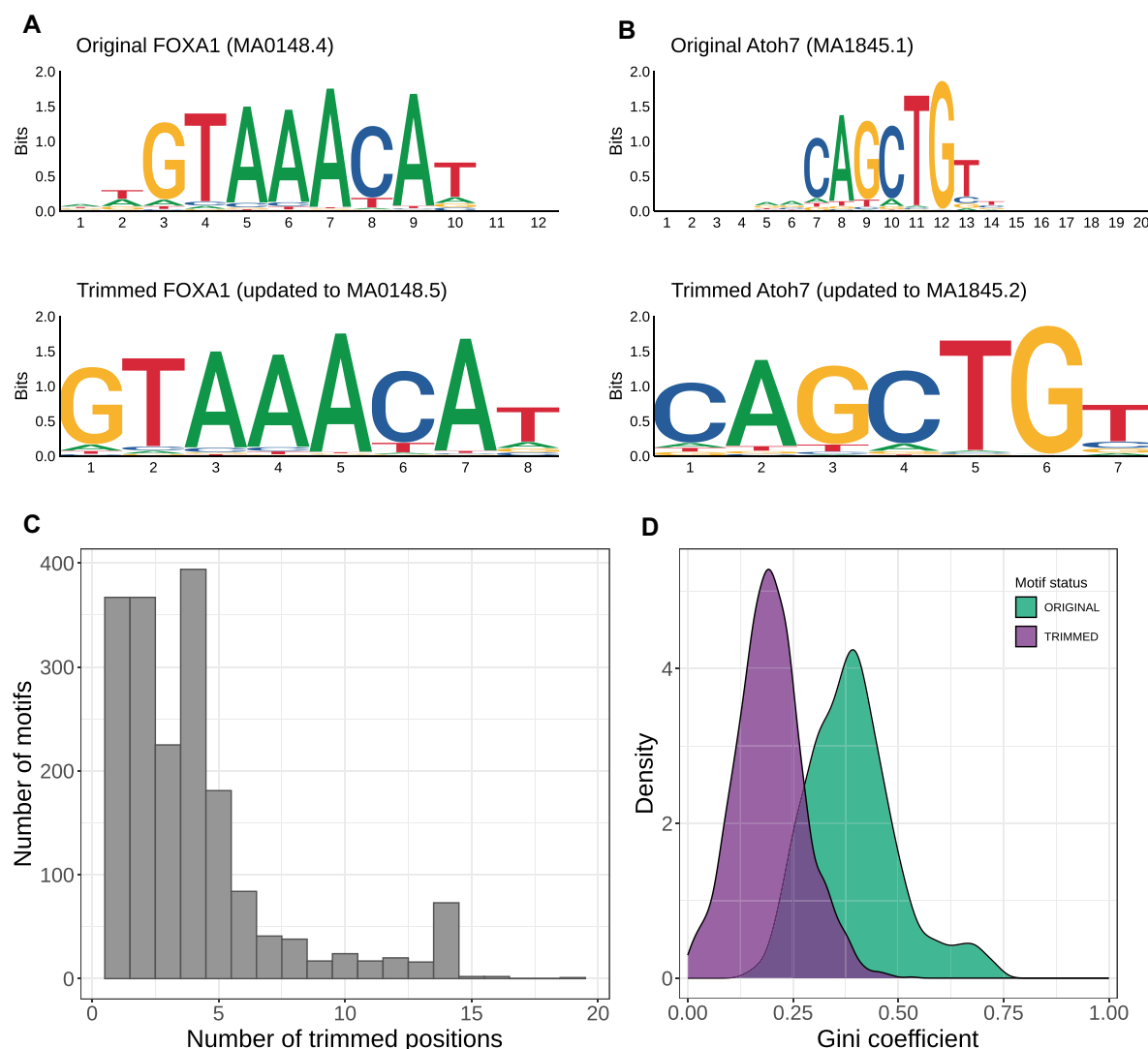


Figure 2. TF binding profile trimming. (A, B) Examples of trimmed PFMs for FOXA1 (A) and Atoh7 (B) TFs with the logos of the original PFMs at the top and the logos of the trimmed PFMs at the bottom. (C) The number of trimmed positions varied from 1 to 19 for PFMs originating from JASPAR 2022. (D) The distribution of Gini coefficients computed on the IC of each position in the original (green) and trimmed (purple) PFMs from JASPAR 2022 exhibits a concentration of informative positions in the updated PFMs.

comparable number of TFBS predictions. However, trimmed profiles predicted more TFBSs overall, with a few exceptions leading to a substantially increased number of predictions (Supplementary Figure S4). Additionally, we relied on the familial binding profiles to merge overlapping TFBSs predicted from similar PFMs. We provide users with the pre-computed TFBS prediction tracks for all TF binding profiles and familial binding profiles for genome visualisation and interpretation; the human and mouse TFBS tracks derived from TF binding profiles in the CORE collection are available as native tracks in the UCSC Genome Browser with prediction scores, logos, and TF names (39).

JASPAR-associated tools

pyJASPAR and R/Bioconductor data package

Beyond the web interface and RESTful API, we provide programmatic access to the data stored in the JASPAR database. Specifically, the users can utilise the pyJASPAR

Python package (<https://github.com/asntech/pyjaspar>) (40) and the JASPAR2024 R/Bioconductor data package (<https://bioconductor.org/packages/JASPAR2024>) to retrieve the data serverless. These packages allow for seamless integration of JASPAR data into Python and R workflows, providing the community with flexible and efficient means of programmatically retrieving and utilising JASPAR data for their research needs.

JASPAR TFBS enrichment tool

We previously introduced a command-line interface to perform TFBS enrichment analyses with JASPAR TFBS predictions in user-provided genomic regions (9). An update of the JASPAR TFBS predictions stored in the underlying LOLA databases for the enrichment tool accompanies the JASPAR 2024 release (41); users can find the LOLA databases on Zenodo at <https://doi.org/10.5281/zenodo.8341374>. Moreover, we now provide a Docker container for the

JASPAR TFBS enrichment tool at https://hub.docker.com/r/cbgr/jaspar_tfbs_enrichment.

JASPAR TFBS extraction tool

This new JASPAR release comes with a new computational tool to extract predicted JASPAR TFBSs intersecting with a user-provided input set of genomic regions. TFBSs can be further filtered by providing TF names, JASPAR matrix IDs and TFBS score thresholds. The software is available as a command-line tool at https://bitbucket.org/CBGR/jaspar_tfbs_extraction and in a Docker container at https://hub.docker.com/r/cbgr/jaspar_tfbs_extraction.

Conclusions and perspectives

For the 10th update of the JASPAR database, we expanded the JASPAR CORE collection by 20% (329 added and 72 upgraded profiles). The new profiles were introduced after manual curation, during which we curated 26 629 TF binding motifs obtained as PFMs or discovered from ChIP-seq/-exo or DAP-seq data. We also revised 2500 profiles from JASPAR 2022 to either promote them to the CORE collection, update the associated metadata, or remove them because of validation inconsistencies or poor quality. The insects and nematodes taxonomic groups received significant additions in the CORE collection (90% and 140% increase, respectively). Preparing this anniversary update, we focussed not only on expanding our profile collections but also on revisiting the quality of current motifs, especially renewing annotations for plant TF families and classes according to the Plant-TFClass (36) and searching for validation for profiles in UNVALIDATED collection.

The continuous expansion of the JASPAR database provides TF binding profiles for an increasing number of TFs from different organisms. With this release, the JASPAR CORE vertebrates collection presents a motif for 53% of the 1435 curated human TFs (58% of the 1118 orthologous mouse TFs) (2), 67% (64% of mouse orthologs) when adding profiles from the UNVALIDATED collection. The JASPAR CORE plants collection presents profiles for 32% of the 1717 reported *A. thaliana* TFs (42), 35% when including profiles from the UNVALIDATED collection. Another example is the JASPAR CORE insects collection, which provides a motif for 43% of the 628 TFs reported for *D. melanogaster* (43) and 50% when including UNVALIDATED profiles. A steady effort from the community to cover all TFs will be necessary to fill the remaining gap.

So far, the JASPAR database has stored and focused mostly on PFMs as the model of choice for TF-DNA interactions. We recognise that the PFMs stored in JASPAR assume nucleotide independence and do not consider the methylation status of nucleotides, which would require DNA methylation data and an expanded alphabet or specific representation (44–46). To account for successive nucleotide dependencies, we introduced transcription factor flexible models (TFFMs) into JASPAR for a set of profiles when data was available to compute them (12,33). Mostly based on convolutional neural networks, deep learning models are now considered state-of-the-art to accurately model data generated from genomic assays such as ChIP-seq, ChIP-nexus, or ATAC-seq (47–49). Some deep learning models have improved performance when initialising their convolutional filters with PWMs derived from JASPAR profiles (50,51), while many models assess derived patterns by comparison with JASPAR profiles

(47,48,52). The high quality of the modelling and improved methods to interpret the deep learning models make them attractive to decipher the *cis*-regulatory code (53). With deep learning approaches becoming critical to studying TF-DNA interactions and discovering the regulatory grammar controlling gene transcription, models based on neural networks will potentially replace PFMs. Similarly to PFMs, deep learning models could be curated and stored in JASPAR. Work remains to incorporate such models in a manner that holds true to the JASPAR ease-of-use principle, consistent with observations from other bioinformatics applications (54,55). Software tools for scanning DNA sequences with diverse deep learning-based motif models are maturing (56), as are methods for understanding motif enrichment and/or combinatorics (51,57–60). In addition to refining efficient motif scanning tools, one important remaining step is to determine how to effectively handle context-specific models (e.g. specific cell types or tissues), as such models can capture motifs for cooperative TFs unavailable in other cell types (61). These next steps demand continued innovation for JASPAR in the years ahead.

While we are pleased with the first 20 years of impact by JASPAR, we recognise that genomics and bioinformatics demand constant observation of the road ahead. We expect that the growing use of intelligent systems to derive inference from large data will continue to accelerate in use. As represented by large language models and deep learning-based image generation technologies, we can expect that bioinformatics methods will increasingly seek to bridge molecular data with structured knowledge. Including word clouds in JASPAR represents an initial movement in this direction. However, ultimately we expect that the binding models will need to be complemented with advanced knowledge representation about TFs, either in the form of knowledge graphs (such as in (62,63)) and/or vectors describing contextual embeddings.

Since the beginning, JASPAR has grown to provide the community with a high-quality, easy-to-use resource that promotes open science. Over the years, JASPAR's scale and scope faithfully accompanied the technological and scientific developments in the field. Striving to ensure the high quality of the database content throughout its continuous expansion meant regularly re-visiting the sourcing, processing, and presentation of JASPAR. This effort was further oriented towards maintaining JASPAR's ease of use, incorporating functionalities and content deliberately to address all user profiles' needs. At the age of 20, the JASPAR team looks ahead to strengthen its contribution within the open science ecosystem to which it contributed, consolidating and supporting the field in deepening our understanding of the role of TF binding in gene regulation.

Data availability

JASPAR is freely available at <https://jaspar.elixir.no/>.

Supplementary data

Supplementary Data are available at NAR Online.

Acknowledgements

As 'research parasites' (64), we processed publicly available datasets made accessible by researchers; we thank all the researchers for making this data available. We thank the

NCMM IT team, Pavel Zarva, Harold Gutch, and Torfinn Nome for IT support, Ingrid Kjelsvik for administrative support, the Kuijjer and Mathelier groups' members for insightful discussions, Stefanie Mantz for her contributions to the data analysis software development, and ELIXIR Norway, especially the Oslo node and Carlos Horro Marcos, for their support in the development and the Norwegian Research and Education Cloud (NREC) for hosting of the web interface.

Author contributions: We follow the Contributor Roles Taxonomy (CRediT) (65). Conceptualisation: I.R., R.R.-P., A.M.; Data curation: I.R., R.R.-P., R.B.L., V.K., J.A.C.-M., J.L., R.B.-M., F.P., A.M.; Formal analysis: I.R., R.R.-P., K.F., J.L.; Funding acquisition: A.M., F.P., W.W.W., B.L., E.H.; Investigation: I.R., R.R.-P., K.F., R.B.L., V.K., J.A.C.-M., J.L., R.B.M.; Methodology: I.R., R.R.-P., K.F., J.A.C.-M.; Project administration: I.R., R.R.-P., K.F., F.P., A.M.; Resources: A.M., E.H.; Software: I.R., R.R.-P., K.F., J.A.C.-M., A.K., D.B., O.F., J.C., S.G., M.J.; Supervision: A.M., F.P., W.W.W., B.L., E.H.; Validation: I.R., R.R.-P., K.F., R.B.L., V.K., J.A.C.-M., J.L., R.B.-M., F.P., A.M.; Visualisation: I.R., R.R.-P., K.F., J.A.C.-M.; Writing - original draft: R.R.-P., I.R., R.B.L., V.K., F.P., A.M.; Writing - review and editing: I.R., R.R.-P., K.F., R.B.L., V.K., J.A.C.-M., A.K., R.B.-M., J.L., O.F., A.S., W.W.W., J.C., S.G., E.H.

Funding

The Research Council of Norway [187 615]; Helse Sør-Øst, and the University of Oslo through the Centre for Molecular Medicine Norway (NCMM) to Mathelier group; Norwegian Cancer Society [197 884, 245 890] to Mathelier group; the Research Council of Norway [288 404] to Mathelier group; Research Council of Norway [322 392] to ELIXIR Norway; GRAL Labex financed within the University Grenoble Alpes graduate school (Ecoles Universitaires de Recherche)CBH-EUR-GS [ANR-17-EURE-0003 to Parcy group]; Novo Nordisk Foundation [NNF20OC0059951 to A.S.]; Danish Cancer Society [R325-A18868 to A.S.]; Natural Sciences and Engineering Research Council of Canada (NSERC) [RGPIN-2017-06824 to W.W.]; Canadian Institutes of Health Research [PJT-16120 to W.W.]. Funding for open access charge: Norges Forskningsråd.

Conflict of interest statement

O.F. is employed by Roche.

References

- Lambert,S.A., Jolma,A., Campitelli,L.F., Das,P.K., Yin,Y., Albu,M., Chen,X., Taipale,J., Hughes,T.R. and Weirauch,M.T. (2018) The Human Transcription Factors. *Cell*, **172**, 650–665.
- Lovering,R.C., Gaudet,P., Acencio,M.L., Ignatchenko,A., Jolma,A., Fornes,O., Kuiper,M., Kulakovskiy,I.V., Lægrend,A., Martin,M.J., *et al.* (2021) A GO catalogue of human DNA-binding transcription factors. *Biochim. Biophys. Acta Gene Regul. Mech.*, **1864**, 194765.
- Reid,J.E., Evans,K.J., Dyer,N., Wernisch,L. and Ott,S. (2010) Variable structure motifs for transcription factor binding sites. *BMC Genomics*, **11**, 30.
- Weirauch,M.T., Yang,A., Albu,M., Cote,A.G., Montenegro-Montero,A., Drewe,P., Najafabadi,H.S., Lambert,S.A., Mann,I., Cook,K., *et al.* (2014) Determination and inference of eukaryotic transcription factor sequence specificity. *Cell*, **158**, 1431–1443.
- Weirauch,M.T. and Hughes,T.R. (2011) A catalogue of eukaryotic transcription factor types, their evolutionary origin, and species distribution. *Subcell. Biochem.*, **52**, 25–73.
- Fornes,O., Gheorghe,M., Richmond,P.A., Arenillas,D.J., Wasserman,W.W. and Mathelier,A. (2018) MANTA2, update of the Mongo database for the analysis of transcription factor binding site alterations. *Sci. Data*, **5**, 180141.
- Fu,Y., Liu,Z., Lou,S., Bedford,J., Mu,X.J., Yip,K.Y., Khurana,E. and Gerstein,M. (2014) FunSeq2: a framework for prioritizing noncoding regulatory variants in cancer. *Genome Biol.*, **15**, 480.
- Stormo,G.D. (2013) Modeling the specificity of protein-DNA interactions. *Quant Biol*, **1**, 115–130.
- Castro-Mondragon,J.A., Riudavets-Puig,R., Rauluseviciute,I., Lemma,R.B., Turchi,L., Blanc-Mathieu,R., Lucas,J., Boddie,P., Khan,A., Manosalva Pérez,N., *et al.* (2022) JASPAR 2022: the 9th release of the open-access database of transcription factor binding profiles. *Nucleic Acids Res.*, **50**, D165–D173.
- Lambert,S.A., Yang,A.W.H., Sasse,A., Cowley,G., Albu,M., Caddick,M.X., Morris,Q.D., Weirauch,M.T. and Hughes,T.R. (2019) Similarity regression predicts evolution of transcription factor sequence specificity. *Nat. Genet.*, **51**, 981–989.
- Kulakovskiy,I.V., Vorontsov,I.E., Yevshin,I.S., Soboleva,A.V., Kasianov,A.S., Ashoor,H., Ba-Alawi,W., Bajic,V.B., Medvedeva,Y.A., Kolpakov,F.A., *et al.* (2016) HOCOMOCO: expansion and enhancement of the collection of transcription factor binding sites models. *Nucleic Acids Res.*, **44**, D116–D125.
- Mathelier,A., Fornes,O., Arenillas,D.J., Chen,C.-Y., Denay,G., Lee,J., Shi,W., Shyr,C., Tan,G., Worsley-Hunt,R., *et al.* (2016) JASPAR 2016: a major expansion and update of the open-access database of transcription factor binding profiles. *Nucleic Acids Res.*, **44**, D110–D115.
- Fornes,O., Castro-Mondragon,J.A., Khan,A., van der Lee,R., Zhang,X., Richmond,P.A., Modi,B.P., Correard,S., Gheorghe,M., Baranašić,D., *et al.* (2020) JASPAR 2020: update of the open-access database of transcription factor binding profiles. *Nucleic Acids Res.*, **48**, D87–D92.
- Pepke,S., Wold,B. and Mortazavi,A. (2009) Computation for ChIP-seq and RNA-seq studies. *Nat. Methods*, **6**, S22–S32.
- Wilkinson,M.D., Dumontier,M., Aalbersberg,I.J.J., Appleton,G., Axton,M., Baak,A., Blomberg,N., Boiten,J.-W., da Silva Santos,L.B., Bourne,P.E., *et al.* (2016) The FAIR Guiding Principles for scientific data management and stewardship. *Sci. Data*, **3**, 160018.
- Lenhard,B. and Wasserman,W.W. (2002) TFBS: computational framework for transcription factor binding site analysis. *Bioinformatics*, **18**, 1135–1136.
- Mathelier,A., Zhao,X., Zhang,A.W., Parcy,F., Worsley-Hunt,R., Arenillas,D.J., Buchman,S., Chen,C.-Y., Chou,A., Ienasescu,H., *et al.* (2014) JASPAR 2014: an extensively expanded and updated open-access database of transcription factor binding profiles. *Nucleic Acids Res.*, **42**, D142–D147.
- Tan,G. and Lenhard,B. (2016) TFBSTools: an R/bioconductor package for transcription factor binding site analysis. *Bioinformatics*, **32**, 1555–1556.
- Khan,A. and Mathelier,A. (2018) JASPAR RESTful API: accessing JASPAR data from any programming language. *Bioinformatics*, **34**, 1612–1614.
- Khan,A., Fornes,O., Stigliani,A., Gheorghe,M., Castro-Mondragon,J.A., van der Lee,R., Bessy,A., Chèneby,J., Kulkarni,S.R., Tan,G., *et al.* (2018) JASPAR 2018: update of the open-access database of transcription factor binding profiles and its web framework. *Nucleic Acids Res.*, **46**, D260–D266.
- Johnson,D.S., Mortazavi,A., Myers,R.M. and Wold,B. (2007) Genome-wide mapping of in vivo protein-DNA interactions. *Science*, **316**, 1497–1502.
- Bartlett,A., O'Malley,R.C., Huang,S.-S.C., Galli,M., Nery,J.R., Gallavotti,A. and Ecker,J.R. (2017) Mapping genome-wide

- transcription-factor binding sites using DAP-seq. *Nat. Protoc.*, **12**, 1659–1672.
23. Berger, M.F. and Bulky, M.L. (2009) Universal protein-binding microarrays for the comprehensive characterization of the DNA-binding specificities of transcription factors. *Nat. Protoc.*, **4**, 393–411.
 24. Isakova, A., Groux, R., Imbeault, M., Rainer, P., Alpern, D., Dainese, R., Ambrosini, G., Trono, D., Bucher, P. and Deplancke, B. (2017) SMiLE-seq identifies binding motifs of single and dimeric transcription factors. *Nat. Methods*, **14**, 316–322.
 25. Roulet, E., Busso, S., Camargo, A.A., Simpson, A.J.G., Mermod, N. and Bucher, P. (2002) High-throughput SELEX SAGE method for quantitative modeling of transcription-factor binding sites. *Nat. Biotechnol.*, **20**, 831–835.
 26. Muir, P., Li, S., Lou, S., Wang, D., Spakowicz, D.J., Salichos, L., Zhang, J., Weinstock, G.M., Isaacs, F., Rozowsky, J., et al. (2016) The real cost of sequencing: scaling computation to keep pace with data generation. *Genome Biol.*, **17**, 53.
 27. Reuter, J.A., Spacek, D.V. and Snyder, M.P. (2015) High-throughput sequencing technologies. *Mol. Cell*, **58**, 586–597.
 28. Hammal, F., de Langen, P., Bergon, A., Lopez, F. and Ballester, B. (2022) ReMap 2022: a database of Human, Mouse, Drosophila and Arabidopsis regulatory regions from an integrative analysis of DNA-binding sequencing experiments. *Nucleic Acids Res.*, **50**, D316–D325.
 29. Kolmykov, S., Yevshin, I., Kulyashov, M., Sharipov, R., Kondrakhin, Y., Makeev, V.J., Kulakovskiy, I.V., Kel, A. and Kolpakov, F. (2021) GTRD: an integrated view of transcription regulation. *Nucleic Acids Res.*, **49**, D104–D111.
 30. Lai, W.K.M., Mariani, L., Rothschild, G., Smith, E.R., Venters, B.J., Blanda, T.R., Kuntala, P.K., Bocklund, K., Mairose, J., Dweikat, S.N., et al. (2021) A ChIP-exo screen of 887 Protein Capture Reagents Program transcription factor antibodies in human cells. *Genome Res.*, **31**, 1663–1679.
 31. Fuxman Bass, J.I., Pons, C., Kozlowski, L., Reece-Hoyes, J.S., Shrestha, S., Holdorf, A.D., Mori, A., Myers, C.L. and Walhout, A.J. (2016) A gene-centered C. elegans protein-DNA interaction network provides a framework for functional predictions. *Mol. Syst. Biol.*, **12**, 884.
 32. Thomas-Chollier, M., Herrmann, C., Defrance, M., Sand, O., Thieffry, D. and van Helden, J. (2012) RSAT peak-motifs: motif analysis in full-size ChIP-seq datasets. *Nucleic Acids Res.*, **40**, e31.
 33. Mathelier, A. and Wasserman, W.W. (2013) The next generation of transcription factor binding site prediction. *PLoS Comput. Biol.*, **9**, e1003214.
 34. Wingender, E., Schoeps, T., Haubrock, M. and Dönitz, J. (2015) TFClass: a classification of human transcription factors and their rodent orthologs. *Nucleic Acids Res.*, **43**, D97–D102.
 35. Wingender, E., Schoeps, T., Haubrock, M., Krull, M. and Dönitz, J. (2018) TFClass: expanding the classification of human transcription factors to their mammalian orthologs. *Nucleic Acids Res.*, **46**, D343–D347.
 36. Blanc-Mathieu, R., Dumas, R., Turchi, L., Lucas, J. and Parcy, F. (2023) Plant-TFClass: a structural classification for plant transcription factors. *Trends Plant Sci.*, <https://doi.org/10.1016/j.tplants.2023.06.023>.
 37. Castro-Mondragon, J.A., Jaeger, S., Thieffry, D., Thomas-Chollier, M. and van Helden, J. (2017) RSAT matrix-clustering: dynamic exploration and redundancy reduction of transcription factor binding motif collections. *Nucleic Acids Res.*, **45**, e119.
 38. Sandelin, A. and Wasserman, W.W. (2004) Constrained binding site diversity within families of transcription factors enhances pattern discovery bioinformatics. *J. Mol. Biol.*, **338**, 207–215.
 39. Navarro Gonzalez, J., Zweig, A.S., Speir, M.L., Schmelter, D., Rosenbloom, K.R., Raney, B.J., Powell, C.C., Nassar, L.R., Maulding, N.D., Lee, C.M., et al. (2021) The UCSC Genome Browser database: 2021 update. *Nucleic Acids Res.*, **49**, D1046–D1057.
 40. Khan, A. (2021) pyJASPAR: a Pythonic interface to JASPAR transcription factor motifs. <https://github.com/asntech/pyjaspar/tree/v2.0.0>.
 41. Sheffield, N.C. and Bock, C. (2016) LOLA: enrichment analysis for genomic region sets and regulatory elements in R and Bioconductor. *Bioinformatics*, **32**, 587–589.
 42. Tian, F., Yang, D.-C., Meng, Y.-Q., Jin, J. and Gao, G. (2020) PlantRegMap: charting functional regulatory maps in plants. *Nucleic Acids Res.*, **48**, D1104–D1113.
 43. Gramates, L.S., Agapite, J., Attrill, H., Calvi, B.R., Crosby, M.A., Dos Santos, G., Goodman, J.L., Goutte-Gattat, D., Jenkins, V.K., Kaufman, T., et al. (2022) FlyBase: a guided tour of highlighted features. *Genetics*, **220**, iyac035.
 44. Xuan Lin, Q.X., Sian, S., An, O., Thieffry, D., Jha, S. and Benoukraf, T. (2019) MethMotif: an integrative cell specific database of transcription factor binding motifs coupled with DNA methylation profiles. *Nucleic Acids Res.*, **47**, D145–D154.
 45. Grau, J., Schmidt, F. and Schulz, M.H. (2023) Widespread effects of DNA methylation and intra-motif dependencies revealed by novel transcription factor binding models. *Nucleic Acids Res.*, **51**, e95.
 46. Viner, C., Ishak, C.A., Johnson, J., Walker, N.J., Shi, H., Sjöberg-Herrera, M.K., Shen, S.Y., Lardo, S.M., Adams, D.J., Ferguson-Smith, A.C., et al. (2023) Modeling methyl-sensitive transcription factor motifs with an expanded epigenetic alphabet. bioRxiv doi: <https://doi.org/10.1101/043794>, 28 February 2023, preprint: not peer reviewed.
 47. Avsec, Ž., Weilert, M., Shrikumar, A., Krueger, S., Alexandari, A., Dalal, K., Fropf, R., McAnany, C., Gagneur, J., Kundaje, A., et al. (2021) Base-resolution models of transcription-factor binding reveal soft motif syntax. *Nat. Genet.*, **53**, 354–366.
 48. Maslova, A., Ramirez, R.N., Ma, K., Schattner, H., Wang, C., Fox, C., Ng, B., Benoist, C., Mostafavi, S. and Immunological Genome Project/Immunological Genome Project (2020) Deep learning of immune cell differentiation. *Proc. Natl. Acad. Sci. U.S.A.*, **117**, 25655–25666.
 49. Brennan, K.J., Weilert, M., Krueger, S., Pampari, A., Liu, H.-Y., Yang, A.W.H., Morrison, J.A., Hughes, T.R., Rushlow, C.A., Kundaje, A., et al. (2023) Chromatin accessibility in the Drosophila embryo is determined by transcription factor pioneering and enhancer activation. *Dev. Cell*, **58**, 1898–1916.
 50. Quang, D. and Xie, X. (2016) DanQ: a hybrid convolutional and recurrent deep neural network for quantifying the function of DNA sequences. *Nucleic Acids Res.*, **44**, e107.
 51. Novakovsky, G., Fornes, O., Saraswat, M., Mostafavi, S. and Wasserman, W.W. (2023) ExplainNN: interpretable and transparent neural networks for genomics. *Genome Biol.*, **24**, 154.
 52. Yuan, H. and Kelley, D.R. (2023) scBasset: sequence-based modeling of single-cell ATAC-seq using convolutional neural networks. *Nat. Methods*, **19**, 1088–1096.
 53. Novakovsky, G., Dexter, N., Libbrecht, M.W., Wasserman, W.W. and Mostafavi, S. (2023) Obtaining genetics insights from deep learning via explainable artificial intelligence. *Nat. Rev. Genet.*, **24**, 125–137.
 54. Sapoval, N., Aghazadeh, A., Nute, M.G., Antunes, D.A., Balaji, A., Baraniuk, R., Barberan, C.J., Dannenfeller, R., Dun, C., Edrisi, M., et al. (2022) Current progress and open challenges for applying deep learning across the biosciences. *Nat. Commun.*, **13**, 1728.
 55. Auslander, N., Gussow, A.B. and Koonin, E.V. (2021) Incorporating machine learning into established bioinformatics frameworks. *Int. J. Mol. Sci.*, **22**, 2903.
 56. Zabardast, A., Tamer, E.G., Son, Y.A. and Yilmaz, A. (2023) An automated framework for evaluation of deep learning models for splice site predictions. *Sci. Rep.*, **13**, 10221.
 57. Kshirsagar, M., Yuan, H., Ferres, J.L. and Leslie, C. (2022) BindVAE: dirichlet variational autoencoders for de novo motif discovery from accessible chromatin. *Genome Biol.*, **23**, 174.
 58. Zhang, S., Ma, A., Zhao, J., Xu, D., Ma, Q. and Wang, Y. (2022) Assessing deep learning methods in cis-regulatory motif finding based on genomic sequencing data. *Brief. Bioinform.*, **23**, bbab374.

59. Yang,J., Ma,A., Hoppe,A.D., Wang,C., Li,Y., Zhang,C., Wang,Y., Liu,B. and Ma,Q. (2019) Prediction of regulatory motifs from human Chip-sequencing data using a deep learning framework. *Nucleic Acids Res.*, **47**, 7809–7824.
60. Ullah,F. and Ben-Hur,A. (2021) A self-attention model for inferring cooperativity between regulatory features. *Nucleic Acids Res.*, **49**, e77.
61. Phuylcharoen,M., Zarrineh,P., Bridoux,L., Amin,S., Losa,M., Chen,K., Bobola,N. and Rattray,M. (2020) Uncovering tissue-specific binding features from differential deep learning. *Nucleic Acids Res.*, **48**, e27.
62. Lobentanzer,S., Aloy,P., Baumbach,J., Bohar,B., Carey,V.J., Charoentong,P., Danhauser,K., Doğan,T., Dreoj,J., Dunham,I., *et al.* (2023) Democratizing knowledge representation with BioCypher. *Nat. Biotechnol.*, **41**, 1056–1059.
63. Wu,Y.-H., Huang,Y.-A., Li,J.-Q., You,Z.-H., Hu,P.-W., Hu,L., Leung,V.C.M. and Du,Z.-H. (2023) Knowledge graph embedding for profiling the interaction between transcription factors and their target genes. *PLoS Comput. Biol.*, **19**, e1011207.
64. Longo,D.L. and Drazen,J.M. (2016) Data sharing. *N. Engl. J. Med.*, **374**, 276–277.
65. Brand,A., Allen,L., Altman,M., Hlava,M. and Scott,J. (2015) Beyond authorship: attribution, contribution, collaboration, and credit. *Learn. Publ.*, **28**, 151–155.

Received: September 15, 2023. Revised: October 20, 2023. Editorial Decision: October 20, 2023. Accepted: October 31, 2023

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Paper III

Identification of transcription factor co-binding patterns with non-negative matrix factorization

IDENTIFICATION OF TRANSCRIPTION FACTOR CO-BINDING PATTERNS WITH NON-NEGATIVE MATRIX FACTORIZATION

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Abstract

Transcription factor (TF) binding to DNA is critical to transcription regulation. Although the binding properties of numerous individual TFs are well-documented, a more detailed comprehension of how TFs interact cooperatively with DNA, forming either complex or co-binding to the same region, is required. Indeed, the combinatorial binding of TFs is essential to cell differentiation, development, and response to external stimuli. We present COBIND, a novel method based on non-negative matrix factorization (NMF) to identify TF co-binding patterns automatically. COBIND applies NMF to one-hot encoded regions flanking known TF binding sites (TFBSs) to pinpoint enriched DNA patterns at fixed distances. We applied COBIND to 5,699 TFBS datasets from UniBind for 401 TFs in seven species. The method uncovered already established co-binding patterns (*e.g.*, between POU5F1 and SOX2 or SOX17) and new co-binding configurations not yet reported in the literature and inferred through motif similarity and protein-protein interaction knowledge. Our extensive analyses across species revealed that 67% of the TFs with discovered co-binding patterns share a co-binding motif with other TFs from the same structural family. The co-binding patterns captured by COBIND are likely functionally relevant as they harbor higher evolutionary conservation than isolated TFBSs. Open chromatin data from matching human cell lines further supported the co-binding predictions. Finally, we used single-molecule footprinting data from mouse embryonic stem cells to confirm that the COBIND-predicted co-binding events associated with some TFs likely occurred on the same DNA molecules.

200 words version:

Transcription factor (TF) binding to DNA is critical to transcription regulation. Although the binding properties of numerous individual TFs are well-documented, a more detailed comprehension of how TFs interact cooperatively with DNA is required. We present COBIND, a novel method based on non-negative matrix factorization (NMF) to identify TF co-binding patterns automatically. COBIND applies NMF to one-hot encoded regions flanking known TF binding sites (TFBSs) to pinpoint enriched DNA patterns at fixed distances. We applied COBIND to 5,699 TFBS datasets from UniBind for 401 TFs in seven species. The method uncovered already established co-binding patterns and new co-binding configurations not yet reported in the literature and inferred through motif similarity and

protein-protein interaction knowledge. Our extensive analyses across species revealed that 67% of the TFs shared a co-binding motif with other TFs from the same structural family. The co-binding patterns captured by COBIND are likely functionally relevant as they harbor higher evolutionary conservation than isolated TFBSs. Open chromatin data from matching human cell lines further supported the co-binding predictions. Finally, we used single-molecule footprinting data from mouse embryonic stem cells to confirm that the COBIND-predicted co-binding events associated with some TFs likely occurred on the same DNA molecules.

1. INTRODUCTION

The interactions between DNA and transcription factors (TFs) are crucial to transcription regulation as they intrinsically determine cell growth, development, and response to stimuli. TFs bind DNA at TF binding sites (TFBSs) in a sequence-specific manner through direct contact between DNA nucleotides and amino acids of the TFs' DNA-binding domain (DBD) (1). Structural similarities between DBDs classify TFs into structural classes and families, and TFs from the same class or family usually recognize similar DNA patterns (or motifs).

The binding properties of individual TFs have been vastly studied (2–4), and several databases store DNA binding profiles for TFs across multiple taxonomic groups (e.g., JASPAR, CIS-BP, and HOCOMOCO (5–7)). While these databases primarily provide TF binding motifs for individual TFs, there is a need to increase our understanding of how TFs cooperatively bind DNA to regulate transcription (8). The cooperative binding of TFs generates many possible binding combinations, thus increasing the complexity of gene regulatory networks (8, 9). Recent studies argue for a flexible grammar of motifs at cis-regulatory regions, where the spacing and orientation between TFBSs would not be a key determinant for transcription regulation (10–13). Nevertheless, some TFs physically cooperate, providing a strict motif syntax important for transcription regulation. For instance, exhaustively testing the combinatorics of liver-associated TFBSs with massively parallel reporter assays revealed that TFBS orientation and order are major drivers of gene regulatory activity (14). A well-characterized example of physical interaction between TFs is POU5F1 (OCT4) cooperation with either SOX2 or SOX17 in pluripotent cells. The spacing between the binding sites will determine POU5F1's partner and the corresponding regulatory effect (15, 16). Hence, the specific spacing and orientation between the canonical motifs recognized by two TFs can characterize their co-binding at given genomic regions (17, 18). However, the cooperative binding of two TFs can also slightly modify their individual DNA sequence preference (19). Consequently, systematically identifying cooperative events genome-wide with strict binding grammar is still challenging.

The CAP-SELEX experimental technique captures co-binding events between predefined sets of TFs *in vitro* (19). However, whether the same co-binding properties will occur *in vivo* remains to be determined. Therefore, computational methods, such as TF-COMB, TACO, SpaMo, and MCOT (20–23), leverage *in vivo* data, such as ChIP-seq (24), to predict co-binding events. These tools rely on identifying the co-occurrence of pre-defined genomic regions bound by the individual TFs or already known individual DNA binding motifs in predefined genomic regions. While these strategies reduce the search space, they restrict discoveries for the pairs of TFs where either bound regions or TF binding motifs exist for both TFs. When relying on already-known TF binding motifs, the quality of the available motif collections is a limiting factor, and this approach cannot discover new DNA sequence patterns. Another approach implemented in the RSAT *dyad-analysis* tool does not rely on

pre-defined motifs and predicts spaced pairs of motifs *de novo* starting from spaced 3-mer patterns (25, 26). Finally, deep learning approaches can infer regulatory patterns from experimental data, with the capacity to pinpoint TF co-binding events (12, 27). Even though the underlying algorithms are advanced and powerful, their complexity and interpretability make it challenging to characterize specific co-binding events without extensive *a posteriori* analyses.

In this study, we aimed to discover TF co-binding patterns in the vicinity of high-quality TFBSs (referred as anchors). The discovery of fixed co-binding patterns can be considered a matrix decomposition (or factorization) problem where an input matrix encodes nucleotides surrounding TFBSs. The ultimate goal is to group nucleotide patterns from this matrix. The non-negative matrix factorization (NMF) technique decomposes a given non-negative matrix into two low-rank, non-negative matrices to reveal underlying patterns and structures within the data (28, 29). This technique has been useful in computational biology to reveal molecular patterns from high-throughput data (28, 30). seqArchR is the first novel application of NMF for the simultaneous identification of sequence features and corresponding clusters (31). The seqArchR tool identifies critical DNA elements in promoter regions by applying NMF to the corresponding one-hot encoded sequences (31). This approach inspired us to use NMF to address the discovery of TF co-binding patterns.

We present COBIND, a Snakemake-based (32) pipeline for the *de novo* discovery of TF co-binding patterns from input sets of TFBSs (https://bitbucket.org/CBGR/cobind_tool). We applied COBIND to 5,699 sets of high-quality TFBSs from seven species (*Arabidopsis thaliana*, *Caenorhabditis elegans*, *Danio rerio*, *Drosophila melanogaster*, *Homo sapiens*, *Mus musculus*, and *Rattus norvegicus*) for 401 unique TFs stored in the UniBind database (33). COBIND recovered established and unreported co-binding events between TFs. Among TFs from the same structural family, the majority shared co-binding patterns. In human and mouse genomes, genomic regions harboring the predicted co-binding events are evolutionarily more conserved than genomic regions without co-binding. Moreover, increased chromatin accessibility in matching human cell lines supported the predicted co-binding events from bulk experimental data. Finally, using single-molecule footprinting data from mouse embryonic stem cells, we confirmed that the anchor and co-binding patterns associated with some TFs are significantly enriched for TF co-occupancy at the single-molecule level. Overall, COBIND can *de novo* discover regions in the genomes that harbor patterns of co-binding events between TFs.

2. MATERIALS AND METHODS

2.1 Predicting co-binding patterns with COBIND

COBIND takes a BED (or FASTA) formatted file providing the genomic coordinates of anchor regions (or DNA sequences centered at the anchor sites), such as TFBS locations, as input. The tool uses NMF to reveal recurring DNA motifs with space constraints in the regions surrounding the input anchors, which are not factorized. The underlying computational pipeline consists of the following steps (Figure 1):

Step 1: Extraction of flanking regions. COBIND extracts the DNA sequences surrounding the anchor TFBSs (n bp upstream and downstream; $n=30$ by default) using bedtools (v2.29.2) (34).

Step 2: One-hot encoding. The flanking sequences are one-hot encoded as vectors of 4 bits per DNA nucleotide (A: 1000, C: 0100, G: 0010, T: 0001; ambiguous nucleotides $\notin [A, C, G, T]$) are encoded as 1111 (**Figure 1 - Step 2**). COBIND constructs two matrices representing the upstream and downstream sequences by combining the vectors of one-hot encoded sequences. Each row corresponds to a sequence flanking an anchor TFBS from the input set.

Step 3: Non-negative matrix factorization. COBIND applies NMF to each matrix using the NMF function of scikit-learn (v0.23.2) (35). The NMF decomposes an input matrix into two matrices: one representing k components (or factors) of nucleotide patterns and one informing the presence of the identified patterns in each row of the input matrix (28, 30). Importantly, COBIND applies NMF with multiple values of k to increase its capacity to capture co-binding patterns with different resolutions (see section “Parameter settings” below) (**Figure 1 - Step 3**). Next, COBIND assigns input sequences to each component following the methodology described by Kim and Park (36). Finally, COBIND builds motifs as positional frequency matrices (PFM) for each component by computing the occurrence frequencies of each nucleotide at each position (**Additional file 1: Figure S1**).

Step 4: Motifs filtering. COBIND filters out PFMs with information content (IC) < 2 to focus on informative patterns. Next, it aims to identify motifs corresponding to positions with local enrichment of high IC to distinguish them from motifs with scattered high IC positions or equal distribution of IC along the flanking region (**Additional file 1: Figure S1**). Specifically, COBIND computes the Gini coefficient g of each PFM to measure the inequality of IC values (37). We discard PFMs with a low Gini coefficient ($g < 0.5$ by default).

Step 5: Motif trimming. COBIND extracts the positions corresponding to the local enrichment of high IC values, revealing co-binding motifs. Specifically, it first computes the smallest window of size n that contains at least half of the IC of the complete flank. Finally, COBIND doubles the size of the window (adding $n/2$ nucleotides up and downstream) to define the co-binding motif.

Step 6: Motif clustering. COBIND can identify similar motifs multiple times since (i) NMF runs with different values for k , (ii) similar motifs can have different spacings with the core motif, and (iii) multiple TFBS sets can be processed in parallel, resulting in the identification of similar motifs (**Figure 1 - Step 3**). COBIND provides non-redundant motifs by clustering identified motifs (**Figure 1 - Step 6**). Specifically, we use the RSAT *matrix-clustering* tool (38, 39) to cluster similar motifs (we provide the used parameters in **Additional file 1: Table S1**). For each motif cluster, COBIND constructs an archetypal motif following the methodology from (40), which summarises motifs by computing the means of the counts from the aligned matrices in each cluster. We discarded successive flanking positions with IC < 0.1 from the archetypal motif.

Similarly, we cluster and summarise the anchor TFBS sequences associated with each co-binding motif (**Additional file 1: Table S1**).

Step 7: Summarise the discovered co-binding events. COBIND computes the relative spacing and orientation to the anchor motif from each co-binding archetypal motif discovered. Finally, it illustrates the corresponding co-binding events as heatmaps (**Figure 1 - Step 7**). The spacing summary plot describes the relationship between the anchor (provided at the center of the plot) and the discovered co-binding patterns (upstream or downstream from the anchor). On the x-axis, one visualizes the number of nucleotides between the anchor and the discovered co-binding pattern motifs, illustrated as sequence logos. The color intensity in the heatmap illustrates the proportion of the input datasets with predicted co-binding motif instances. The inserted number indicates the proportion of unique sequences in the datasets with the corresponding co-binding pattern, orientation, and spacing, that are predicted to harbor the corresponding co-binding configurations.

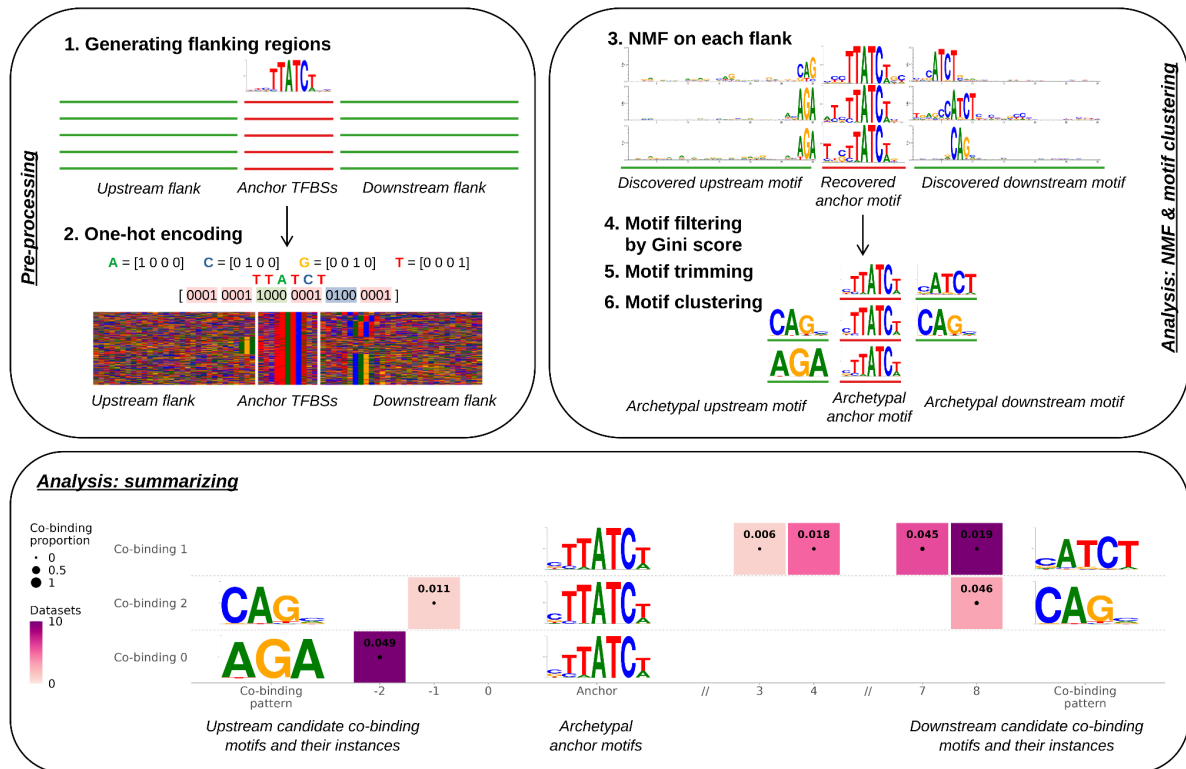


Figure 1. Schematic workflow of COBIND. 1) COBIND takes one or multiple **input** file(s) that provide the genomic location of regions of interest, such as TFBSs. The tool extracts upstream and downstream DNA sequences (default: 30 nt each). 2) The sequences are one-hot encoded to construct encoded matrices of upstream and downstream sequences. 3) COBIND applies NMF on each matrix to extract DNA patterns. 4) The tool computes the Gini coefficient of each DNA pattern and discards the uninformative patterns. 5) COBIND trims the identified flanking patterns to reveal possible TF co-binding motifs. 6) RSAT *matrix-clustering* groups redundant motifs, and COBIND constructs archetypal motifs summarising each cluster of similar motifs. 7) The predicted co-binding patterns are visualized through a heatmap representing the predicted spacing between the anchor (at the center) and the co-binding patterns with the corresponding orientation. The predicted co-binding patterns are provided on the left and/or right of the plots, depending on whether they were found upstream and/or downstream of the anchor sites. The position of the heatmap tiles provides, for each pattern, the number of nucleotides between the co-binding and anchor sites. The color of a heatmap tile represents the number of input datasets in which the co-binding pattern was predicted. The dot size and number within the heatmap tileblocks correspond to the proportion of input sequences in which a co-binding instance with a specific spacing was predicted.

2.2 Parameter settings

The hyperparameters of COBIND are the length of the flanking regions, the number of components for the different runs of the NMF, and the Gini coefficient threshold. Below, we describe the values used in this study for each parameter and why we selected these values.

Length of the flanking sequences. We considered 30 bp upstream and downstream of the input TFBSs to report co-binding events likely associated with physical interactions between the TFs. The longest TFBS motif used in UniBind is about 20 bp-long (33), and cooperative TFs bind TFBSs 9-10 bp away from each other on average (19). Furthermore, larger flanking regions resulted in noisier co-binding motifs and could miss co-binding motif configurations (**Additional file 1: Figure S2**).

Number of components. COBIND runs NMF several times with different numbers of components (k). We used synthetic data to estimate a suitable range of values for k . Specifically, we implanted different proportions of instances of predefined motifs in random sequences. We considered sets of 800, 1,000, 4,000, 8,000, 10,000, 40,000, and 80,000 synthetic sequences. For each set of synthetic sequences, we injected an instance of the considered motif in 0.5, 1, 3, 5, and 10% of the sequences. We ran COBIND with $k \in [2, 17]$ for each synthetic dataset. For each k value, we evaluated specificity and sensitivity by computing the F1 score and Matthew's correlation coefficient (MCC) for each discovered motif. Moreover, we reported the proportion of correctly predicted motifs. We considered that COBIND predicted the correct motif if it was similar to the injected motif - as assessed by Tomtom (41) with a p-value < 0.05 (Additional file 1: Table S1). We used the results from the synthetic data to select $k \in [3, 6]$ for further analyses (Additional file 1: Figure S3-15).

Gini coefficient threshold. To empirically determine the Gini coefficient threshold, we compared the distributions of Gini coefficients for the motifs predicted by COBIND on the TFBS datasets from UniBind (see below) with those of predictions from random sequences. For each UniBind set of TFBSs, we constructed a set of synthetic sequences by shuffling the DNA sequences flanking the TFBSs with the *kmer shuffling* module of the *BiasAway* tool with $k=1$ (42). For each species considered, we selected the Gini coefficient threshold corresponding to a 1% false discovery rate (Additional file 1: Figure S16). This strategy resulted in the following Gini coefficient thresholds: 0.50 for *Arabidopsis thaliana*, 0.46 for *Caenorhabditis elegans*, 0.48 for *Danio rerio*, 0.49 for *Drosophila melanogaster*, 0.50 for *Homo sapiens*, 0.54 for *Mus musculus*, and 0.53 for *Rattus norvegicus*. Finally, we estimated the stability of the thresholds obtained with this strategy by performing the same analysis multiple times and subsampling the number of datasets from *H. sapiens* and *M. musculus*. The Gini coefficient thresholds were stable across the different runs (Additional file 1: Figure S17).

2.3 Transcription factor binding site datasets

We used the robust collection of TFBS sets from the UniBind database (2021 version) (<https://unibind.uio.no/>) derived from 7 species (*Arabidopsis thaliana*, *Caenorhabditis elegans*, *Danio rerio*, *Drosophila melanogaster*, *Homo sapiens*, *Mus musculus*, *Rattus norvegicus*) (33). UniBind TFBS datasets derive from pairs of ChIP-seq peak datasets and JASPAR TF binding profiles. When multiple TF binding profiles exist for a given TF, we selected, for each corresponding ChIP-seq dataset, the profile providing the best centrality p-value as assessed in UniBind. We did not consider JASPAR TF binding profiles associated with TF dimers. Datasets with less than 1000 TFBSs were filtered out (Additional file 1: Table S2).

2.4 Benchmarking

Synthetic datasets

We generated the synthetic datasets following the methodology mentioned above. We constructed synthetic datasets with 800, 1000, 4000, 8000, 10,000, 40,000, and 80,000 sequences with instances of a known motif inserted in 0.5, 1, 3, 5, and 10% of the sequences. We considered 13 motifs (3- to 19-bp long) from the JASPAR database (7), which were synthetically inserted into random sequences with varying spacing (6- to 22-bp) from the anchor sites. To run SpaMO, we inserted the CTCF

(MA0139.1) motif at the center of the flanking sequences since SpaMo requires a motif to anchor its search for co-binding patterns.

Comparisons between COBIND and other tools

We used the synthetic data to compare COBIND with **STREME** (43), RSAT peak-motifs *dyad-analysis* (25, 26, 44, 45), RSAT peak-motifs *oligo-analysis* (26, 44, 45), RSAT peak-motifs *position-analysis* (44, 45), and SpaMo (21). We compared the motifs predicted by the different tools with the original inserted motifs using Tomtom (41) (**Additional file 1: Table S1**). We considered a discovered motif as a match to the injected motif if Tomtom predicted them to be similar with a p-value < 0.05. **We provide the number of incorrectly predicted motifs.** To run SpaMo, we used the JASPAR 2020 CORE collection of non-redundant profiles (46). **We computed the F1 score and Matthew's correlation coefficient (MCC) to assess the capacity of the tools to retrieve the correct sequences where the motifs were injected. When the correct motif was predicted multiple times, we reported the one providing the highest F1 score.**

STREME and RSAT algorithms were run to discover a maximum of 18 motifs to match the number of possible motifs to be predicted by COBIND. All tools were run with default parameters otherwise. When estimating the running time, we used the parallelizable version of the NMF for COBIND.

2.5 Inference of the co-binding transcription factors from the discovered co-binding motifs

We assessed the similarities between a predicted co-binding motif and known motifs associated with 5,272 TFs - collected from JASPAR 2022 non-redundant CORE, JASPAR 2022 non-redundant Unvalidated taxon-specific, and CIS-BP (5, 7) - using Tomtom (**Additional file 1: Table S1**) (41). We predicted TF_B to bind the predicted co-binding motif associated with anchor TFBSs of TF_A if Tomtom reported a similarity p-value < 0.05 with the known canonical motif bound by TF_B (criteria I). To assess possible physical interactions between TF_A and TF_B, we retrieved protein-protein interaction (PPI) data from STRING v11.0 (47). We considered TF_A and TF_B to interact physically when their STRING PPI score was above or equal to 500 (criteria II). We reported the co-binding pair TF_A-TF_B to be already “known” when they met criteria I and II. We reported TF_A-TF_B as a novel co-binding pair when they met criteria I but not II.

2.6 Shared co-binding motifs across transcription factor structural families

To assess the conservation of co-binding motifs across TFs, we ran COBIND (up to Step 5) on all UniBind datasets. We independently clustered all the predictions (COBIND Step 6) for each TF structural family; note that this step considered TFs from multiple species. **TFs without known structural families were excluded from this analysis.** When the co-binding motifs associated with two (or more) TFs from the same family were similar (i.e., belonged to the same cluster), we considered the TFs to share the same co-binding motif.

2.7 Evolutionary conservation

We downloaded the following conservation tracks in bigwig format from the UCSC genome browser: phastCons100way (human; hg38), phastCons60way (mouse; mm10), phastCons135way (*C. elegans*; WBcel235), and phastCons27way (*D. melanogaster*; dm6) (48). We obtained the 63 flowering plants PhastCons conservation track from PlantRegMap (49) for analyzing datasets from *A. thaliana*. We computed the conservation scores of a genomic region using the *aggregate* and *extract* functions of bwtool (v1.0) (50).

We compared the distributions of conservation scores between two regions with one-sided Wilcoxon signed-rank tests. Specifically, we compared the conservation values at the anchor (or co-binding) sites between the genomic regions predicted to harbor a co-binding pattern or not. These comparisons result in two p-values per co-binding pattern - one comparing the anchor sites and one comparing the co-binding sites. Histograms were used to summarise the p-values across datasets, considering the co-binding pattern with the lowest p-value per dataset. We compared the p-values to random expectation. Specifically, we randomly assigned, for each dataset, each genomic region to harbor or not a co-binding pattern (keeping the same number of regions in each category as the number predicted by COBIND). We performed five randomizations per dataset and reported the combined results.

2.8 DNase I hypersensitive footprinting analysis

We downloaded DNase I hypersensitivity (DHS) data generated by the ENCODE consortium for multiple human cell types in bigwig format at https://resources.altius.org/~jvierstra/projects/footprinting_2020/ (51). We considered the 53 cell lines or tissues for which both DHS and COBIND predictions were available; we analyzed 66 DHS datasets associated with 70 co-binding configurations for 44 TF binding profiles as anchors. We followed the methodology described above to compare evolutionary conservation scores when comparing the DHS signal between two genomic regions (with or without a co-binding event predicted by COBIND).

2.9 Single-molecule footprinting analysis

We downloaded 12 double enzyme single-molecule footprinting (SMF) datasets produced from tripleDNA methyltransferases knockout lines of mouse embryonic stem cells (mESCs) from ArrayExpress with accession numbers E-MTAB-9033 and E-MTAB-9123 (52). Whenever applicable, we performed the analyses of SMF data with the SingleMoleculeFootprinting R package following the instructions previously described (53). To compare the SMF data in the genomic regions predicted by COBIND to contain or not a co-binding event in mESC datasets, we modified the SortReadsByTFCluster_MultiSiteWrapper functions from the development version of the SingleMoleculeFootprinting package (54) (https://bitbucket.org/CBGR/cobind_manuscript/src/master/bin/single_molecule/analyse_sm/sort_tf_reads_by_cobind_clusters.R). Using the methylation status provided by the SMF data, we determined each genomic region to be either (i) free of nucleosomes (“Accessible”), (ii) occupied by nucleosomes (“Nucleosome”), (iii) bound only by the anchor (“Anchor”), (iv) bound only by the co-binding TF

(“Co-binding”), or (v) bound by both the anchor and co-binding TF (“Anchor + Co-binding”). We summed the number of molecules for each state in different samples and computed the proportions of molecules. We summarised the proportions of each state for all predictions and all anchors together. We compared the proportions of each state in genomic regions predicted to contain a co-binding event or not using a two-sided Wilcoxon signed-rank test. We compared the same type of regions in the “Anchor + Co-binding” state for individual anchors using a one-sided Wilcoxon signed-rank test.

3. RESULTS

3.1 COBIND discovers co-binding patterns *de novo*

Comparisons with other tools on simulated data

We report a new computational framework, COBIND, to predict **space-fixed** co-binding patterns in the vicinity of TFBSs provided as input. COBIND relies on applying NMF to the one-hot encoded regions flanking the **user-provided** TFBSs to predict co-occurring DNA motifs with fixed spacing (Materials and Methods). We compared COBIND to other tools discovering motifs *de novo* (STREME, RSAT *dyad-analysis*, RSAT *oligo-analysis*, and RSAT *position-analysis*). Additionally, we compared COBIND to SpaMO, which predicts spatially co-occurring instances of known motifs (Additional file 1: Table S3). For comparisons, we generated synthetic data by injecting instances of **known motifs (13 distinct motifs were used)** at different frequencies in random sets of **800 to 80,000** DNA sequences (Materials and Methods).

COBIND predicted the correct motif inserted when considering a minimum of **800** sequences **for 10 of the 13 used inserted motifs**. For example, COBIND accurately predicted the injected ATF4 motif in datasets of varying sizes without incorrect motif predictions (Figure 2). When considering the shorter TAL1 motif (3-bp known to be co-bound by the dimer GATA:TAL1), COBIND provides the most accurate predictions in multiple datasets when other tools obtain lower F1 scores and MCC values and predict incorrect motifs (Additional file 1: Figure S18-29). Overall, COBIND predicted no false positive motif across most configurations (see the number of **incorrect motifs discovered in Additional file 1: Figure S18-29**). In most cases, COBIND discovered the injected motifs in the correct sequences, as illustrated by high F1 scores and MCC values. Comparably, STREME discovered only correct motifs but failed to predict any motifs in some configurations (especially when the number of sequences with the injected motif was low). When considering many sequences (datasets of 40,000 and 80,000 sequences), COBIND and STREME recovered the correct motifs. SpaMo discovered the correct inserted motif in most cases, but it came at the cost of predicting many false positives. One should note that SpaMo requires a reference set of known motifs to predict co-occurrences, while COBIND predicts co-binding motifs *de novo*.

COBIND exhibited a similar run time to the other tools on datasets with up to 10,000 sequences. Nevertheless, it was slower than the other tools, except STREME, with larger datasets (>40,000 sequences) due to the motif clustering step applied to the results of the NMF with 3 to 6 components. Notwithstanding, the COBIND workflow (from motif discovery with NMF to co-binding summary) takes ~500 seconds on 40,000 sequences as it allows for parallelization across CPU cores (Additional file 1: Figure S30).

co-binds with SOX17 to control endoderm differentiation at other regulatory elements (11, 16). While POU5F1 and SOX2 cooperate through binding at instances of their respective canonical motifs, POU5F1 can associate with SOX17 to bind a compressed motif (**Figure 3A**). COBIND successfully discovered the two co-binding patterns - canonical and compressed - recognized by POU5F1 when applied to the SOX2 and SOX17 TFBS datasets from *H. sapiens* and *M. musculus* (**Figure 3B**). COBIND retrieved the correct pattern in 11 human SOX2 datasets (58%), with ~7% of the sequences predicted to harbor the pattern, and in 20 mouse datasets (27%), with ~6% of the sequences harboring the pattern (**Additional file 1: Figure S31A-B**). In agreement with previous studies, the datasets where COBIND predicted the pattern derived from human and mouse ESC and mouse embryonic fibroblast (16, 18, 55). When considering the SOX17 datasets, COBIND successfully discovered the canonical co-binding pattern and the compressed motif. It predicted the compressed co-binding pattern in 2 datasets (50%; derived from mouse ESC), with 5% of the sequences harboring the co-binding motif (**Additional file 1: Figure S31C**).

In this proof-of-concept example, we knew *a priori* that POU5F1 was the binding partner of SOX2 and SOX17. Nevertheless, COBIND discovered *de novo* the co-binding motif. In other settings, one would not know the TF binding to the co-binding motif discovered. To address this challenge, we combined motif similarity to already known motifs from JASPAR and CIS-BP with protein-protein interaction (PPI) data from the STRING database to infer the TFs potentially binding the motifs revealed by COBIND (Materials and Methods). This strategy confirmed that the framework inferred POU5F1 as the binding partner of SOX2 and SOX17 (**Figure 3C-D**; **Additional file 1: Figure S31**).

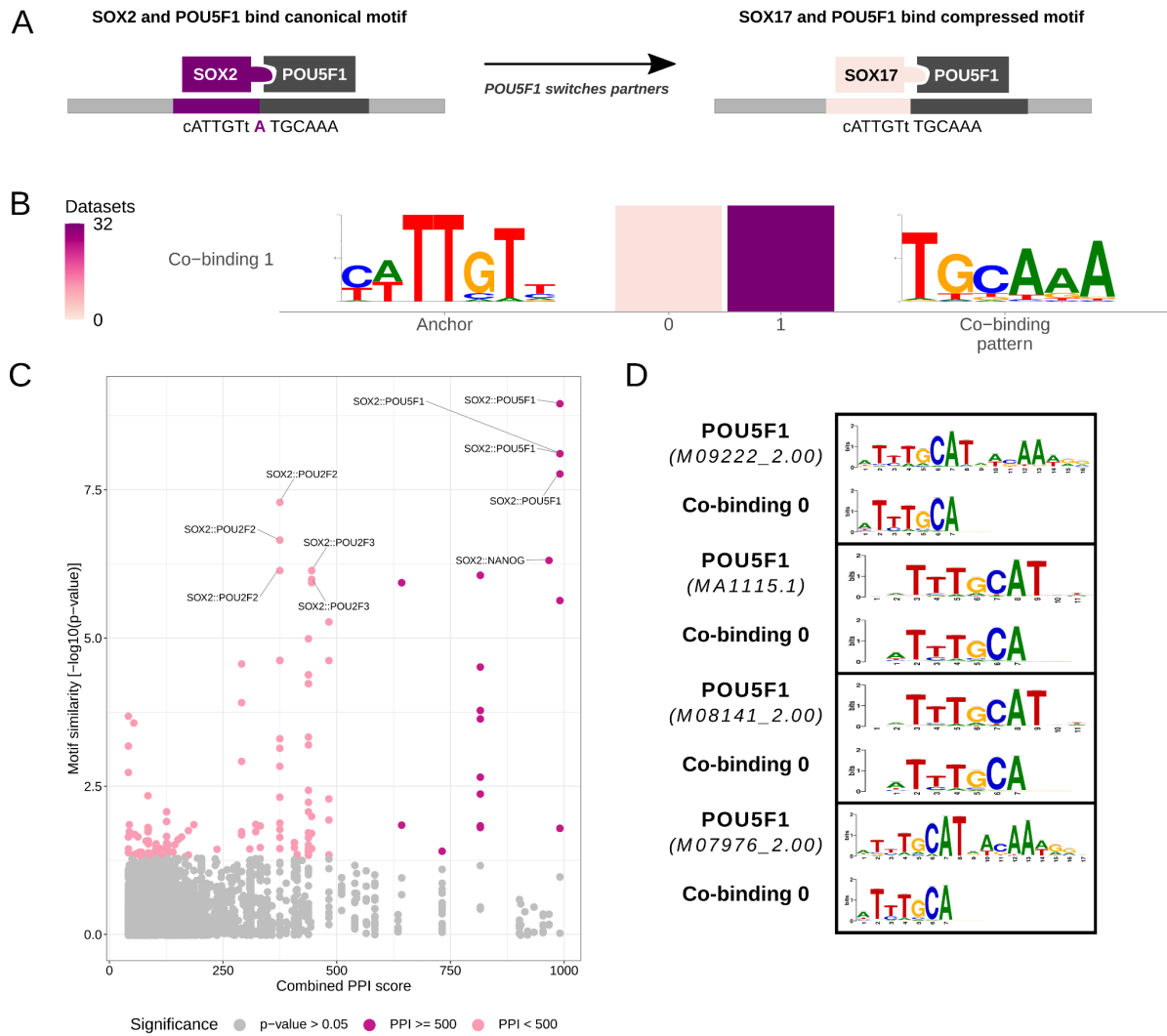


Figure 3. Application of COBIND to SOX2 and SOX17 TFBS datasets. COBIND predicts POU5F1 to co-bind with SOX2 (human and mouse) and SOX17 (mouse), upholding the previously known mechanism of POU5F1 switching partner TFs and binding motifs with different syntax. **A.** POU5F1 partners with SOX2 to bind its canonical motif. In contrast, POU5F1 associates with SOX17 to bind a compressed motif. **B.** The co-binding patterns discovered in SOX2 and SOX17 datasets correspond to the expected canonical and compressed motifs. **One co-binding pattern was discovered downstream of the anchor with either zero or one nucleotide between them.** **C.** The co-binding motif found in the SOX2 datasets was associated with POU5F1 using motif similarity and PPI data (significant motif similarity p-value < 0.05 and PPI combined score > 500). **D.** Visual representation of the similarity between the discovered co-binding motif and motifs recognized by POU5F1 in JASPAR and CIS-BP.

COBIND discovers known and novel co-binding patterns in a large-scale analysis across several species

We ran COBIND on 5,699 UniBind TFBS datasets associated with 401 unique TFs from 7 species (Materials and Methods; **Additional file 1: Table S2**). Beyond predicting co-binding patterns with COBIND, we inferred the TFs likely binding to the discovered patterns following the strategy described above (also see Materials and Methods). Altogether, COBIND revealed 591 co-binding patterns for 224 unique TFs (22 in *A. thaliana* datasets, 7 in *C. elegans*, 3 in *D. rerio*, 22 in *D. melanogaster*, 309 in *H. sapiens*, 217 in *M. musculus*, and 11 in *R. norvegicus*). For 78% of the reported co-binding motifs (462 out of 591), we found motif similarity and PPI data to support the

inferred pair of co-binding TFs (**Figure 4**). Specifically, the data supported all co-binding patterns for *D. rerio* and *C. elegans*, more than half of the patterns for *R. norvegicus*, *H. sapiens*, and *M. musculus*, 18.2% for *A. thaliana*, and 31.8% for *D. melanogaster*. We provide all predictions for the community to explore through a dedicated website at https://cbgr.bitbucket.io/COBIND_results_page/.

Next, we evaluated the frequency at which a TF_A is predicted as a partner of TF_B and, conversely, TF_B as a partner of TF_A. We analyzed co-binding patterns predicted in a specific cell type for each TF_A, provided data was available for a predicted co-binding TF_B in the same cell type (noting that multiple TFs can recognize the same co-binding pattern). When examining datasets linked to TF_B, COBIND successfully predicted back TF_A in 77% of the co-binding patterns from human datasets and 84% from mouse datasets.

Furthermore, the analysis of motif similarity between COBIND's predictions revealed that 67% (143 out of 214) of the TFs shared a co-binding motif with another member of the same TF structural family (Materials and Methods), which confirmed the conservation of co-binding patterns within families of TFs across species. It is noteworthy that anchor regions bound by TFs from the same structural family tend to be more similar (Mann-Whitney U test p-value < 0.05) than those bound by different TF families, even though they typically show low Jaccard index values (**Additional file 1: Figure S32**). This observation was consistent across different species, covering all regions with and without predicted co-binding patterns (**Additional file 1: Figure 32**). However, no statistically significant differences were noted (Mann-Whitney U test p-value > 0.05) when comparing regions with predicted co-binding patterns to those without, irrespective of whether the same or different TF family members bind them.

As a case example, COBIND predicted two co-binding patterns around TEAD1 TFBSs. Specifically, COBIND revealed the same motif with different spacings from the provided anchor TFBSs (2nt upstream or 3nt downstream of TEAD1 TFBSs; **Additional file 1: Figure S33A**). We found that the two co-binding patterns occurred in ~6% (3-bp downstream pattern) and ~7% (2-bp upstream pattern) of the input sequences from mouse nerve tumor cells and in ~3% (2-bp upstream and 3-bp downstream patterns each) of the sequences from human pancreas and astrocytoma cells. Our approach inferred that TEAD2 and TEAD4 cooperate with TEAD1 (**Additional file 1: Figure S33B-C**). Previous studies reported that TEAD TFs bind either an isolated M-CAT element or direct DNA repeats with spacing ranging from 0 to 6 bp (56–58). These studies further support the co-binding patterns predicted by COBIND. The predictions around TEAD1 TFBSs provide an example of COBIND's capacity to predict co-binding events supported by motif similarity and PPI data.

To exemplify predictions of new co-binding TFs, COBIND revealed a co-binding motif located 4 bp upstream of HY5 anchor TFBSs in *A. thaliana* (**Additional file 1: Figure S34A**). The analyses of TFBS datasets associated with the TFs ABF1, ABF3, ABF4, GBF2, and GBF3 displayed the same co-binding pattern. We found that the canonical motifs recognized by human NF-Y TFs (NF-YC and NF-YA) were similar to the co-binding motif discovered by COBIND (**Additional file 1: Figure S34B-C**). However, no PPI data currently support the physical interactions between the anchor TFs and NF-Y orthologs in plants. Nevertheless, the bZIP67 TF, which belongs to the same basic leucine zipper (bZIP) family as HY5, ABF1, ABF3, ABF4, GBF2, and GBF3, is known to form a transcriptional complex together with NF-YC2 and bind ER stress response elements (ERSEs) to

regulate omega-3 fatty acid content in *A. thaliana* (59). In agreement with this observation, we noted that the anchor motif combined with the co-binding motif discovered by COBIND forms the motif associated with ERSEs (60). Furthermore, HY5 competes with bZIP28, another member of the bZIP family, to bind ERSEs on promoters of unfolded protein response genes, and bZIP28 interacts with NF-Y protein complexes (60, 61). Consistent with this knowledge, we found that most of the genomic regions predicted by COBIND to harbor the co-binding events in the HY5 dataset were in promoter regions (**Additional file 1: Figure S34D**). Altogether, these multiple lines of evidence strongly support the co-binding patterns predicted by COBIND between bZIP and NF-Y TFs in *A. thaliana*.

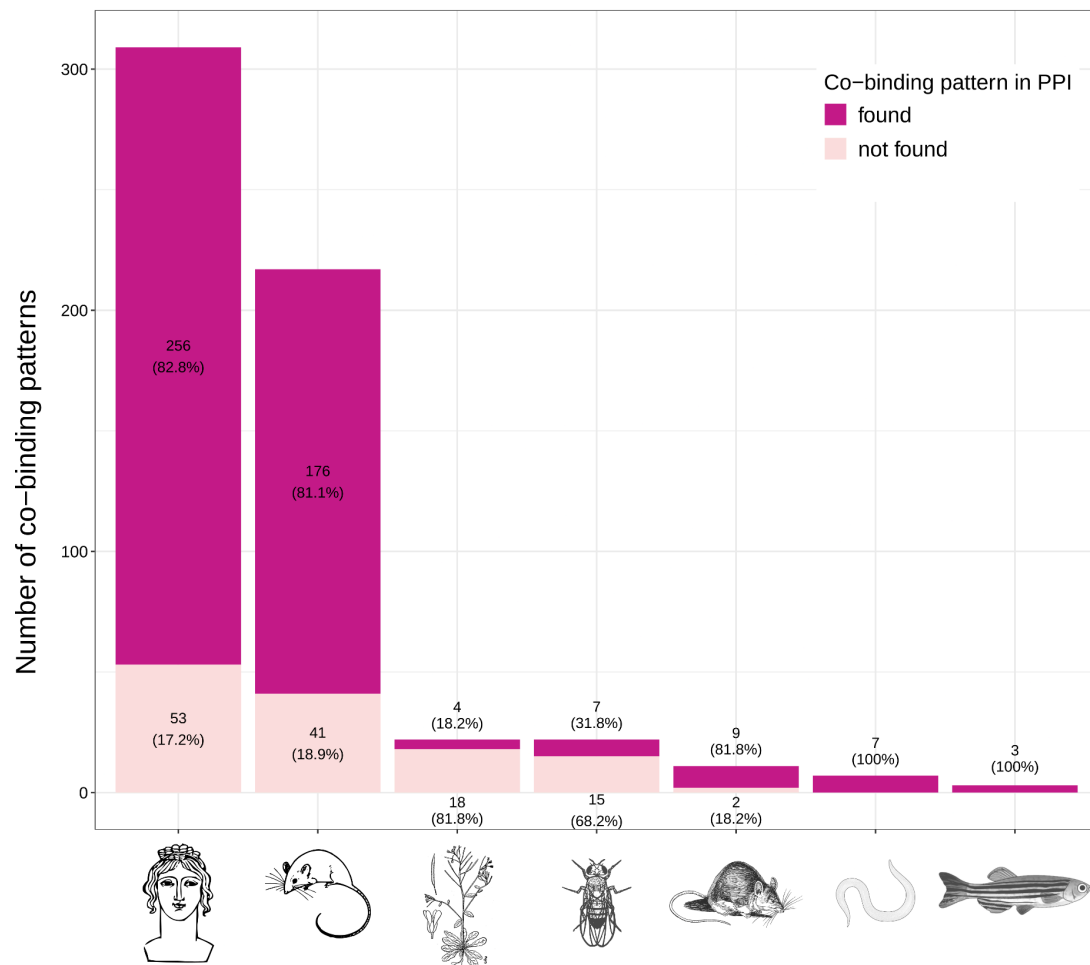


Figure 4. Overview of COBIND predicted co-binding patterns. The bar plot provides the number of co-binding patterns (y-axis) discovered by COBIND with (dark pink) or without (light pink) support from motif similarity and PPI data. The bars contain the corresponding numbers and percentages. Each bar summarises the results for a species (x-axis).

3.2 COBIND discovers the extended motif bound by CTCF

We investigated another example where COBIND predicted a co-binding pattern unsupported by motif similarity and PPI data. Specifically, COBIND identified a co-binding motif in the proximity of CTCF TFBSs for several datasets (**Figure 5A**). We observed two spacing configurations between the anchor CTCF TFBSs and the predicted co-binding events. Across the datasets, COBIND predicted from 2.6 to 5% of the sequences to harbor the co-binding patterns. All datasets from human and zebrafish and 25% of the mouse datasets (95 out of 382) contained the identified co-binding patterns (**Additional file 1: Figure S35**). Previous studies have reported the motif predicted by COBIND as an

extension of the canonical CTCF motif (62, 63). These studies revealed that the eighth zinc finger of CTCF acts as a linker instead of a clamp to allow zinc fingers 9-11 to bind the extended motif (**Figure 5B**), affecting the binding efficiency, residence time, and binding off-rate of CTCF (62, 63). Consequently, COBIND did not predict co-binding between distinct TFs, but CTCF seldomly bound an extended motif through different combinations of zinc finger contacts with the DNA.

Since evolutionary conservation is a hallmark of functional importance (48), we assessed the functional relevance of the genomic regions harboring the discovered binding patterns by examining their evolutionary conservation across vertebrates (Materials and Methods). We observed that the **anchor and extended motif sites (Figure 5C - right, red and green segments, respectively) were more conserved in the regions predicted to harbor the extended motif than in the other regions** (Wilcoxon test p -value < 0.001 for both segments; **Figure 5C right versus left**). The increased evolutionary conservation was consistent across all spacing configurations observed in human and mouse. These results exemplify how COBIND can predict relevant DNA patterns that do not correspond to the co-binding of two proteins but to binding variants for the same TF, a particular case of the zinc finger families.

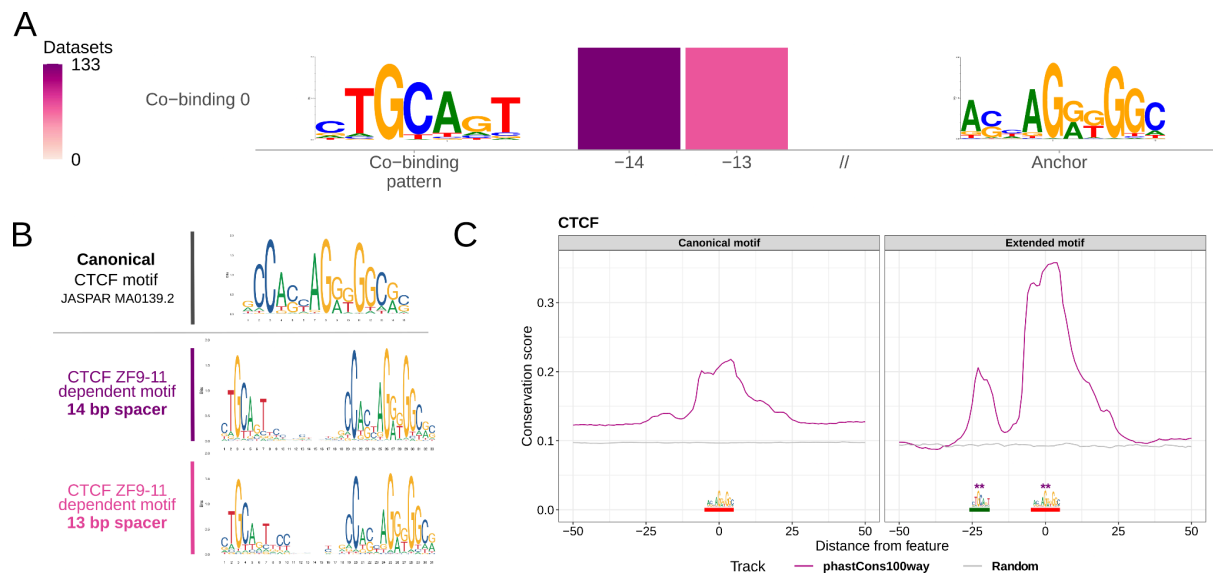


Figure 5. CTCF binds a conserved extended motif. (A) COBIND predicted two binding configurations through an extended motif for CTCF in human and mouse datasets. **The extended binding pattern was discovered upstream of the anchor with either thirteen or fourteen nucleotides between them.** (B) The two binding configurations derive from contacts between the zinc fingers (ZF) 9-11 and the DNA at the upstream motif. (C) Comparison between vertebrate evolutionary conservation of *H. sapiens* genomic regions harbouring the CTCF canonical motif **only** (left) and regions with the extended motif revealed higher conservation of the extended motif. The purple lines provide the mean conservation score across the regions considered. The grey lines provide the mean conservation score across the same number of random genomic regions in the human genome. ** indicates a Wilcoxon test p -value < 0.001 **at the anchor (red) and motif sites (green).**

3.3 Genomic regions harboring a co-binding pattern are evolutionarily more conserved than regions without co-binding

We assessed the functional relevance of all the co-binding patterns discovered by COBIND across species by analyzing their evolutionary conservation (Materials and Methods). Specifically, we compared the evolutionary conservation of the **anchor and co-binding positions (Figure 5C red and**

green, respectively) in genomic regions harboring a co-binding pattern, named co-bound regions for simplicity, to those without a predicted co-binding pattern. We first describe the analyses of the case studies presented above as examples. We found that the co-bound regions associated with SOX2::POU5F1 and TEAD1::TEAD in the human and mouse genomes exhibited a significantly increased evolutionary conservation compared to those without predicted co-binding (Wilcoxon test p -value < 0.001 ; **Figure 6A**; **Additional file 1: Figure S36A**). Notably, we observed increased conservation at the co-binding and the anchor motifs. When considering the co-binding pattern associated with HY5 in *A. thaliana*, we did not observe a significant difference in conservation between the regions with and without the co-binding pattern. Nevertheless, both the anchor and the co-binding motifs exhibited peaks of evolutionary conservation, confirming the likely functional relevance of the predicted co-binding pattern across the flowering plant kingdom (**Additional file 1: Figure S37A**).

Across all datasets and species, we systematically compared the evolutionary conservation at locations of the anchor and predicted co-binding sites discovered by COBIND in co-bound regions versus regions without co-binding sites. Altogether, we observed across species that 23% (for *A. thaliana*), 55% (for *D. melanogaster*), 60% (for *C. elegans*), 82% (for *H. Sapiens*), and 83% (for *M. musculus*) of the datasets with predicted co-binding patterns exhibited a co-binding motif with significantly more conservation in co-bound regions than in the other regions (**Figure 6B**; **Additional file 1: Figure S36B-S38**). Furthermore, the associated anchor sites were more conserved in the co-bound regions for 31% (*A. thaliana*), 55% (for *D. melanogaster*), 60% (for *C. elegans*), 64% (for *M. musculus*), and 67% (*H. sapiens*) of the datasets (**Figure 6B**; **Additional file 1: Figure S36B, S37B, S38**). Finally, we observed for 64% of the human datasets that the locations of both the anchor and a co-binding motif were more conserved in the co-bound regions than in regions without co-binding sites (61% for *M. musculus*, 60% for *C. elegans*, 45% for *D. melanogaster*, and 23% for *A. thaliana*). We compared with random expectation to further support the higher conservation of the anchor and co-binding sites in co-bound regions than similar locations in regions without a predicted co-binding pattern. Specifically, we randomly assigned, for each dataset, each genomic region to harbor or not a co-binding pattern (keeping the same number of regions in each category as the number predicted by COBIND). **Figure 6B** and **Additional file 1: Figures S36B and S37B** confirm that more datasets exhibit increased conservation in co-bound regions than expected by chance. The consistent increased evolutionary conservation of the co-binding patterns supports the functional importance of COBIND's predictions across species. Furthermore, our results suggest that the underlying genomic regions harboring a fixed binding motif syntax are evolutionarily important.

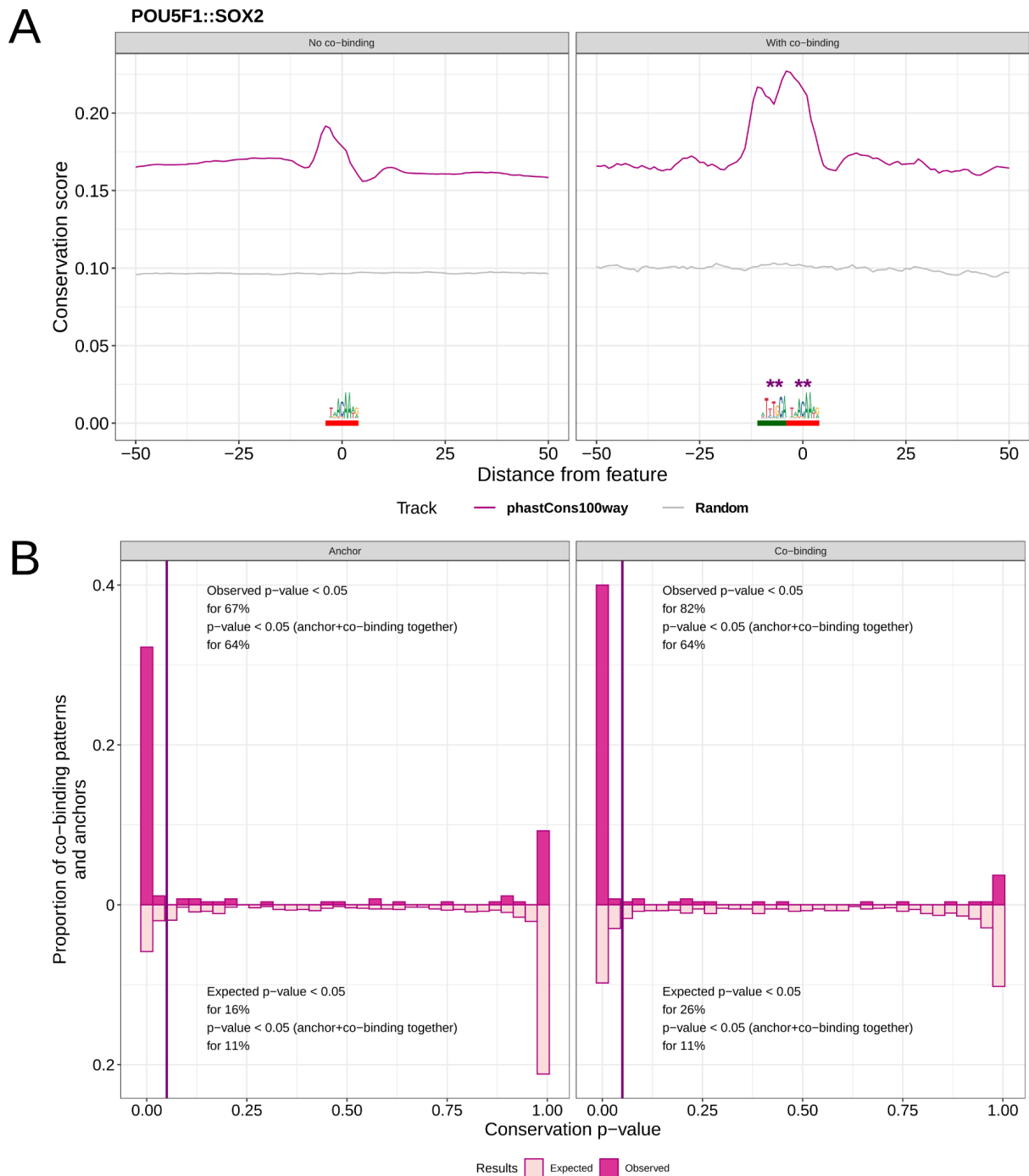


Figure 6. Analysis of the evolutionary conservation of co-binding patterns discovered by COBIND in the human genome. (A) Comparison between vertebrate evolutionary conservation of genomic regions where COBIND predicted a co-binding pattern (right) or not (left). The purple lines provide the mean conservation score across the regions considered. The grey lines provide the mean conservation score across the same number of random genomic regions in the human genome. ** indicates a Wilcoxon test p-value < 0.001. The anchor SOX2 anchor motif is underlined in red, and the co-binding motif is underlined in green. (B) We compared the evolutionary conservation scores at the anchors' motif locations in genomic regions harboring a co-binding pattern or not. The left panel represents the histogram of the proportion of datasets with the anchor sites harboring higher evolutionary conservation in co-bound regions than in other regions. The right panel represents the corresponding histogram when comparing genomic locations of the co-binding motif. Dark pink represents observed values, and light pink the expected values computed from random assignment of genomic regions predicted as co-bound or not.

3.4 DNase I hypersensitive footprints support the predicted co-binding patterns

We further assessed the co-binding patterns discovered by COBIND by analyzing SKJEMA FOR OPPLYSNINGER OM ANGRERETT orthogonal experimental data probing chromatin openness. The DNase-seq assay captures open chromatin regions by revealing DNase I hypersensitive sites (DHS) (64). Importantly, TFs interacting with the DNA in open chromatin regions protect their TFBSs from the DNase cleavage, which leaves a footprint on the corresponding TFBSs (51, 65). We retrieved 66 DHS footprint datasets from 53 human cell types (51) that matched some of the UniBind TFBS datasets used in this study. This data allowed us to investigate DHS footprints at the discovered co-binding motifs for 70 co-binding patterns associated with 44 TF binding profiles as anchors. For each co-binding pattern, we compared the depth of the DHS footprint at the locations of the co-binding motifs between the co-bound regions and the other regions (Materials and Methods).

As we ran COBIND on regions surrounding TFBSs predicted as high-quality direct TF-DNA interactions with ChIP-seq and computational evidence from UniBind, we expected DHS footprints at the anchor TFBSs. Indeed, we observed footprints of TF-DNA interactions; for instance, the DHS footprints observed for the CTCF datasets (**Figure 7A**). Notably, the DHS footprints at the CTCF anchor locations were significantly deeper in regions predicted with the extended motif than in the other regions (**Figure 7A**, red segments). Furthermore, the analyses exhibited deep DHS footprints at the location of the co-binding motifs (**Figure 7A**, green segment on the right compared to the equivalent locations on the left panel). Overall, we found significantly deeper DHS footprints at the locations of a co-binding motif in the predicted co-bound regions than in the other regions for 85% of the co-binding pattern - DHS dataset pairs. In comparison, 26% was expected by chance when randomly assigned the genomic regions as co-bound or not (**Figure 7B**, right panel). The anchor TFBSs at co-bound regions exhibited deeper DHS footprints than the TFBSs in other regions for 78% of the pairs, while 21% were expected by chance (**Figure 7B**, left panel). When considering deeper DHS footprints at both the anchor and a co-binding pattern sites, it was observed for 73% of the pairs, while it was never observed with the random assignments (**Figure 7B**). As CTCF datasets represented 43% of the total datasets analyzed here, we performed the same analysis, excluding the CTCF datasets. We found that 53% (25% expected by chance) of the co-binding pattern-DHS dataset pairs showed significant DHS footprints at the anchor TFBSs, and 68% (23% expected by chance) exhibited significantly deeper DHS footprints at the co-binding motif locations in co-bound regions than in the other regions. Altogether, 45% of the pairs exhibited deeper DHS footprints at both the co-bound and anchor TFBS locations. Overall, the DHS footprint analyses supported the co-occupancy of the anchor and the co-binding motifs at the co-bound regions.

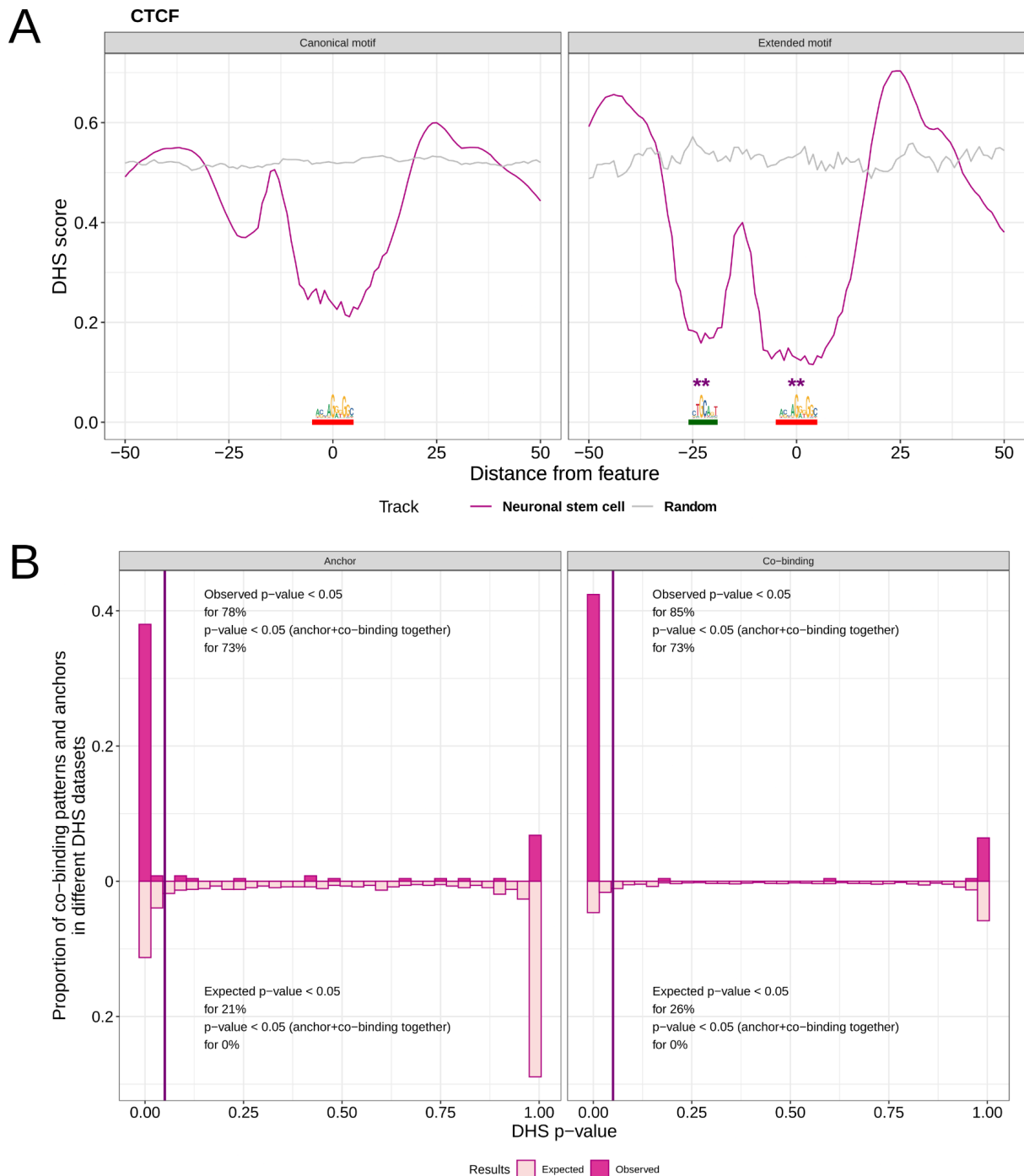


Figure 7. DHS footprinting analyses at anchor and co-binding locations. (A) The plots represent the average DHS score in regions surrounding CTCF TFBSs in neuronal stem cells (purple lines) or at random regions (grey lines). The left plot provides DHS scores at genomic regions where COBIND did not predict the CTCF extended motif; the right plot provides the DHS scores at genomic regions harbouring the CTCF extended motif. ** indicates a Wilcoxon test p-value < 0.001. The anchor CTCF motif is underlined in red and the upstream portion of the extended motif is in green. (B) We compared DHS scores at the **anchor motif locations** in genomic regions harbouring a co-binding pattern or not. The left panel represents the histogram of the datasets where the DHS footprint was deeper in genomic regions harbouring a co-binding pattern. The right panel represents the corresponding histogram when comparing genomic locations of the co-binding motif **locations**. **Dark pink** represents observed values, and **light pink** the expectation computed from randomly assigning regions as containing or not a co-binding pattern.

3.5 Single-molecule footprints support TF co-occupancy at single-molecule resolution

The DHS footprinting analysis described above relied on bulk DNase-seq data. Consequently, the results provided average estimations of the co-occupancy of TFs across cells. We aimed to investigate the co-occupancy of TFs at the co-binding patterns predicted by COBIND at single-molecule resolution. To this end, we considered single-molecule footprinting (SMF) data, which probed the co-occupancy of TFs and nucleosomes at single-molecule resolution for accessible genomic regions in mouse embryonic stem cells (mESC) (52). The SMF data overlapped 1,694 regions with predicted co-binding patterns (483,453 reads) and 54,727 regions (14,457,475 reads) not predicted to contain any co-binding pattern for 17 TFs in mESC (Materials and Methods).

For each genomic region predicted by COBIND to contain a co-binding pattern or not, we determined the fractions of molecules in each of the following five states: (i) accessible, (ii) occupied by nucleosomes, (iii) only occupied at the anchor motif, (iv) only occupied at the co-binding motif, or (v) co-occupied at both the anchor and the co-binding motifs. This is done by evaluating the binary methylation status of each molecule for each cytosine in and around the anchor and co-binding pattern sites (**Figure 8A**). **Figure 8B-C** illustrates the SMF data analysis at two genomic regions predicted to harbour co-binding events for SOX2 and POU5F1 (**Figure 8B**) and the extended CTCF motif (**Figure 8C**). We observed footprints of co-occupancy at both the anchor motif and the co-binding motif locations. Specifically, the analysis of 273 molecules across five replicates revealed co-occupancy for 27% of the molecules when considering the SOX2-associated region (**Figure 8B**). Five hundred ninety-two molecules across six replicates revealed co-occupancy of the extended CTCF motif for 59% of the molecules (**Figure 8C**). In contrast, the example regions where COBIND did not predict a co-binding pattern for SOX2 or CTCF, we only observed occupancy for the SOX2 anchor (17% of 89 molecules) or the canonical CTCF sites (46% of 297 molecules) while co-binding sites were unoccupied (**Additional file 1: Figure S39A-B**). When considering all the regions with SMF data, we found a larger fraction of co-occupied anchor and co-binding motif locations on the same molecule in COBIND-predicted co-bound regions than in other regions (p-value < 0.001; **Additional file 1: Figure S39C**). There were significant differences in all state abundances, except for the specific occupancy of the co-binding sites, which is in agreement with these sites not being occupied when the anchor sites are not either. However, not all anchor TF binding profiles satisfied this observation (**Figure 8D**). Moreover, we found that genomic regions predicted by COBIND to harbour a co-binding pattern were significantly more occupied (for “Anchor + Co-binding” occupancy state independently) than the other genomic regions (p-value < 0.05; **Figure 8D**). Altogether, the results confirmed that the genomic regions predicted by COBIND to harbour a co-binding pattern were overall more accessible or co-occupied at single-molecule resolution.

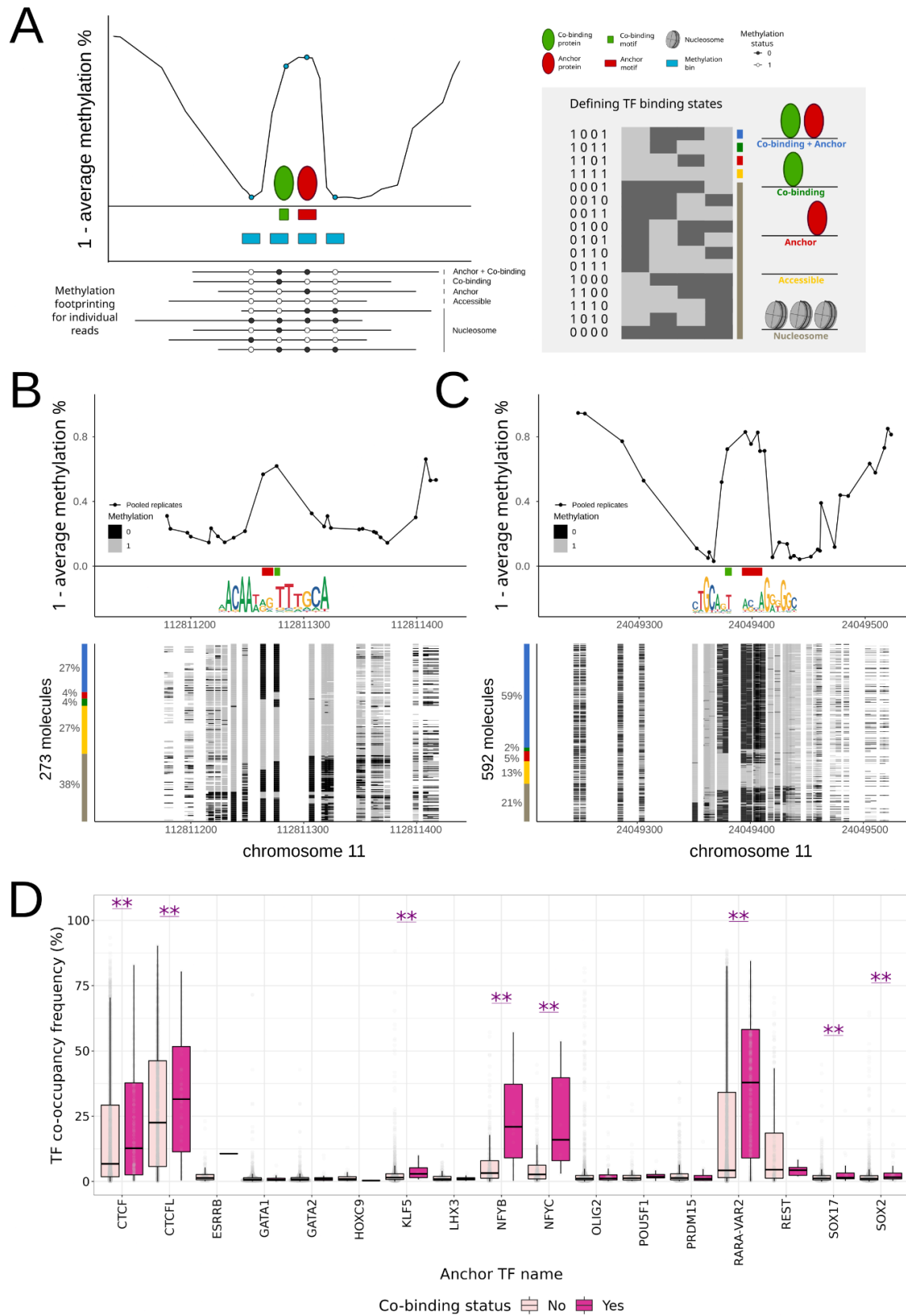


Figure 8. Single-molecule footprinting analysis. (A) For each region, single molecules are grouped and assigned to one of the five states (“Co-binding + Anchor”, “Co-binding”, “Anchor”, “Accessible”, and “Nucleosome”) based on the binary cytosine methylation status in and around anchor and co-binding patterns (for two regions of interest, four bins are

analysed). (B-C) Single-locus examples of predicted co-bound regions where the anchor TFBSs associate with SOX2 (B) and CTCF (C). Line plots in the top panels represent the average methylation levels. We provide logos and locations of the anchor motifs (red) and the predicted co-binding motifs (green). Stacks of sorted single molecules are visualised in the lower panels (light grey indicates methylated Cs, accessible positions; black - unmethylated Cs, protected). The y-axis corresponds to the total number of molecules. Coloured bars and percentages show the proportion of molecules in each state (colours corresponding to A). (D) We provide boxplots and dotplots showing the distribution of “Anchor+Co-binding” state frequencies of molecules that overlap the regions in two groups: one with predicted co-binding (shown in dark pink) and one with only an anchor TFBS (shown in light pink). The regions with co-binding events show a significantly higher frequency of molecules as co-occupied (“Anchor + Co-binding”). The differences between groups persist for molecules in other states. ** represents a Wilcoxon test p-value < 0.05.

4. DISCUSSION

We introduced COBIND, a computational framework for discovering *de novo* DNA motif patterns in the vicinity of predetermined genomic regions. We applied COBIND to sets of TFBSs from UniBind to identify locally enriched DNA motifs that define co-binding patterns of cooperative TFs. The method uncovered both established and new co-binding patterns, with most TFs sharing a co-binding motif with other TFs from the same family. Additionally, we inferred, when possible, the TFs most likely co-binding the discovered patterns. We make the collection of co-binding patterns revealed by COBIND available to the scientific community for each of the 214 TFs and 58 TF families across seven species. The co-binding events captured by COBIND are likely functionally relevant since they exhibit higher evolutionary conservation than isolated TFBSs. Furthermore, chromatin and single-molecule footprinting data support the occurrence of the identified co-binding events on the same DNA molecules.

We used a combination of motif similarity and prior knowledge of PPIs to identify TFs that may cooperatively bind to the patterns discovered by COBIND. Identifying well-known TF co-binding events, such as POU5F1 with SOX2 or SOX17, essential pluripotency regulators (15, 16, 66), confirmed the approach's efficacy. Importantly, we used a multi-species collection of DNA binding motifs to predict new co-binding configurations for pairs of TFs currently unsupported by PPIs. We discovered that NF-Y proteins are potential partners for HY5, ABF1,3-4, and GBF2-3 TFs. Previous studies identified NF-Y proteins as co-binders for bZIP67 and bZIP28 proteins belonging to the same TF family (59–61). More specifically, other studies also pointed to physical interactions of either HY5 or ABF1,3-4 with NF-YC9 proteins (67, 68). We identified the NF-Y-bound co-binding motifs using motifs associated with human TFs because DNA binding profiles for NF-Y are limited in *A. thaliana*. As HY5 competes with bZIP28, we acknowledge that the NF-Y binding pattern observed close to HY5 TFBSs might be occupied only when bZIP28 is binding. Nevertheless, using multi-species libraries generated potential co-binding predictions that will require further validations to decipher the binding cooperation of TFs at the corresponding genomic regions.

We observed that the COBIND predictions associated with CTCF anchor TFBSs exhibited a binding pattern that did not result from the co-binding of two TFs. Previous studies have discussed this binding pattern and have suggested that CTCF co-binds with ZBTB3 in mouse liver and various human cell lines, including embryonic stem cells and K562 (69, 70). Based on motif similarity, we inferred that ZBTB3 could also bind the additional motif identified by COBIND. Instead, the co-binding pattern identified by COBIND corresponds to an extended binding motif for CTCF, involving zinc fingers 9-11 (62). This orthogonal evidence allowed the two variants of extended

CTCF motifs to complement the JASPAR database (MA1930.1 and MA1929.1) (7, 71). Furthermore, this binding pattern is critical for optimal CTCF protein residence time on DNA (62). Importantly, we found that the genomic regions harbouring the extended motif were conserved in both mouse and human genomes, supporting the functional relevance of this binding pattern. Studies frequently reported that zinc-finger proteins bind DNA through a subset of their zinc fingers, but it remains unclear whether the other fingers have additional binding preferences (72). Thus, binding to extended motifs with additional zinc fingers may be a common characteristic of zinc finger proteins that *de novo* motif prediction tools like COBIND could capture.

The conservation of TFBSs is an essential indicator of their functional relevance in gene regulation (48). Furthermore, the combinatorial binding of TFs is important for evolutionary stability, and increased co-binding of TFs associates with a higher probability of a regulatory region being conserved (73). This study analysed the conservation of co-binding patterns across different species. The regions predicted to harbour TFBSs with fixed spacing were more evolutionarily conserved than those with a single TFBS. Interestingly, not only were the predicted co-bound motif locations conserved, but the anchor motif locations also showed an increase in conservation. We found most co-binding patterns with significant conservation from the human and mouse datasets, which may reflect the more significant number of datasets analysed for these species. **Overall, the increased conservation of both the anchor and co-binding sites was found in more datasets than expected by chance. However, this was not observed for *A. thaliana*; we hypothesise that this is due to the limited number of conservation tracks, which was restricted to flowering plants** (49).

COBIND is a tool for the *de novo* discovery of co-occurring DNA patterns. It does not require *a priori* knowledge of DNA motifs, such as a known DNA motif binding library, to perform the discovery. The requirement of known motif collections as an input is a standard limitation for other tools dealing with TF binding analysis, such as SpaMo or TACO (20, 21). This reliance on existing collections of DNA binding motifs can be problematic as many TF binding motifs still need to be discovered or included. The only input required for COBIND is a set of genomic reference regions to "anchor" the analysis and generate regions where it will search for patterns. The *de novo* discovery can reveal potential new or unannotated DNA pattern motifs, thereby circumventing the constraints of other tools. As such, COBIND offers a powerful alternative to other motif discovery tools that rely on existing collections of DNA binding motifs. Nonetheless, a limitation is that additional analysis will be required to interpret newly discovered motifs.

In this study, we applied COBIND to genomic regions near TFBSs. However, one can use COBIND in different biological contexts. As a test case, we generated anchor regions for analysis by taking two nucleotides at the donor and acceptor sites of intron-exon boundaries in the human genome. Analysis with COBIND recovered known donor and acceptor motifs (**Additional File 2: Figure S40**), demonstrating the versatility of the approach for DNA pattern discovery in other types of biological problems.

We utilised the available SMF data to assess the co-occupancy of the anchor and co-binding motif instances on the same DNA molecules (53). A limitation of the assay is that one can only apply it to organisms and cell lines that can survive without endogenous methylation. We restricted our analyses to mouse ESCs, for which SMF was already available. A recent study assessed chromatin openness

using methylation in the GpC context, allowing applications to different cell types with larger genomic coverage (74).

We recognise that COBIND and this study have several limitations. A fundamental limitation of COBIND is the restricted prediction of co-binding motifs with fixed spacing with the anchor motif. Because of the strict spacing requirements and the limitations of the NMF, COBIND restricts the search space to the close proximity of TFBSs. Previous studies have hypothesised the existence of two distinct sets of enhancers with distinct information-processing mechanisms (10). One proposed mechanism distinguishes between enhancers where TF binding is highly cooperative and coordinated with fixed spacing, referred to as the 'enhanceosome' model, and enhancers where TF cooperation is flexible, known as the 'billboard' model (75). Regardless, these mechanisms are compatible with the presence of both fixed and flexible cooperation of TFs at regulatory elements, depending on the type of TF cooperation. Recent studies support both mechanisms, and even though some suggest that TFBS spacing and orientation are not key determinants of transcription regulation, other studies contradict this statement (10–14). The evolutionary conservation of the co-binding patterns revealed by COBIND further argues for the functional relevance of strict grammar.

COBIND relies on the NMF algorithm to discover the co-binding patterns, so it requires enough sequences harbouring the fixed pattern. Our simulated data showed that we obtained reliable results when at least 10% of at least 800 sequences (also 5% of 1000 sequences) contained the pattern to be predicted by COBIND. Since we considered the anchor TFBS and the flanks independently in the COBIND processing, discovering overlapping motifs represents a challenge. COBIND could detect such overlaps by revealing a small co-binding motif, but interpreting such results becomes increasingly difficult as the partial motif becomes small. Furthermore, we used anchor TFBSs predicted from TF binding profiles corresponding to the canonical motifs recognised by the corresponding TFs. This approach prohibits the identification of co-binding patterns where the anchor TFs would recognise altered motifs when cooperating with other proteins. Finally, another limitation lies in the computational time necessary for motif clustering when the NMF identifies many possible motifs. Our comparison to other tools revealed that COBIND's computational time was not prohibitive.

DATA AVAILABILITY

COBIND is implemented in Python, R, and C++. The COBIND source code and documentation are available at https://bitbucket.org/CBGR/cobind_tool/src/master/. The source code and data to reproduce the results outlined in this report are available at https://bitbucket.org/CBGR/cobind_manuscript/src/master/ and pre-processed data deposited at <https://doi.org/10.5281/zenodo.7681482>. All results presented here are available to the community through a webpage at https://cbgr.bitbucket.io/COBIND_results_page/.

AUTHORS' CONTRIBUTIONS

We follow here the Contributor Roles Taxonomy (CRediT) (76). Ieva Rauluseviciute: methodology, software, validation, investigation, formal analysis, writing - original draft, visualisation; Timothée Launay: methodology, investigation, software, writing - review & editing; Guido Barzaghi: resources,

writing - review & editing; Sarvesh Nikumbh: writing - review & editing; Boris Lenhard: writing - review & editing, funding acquisition; Arnaud Regis Krebs: writing - review & editing, funding acquisition; Jaime A. Castro-Mondragon: conceptualisation, methodology, writing - review & editing; Anthony Mathelier: conceptualisation, methodology, writing - review & editing, supervision, project administration, funding acquisition.

ACKNOWLEDGMENTS

As “research parasites” (77), we thank all the researchers who made their data available. We thank Marcel Schulz for his suggestion on protein-protein interaction data analysis, François Parcy for fruitful discussion on the plant co-binding motifs and his help finding the plant evolutionary conservation data, Judith Zaugg for providing feedback on the project, Roza Berhanu Lemma, Vipin Kumar, Ladislav Hovan, and Rafael Riudavets Puig for reading the manuscript and providing valuable feedback, Harold Gutch, Torfinn Nome, and the NCMM IT team for their IT support, Ingrid Kjelsvik for administrative support, and the members of the Kuijjer and Mathelier groups for insightful discussions.

FUNDING

The Research Council of Norway [187615], Helse Sør-Øst, and the University of Oslo through the Centre for Molecular Medicine Norway (NCMM) (to Mathelier group); the Norwegian Cancer Society [197884] (to Mathelier group); the Research Council of Norway [288404] (to Mathelier group); the Nordic EMBL Partnership Hub for Molecular Medicine, NordForsk Grant [96782] (to I.R.); the Deutsche Forschungsgemeinschaft [KR 5247/1-1] (salary of G.B.); core funding from the EMBL and the Deutsche Forschungsgemeinschaft [KR 5247/1-2] (to support research in the laboratory of A.R.K.); Wellcome Trust Joint-Investigator award [106955/Z/15/Z] (to B.L.), core funding from the MRC LMS (salary for S.N.).

REFERENCES

1. Wingender,E., Schoeps,T., Haubrock,M., Krull,M. and Dönitz,J. (2018) TFClass: expanding the classification of human transcription factors to their mammalian orthologs. *Nucleic Acids Res.*, **46**, D343–D347.
2. Lambert,S.A., Jolma,A., Campitelli,L.F., Das,P.K., Yin,Y., Albu,M., Chen,X., Taipale,J., Hughes,T.R. and Weirauch,M.T. (02/2018) The Human Transcription Factors. *Cell*, **172**, 650–665.
3. Zeitlinger,J. (2020) Seven myths of how transcription factors read the cis-regulatory code. *Current Opinion in Systems Biology*, **23**, 22–31.
4. Suter,D.M. (2020) Transcription Factors and DNA Play Hide and Seek. *Trends Cell Biol.*, **30**, 491–500.
5. Weirauch,M.T., Yang,A., Albu,M., Cote,A.G., Montenegro-Montero,A., Drewe,P., Najafabadi,H.S., Lambert,S.A., Mann,I., Cook,K., *et al.* (2014) Determination and inference of eukaryotic transcription factor sequence specificity. *Cell*, **158**, 1431–1443.
6. Kulakovskiy,I.V., Vorontsov,I.E., Yevshin,I.S., Sharipov,R.N., Fedorova,A.D., Rumynskiy,E.I., Medvedeva,Y.A., Magana-Mora,A., Bajic,V.B., Papatsenko,D.A., *et al.* (2018) HOCOMOCO: towards a complete collection of transcription factor binding models for human and mouse via large-scale ChIP-Seq analysis. *Nucleic Acids Res.*, **46**, D252–D259.
7. Castro-Mondragon,J.A., Riudavets-Puig,R., Rauluseviciute,I., Berhanu Lemma,R., Turchi,L., Blanc-Mathieu,R., Lucas,J., Boddie,P., Khan,A., Manosalva Pérez,N., *et al.* (2021) JASPAR 2022: the 9th release of the open-access database of

transcription factor binding profiles. *Nucleic Acids Res.*

8. Reiter,F., Wienerroither,S. and Stark,A. (2017) Combinatorial function of transcription factors and cofactors. *Curr. Opin. Genet. Dev.*, **43**, 73–81.
9. Zhou,M., Li,H., Wang,X. and Guan,Y. (2020) Evidence of widespread, independent sequence signature for transcription factor cobinding. *Genome Res.*
10. Arnosti,D.N. and Kulkarni,M.M. (2005) Transcriptional enhancers: Intelligent enhanceosomes or flexible billboards? *J. Cell. Biochem.*, **94**, 890–898.
11. King,D.M., Hong,C.K.Y., Shepherdson,J.L., Granas,D.M., Maricque,B.B. and Cohen,B.A. (2020) Synthetic and genomic regulatory elements reveal aspects of cis-regulatory grammar in mouse embryonic stem cells. *Elife*, **9**, e41279.
12. Avsec,Ž., Weilert,M., Shrikumar,A., Krueger,S., Alexandari,A., Dalal,K., Fropf,R., McAnany,C., Gagneur,J., Kundaje,A., *et al.* (2021) Base-resolution models of transcription-factor binding reveal soft motif syntax. *Nat. Genet.*, **53**, 354–366.
13. Sahu,B., Hartonen,T., Pihlajamaa,P., Wei,B., Dave,K., Zhu,F., Kaasinen,E., Lidschreiber,K., Lidschreiber,M., Daub,C.O., *et al.* (2022) Sequence determinants of human gene regulatory elements. *Nat. Genet.*, **54**, 283–294.
14. Georgakopoulos-Soares,I., Deng,C., Agarwal,V., Chan,C.S.Y., Zhao,J., Inoue,F. and Ahituv,N. (2023) Transcription factor binding site orientation and order are major drivers of gene regulatory activity. *Nat. Commun.*, **14**, 2333.
15. Li,M. and Belmonte,J.C.I. (2018-4) Deconstructing the pluripotency gene regulatory network. *Nat. Cell Biol.*, **20**, 382–392.
16. Aksoy,I., Jauch,R., Chen,J., Dyla,M., Divakar,U., Bogu,G.K., Teo,R., Leng Ng,C.K., Herath,W., Lili,S., *et al.* (2013) Oct4 switches partnering from Sox2 to Sox17 to reinterpret the enhancer code and specify endoderm. *EMBO J.*, **32**, 938–953.
17. Nagy,G. and Nagy,L. (2020) Motif grammar: The basis of the language of gene expression. *Comput. Struct. Biotechnol. J.*, **18**, 2026–2032.
18. Jauch,R., Aksoy,I., Hutchins,A.P., Ng,C.K.L., Tian,X.F., Chen,J., Palasingam,P., Robson,P., Stanton,L.W. and Kolatkar,P.R. (2011) Conversion of Sox17 into a pluripotency reprogramming factor by reengineering its association with Oct4 on DNA. *Stem Cells*, **29**, 940–951.
19. Jolma,A., Yin,Y., Nitta,K.R., Dave,K., Popov,A., Taipale,M., Enge,M., Kivioja,T., Morgunova,E. and Taipale,J. (2016) DNA-dependent formation of transcription factor pairs alters their binding specificity. *Nature*, **534**, S15–S16.
20. Jankowski,A., Prabhakar,S. and Tiurnyn,J. (2014) TACO: a general-purpose tool for predicting cell-type-specific transcription factor dimers. *BMC Genomics*, **15**, 208.
21. Whittington,T., Frith,M.C., Johnson,J. and Bailey,T.L. (2011-8) Inferring transcription factor complexes from ChIP-seq data. *Nucleic Acids Res.*, **39**, e98.
22. Levitsky,V., Zemlyanskaya,E., Oshchepkov,D., Podkolodnaya,O., Ignatieva,E., Grosse,I., Mironova,V. and Merkulova,T. (2019) A single ChIP-seq dataset is sufficient for comprehensive analysis of motifs co-occurrence with MCOT package. *Nucleic Acids Res.*, **47**, e139.
23. Bentsen,M., Heger,V., Schultheis,H., Kuenne,C. and Looso,M. (2022) TF-COMB - Discovering grammar of transcription factor binding sites. *Comput. Struct. Biotechnol. J.*, **20**, 4040–4051.
24. Park,P.J. (10/2009) ChIP-seq: advantages and challenges of a maturing technology. *Nat. Rev. Genet.*, **10**, 669–680.
25. van Helden,J., Rios,A.F. and Collado-Vides,J. (2000) Discovering regulatory elements in non-coding sequences by analysis of spaced dyads. *Nucleic Acids Res.*, **28**, 1808–1818.
26. Defrance,M., Janky,R. 's, Sand,O. and van Helden,J. (2008) Using RSAT oligo-analysis and dyad-analysis tools to discover regulatory signals in nucleic sequences. *Nat. Protoc.*, **3**, 1589–1603.
27. Minnoye,L., Taskiran,I.I., Mauduit,D., Fazio,M., Van Aerschot,L., Hulselmans,G., Christiaens,V., Makhzami,S., Seltenthaler,M., Karras,P., *et al.* (2020) Cross-species analysis of enhancer logic using deep learning. *Genome Res.*, **30**, 1815–1834.
28. Stein-O'Brien,G.L., Arora,R., Culhane,A.C., Favorov,A.V., Garmire,L.X., Greene,C.S., Goff,L.A., Li,Y., Ngom,A., Ochs,M.F., *et al.* (2018) Enter the Matrix: Factorization Uncovers Knowledge from Omics. *Trends Genet.*, **34**, 790–805.

29. Lee,D.D. and Seung,H.S. (1999) Learning the parts of objects by non-negative matrix factorization. *Nature*, **401**, 788–791.
30. Devarajan,K. (2008) Nonnegative Matrix Factorization: An Analytical and Interpretive Tool in Computational Biology. *PLoS Comput. Biol.*, **4**, e1000029.
31. Nikumbh,S. and Lenhard,B. (2023) Identifying promoter sequence architectures via a chunking-based algorithm using non-negative matrix factorisation. *bioRxiv*, 10.1101/2023.03.02.530868.
32. Mölder,F., Jablonski,K.P., Letcher,B., Hall,M.B., Tomkins-Tinch,C.H., Sochat,V., Forster,J., Lee,S., Twardziok,S.O., Kanitz,A., *et al.* (2021) Sustainable data analysis with Snakemake. *F1000Res.*, **10**, 33.
33. Puig,R.R., Boddie,P., Khan,A., Castro-Mondragon,J.A. and Mathelier,A. (2021) UniBind: maps of high-confidence direct TF-DNA interactions across nine species. *BMC Genomics*, **22**, 482.
34. Quinlan,A.R. and Hall,I.M. (2010) BEDTools: a flexible suite of utilities for comparing genomic features. *Bioinformatics*, **26**, 841–842.
35. Pedregosa,F. (2011) Scikit-learn: Machine Learning in Python. *J. Mach. Learn. Res.*, **12**, 2825–2830.
36. Kim,H. and Park,H. (2007) Sparse non-negative matrix factorizations via alternating non-negativity-constrained least squares for microarray data analysis. *Bioinformatics*, **23**, 1495–1502.
37. Gini,C. (1912) Variabilità e mutabilità: contributo allo studio delle distribuzioni e delle relazioni statistiche. [Fasc. I.] Tipogr. di P. Cuppini.
38. Castro-Mondragon,J.A., Jaeger,S., Thieffry,D., Thomas-Chollier,M. and van Helden,J. (2017) RSAT matrix-clustering: dynamic exploration and redundancy reduction of transcription factor binding motif collections. *Nucleic Acids Res.*, **45**, e119.
39. Castro-Mondragon,J. (2022) matrix-clustering_stand-alone.
40. Shrikumar,A., Tian,K., Avsec,Ž., Shcherbina,A., Banerjee,A., Sharmin,M., Nair,S. and Kundaje,A. (2020) Technical Note on Transcription Factor Motif Discovery from Importance Scores (TF-MoDISco) version 0.5.6.5.
41. Gupta,S., Stamatoyannopoulos,J.A., Bailey,T.L. and Noble,W.S. (2007) Quantifying similarity between motifs. *Genome Biol.*, **8**, R24.
42. Khan,A., Riudavets Puig,R., Boddie,P. and Mathelier,A. (2020) BiasAway: command-line and web server to generate nucleotide composition-matched DNA background sequences. *Bioinformatics*, 10.1093/bioinformatics/btaa928.
43. Bailey,T.L. (2021) STREME: accurate and versatile sequence motif discovery. *Bioinformatics*, **37**, 2834–2840.
44. Thomas-Chollier,M., Darbo,E., Herrmann,C., Defrance,M., Thieffry,D. and van Helden,J. (2012) A complete workflow for the analysis of full-size ChIP-seq (and similar) data sets using peak-motifs. *Nat. Protoc.*, **7**, 1551–1568.
45. Thomas-Chollier,M., Herrmann,C., Defrance,M., Sand,O., Thieffry,D. and van Helden,J. (2012) RSAT peak-motifs: motif analysis in full-size ChIP-seq datasets. *Nucleic Acids Res.*, **40**, e31–e31.
46. Fornes,O., Castro-Mondragon,J.A., Khan,A., van der Lee,R., Zhang,X., Richmond,P.A., Modi,B.P., Correard,S., Gheorghe,M., Baranašić,D., *et al.* (2020) JASPAR 2020: update of the open-access database of transcription factor binding profiles. *Nucleic Acids Res.*, **48**, D87–D92.
47. Szklarczyk,D., Gable,A.L., Lyon,D., Junge,A., Wyder,S., Huerta-Cepas,J., Simonovic,M., Doncheva,N.T., Morris,J.H., Bork,P., *et al.* (2019) STRING v11: protein-protein association networks with increased coverage, supporting functional discovery in genome-wide experimental datasets. *Nucleic Acids Res.*, **47**, D607–D613.
48. Siepel,A., Bejerano,G., Pedersen,J.S., Hinrichs,A.S., Hou,M., Rosenbloom,K., Clawson,H., Spieth,J., Hillier,L.W., Richards,S., *et al.* (2005) Evolutionarily conserved elements in vertebrate, insect, worm, and yeast genomes. *Genome Res.*, **15**, 1034–1050.
49. Tian,F., Yang,D.-C., Meng,Y.-Q., Jin,J. and Gao,G. (2020) PlantRegMap: charting functional regulatory maps in plants. *Nucleic Acids Res.*, **48**, D1104–D1113.
50. Pohl,A. and Beato,M. (2014) bwtool: a tool for bigWig files. *Bioinformatics*, **30**, 1618–1619.
51. Vierstra,J., Lazar,J., Sandstrom,R., Halow,J., Lee,K., Bates,D., Diegel,M., Dunn,D., Neri,F., Haugen,E., *et al.* (2020) Global reference mapping of human transcription factor footprints. *Nature*, **583**, 729–736.

52. Sönmezer,C., Kleinendorst,R., Imanci,D., Barzaghi,G., Villacorta,L., Schübeler,D., Benes,V., Molina,N. and Krebs,A.R. (2021) Molecular Co-occupancy Identifies Transcription Factor Binding Cooperativity In Vivo. *Mol. Cell*, **81**, 255–267.e6.
53. Kleinendorst,R.W.D., Barzaghi,G., Smith,M.L., Zaugg,J.B. and Krebs,A.R. (2021) Genome-wide quantification of transcription factor binding at single-DNA-molecule resolution using methyl-transferase footprinting. *Nat. Protoc.*, **16**, 5673–5706.
54. Barzaghi,G., Krebs,A. and Smith,M. (2022) SingleMoleculeFootprinting: Analysis tools for Single Molecule Footprinting (SMF) data Bioconductor version: Release (3.15).
55. Mistri,T.K., Devasia,A.G., Chu,L.T., Ng,W.P., Halbritter,F., Colby,D., Martynoga,B., Tomlinson,S.R., Chambers,I., Robson,P., *et al.* (2015) Selective influence of Sox2 on POU transcription factor binding in embryonic and neural stem cells. *EMBO Rep.*, **16**, 1177–1191.
56. Jiang,S.W., Desai,D., Khan,S. and Eberhardt,N.L. (2000) Cooperative binding of TEF-1 to repeated GGAATG-related consensus elements with restricted spatial separation and orientation. *DNA Cell Biol.*, **19**, 507–514.
57. Anbanandam,A., Albarado,D.C., Nguyen,C.T., Halder,G., Gao,X. and Veeraraghavan,S. (2006) Insights into transcription enhancer factor 1 (TEF-1) activity from the solution structure of the TEA domain. *Proc. Natl. Acad. Sci. U. S. A.*, **103**, 17225–17230.
58. Lee,D.-S., Vonnrhein,C., Albarado,D., Raman,C.S. and Veeraraghavan,S. (2016) A potential structural switch for regulating DNA-binding by TEAD transcription factors. *J. Mol. Biol.*, **428**, 2557–2568.
59. Mendes,A., Kelly,A.A., van Erp,H., Shaw,E., Powers,S.J., Kurup,S. and Eastmond,P.J. (2013) bZIP67 Regulates the Omega-3 Fatty Acid Content of Arabidopsis Seed Oil by Activating FATTY ACID DESATURASE3. *Plant Cell*, **25**, 3104–3116.
60. Pastor-Cantizano,N., Ko,D.K., Angelos,E., Pu,Y. and Brandizzi,F. (2020-2) Functional diversification of ER stress responses in Arabidopsis. *Trends Biochem. Sci.*, **45**, 123–136.
61. Nawkar,G.M., Kang,C.H., Maibam,P., Park,J.H., Jung,Y.J., Chae,H.B., Chi,Y.H., Jung,I.J., Kim,W.Y., Yun,D.-J., *et al.* (2017) HY5, a positive regulator of light signaling, negatively controls the unfolded protein response in Arabidopsis. *Proc. Natl. Acad. Sci. U. S. A.*, **114**, 2084–2089.
62. Soochit,W., Sleutels,F., Stik,G., Bartkuhn,M., Basu,S., Hernandez,S.C., Merzouk,S., Vidal,E., Boers,R., Boers,J., *et al.* (2021) CTCF chromatin residence time controls three-dimensional genome organization, gene expression and DNA methylation in pluripotent cells. *Nat. Cell Biol.*, **23**, 881–893.
63. Nakahashi,H., Kieffer Kwon,K.-R., Resch,W., Vian,L., Dose,M., Stavreva,D., Hakim,O., Pruett,N., Nelson,S., Yamane,A., *et al.* (2013) A genome-wide map of CTCF multivalency redefines the CTCF code. *Cell Rep.*, **3**, 1678–1689.
64. Thurman,R.E., Rynes,E., Humbert,R., Vierstra,J., Maurano,M.T., Haugen,E., Sheffield,N.C., Stergachis,A.B., Wang,H., Vernot,B., *et al.* (2012) The accessible chromatin landscape of the human genome. *Nature*, **489**, 75–82.
65. Funk,C.C., Casella,A.M., Jung,S., Richards,M.A., Rodriguez,A., Shannon,P., Donovan-Maiye,R., Heavner,B., Chard,K., Xiao,Y., *et al.* (2020) Atlas of Transcription Factor Binding Sites from ENCODE DNase Hypersensitivity Data across 27 Tissue Types. *Cell Rep.*, **32**, 108029.
66. Pan,X., Cang,X., Dan,S., Li,J., Cheng,J., Kang,B., Duan,X., Shen,B. and Wang,Y.-J. (2016) Site-specific Disruption of the Oct4/Sox2 Protein Interaction Reveals Coordinated Mesendodermal Differentiation and the Epithelial-Mesenchymal Transition. *J. Biol. Chem.*, **291**, 18353–18369.
67. Kumimoto,R.W., Siriwardana,C.L., Gayler,K.K., Risinger,J.R., Siefers,N. and Holt,B.F.,3rd (2013) NUCLEAR FACTOR Y transcription factors have both opposing and additive roles in ABA-mediated seed germination. *PLoS One*, **8**, e59481.
68. Myers,Z.A., Kumimoto,R.W., Siriwardana,C.L., Gayler,K.K., Risinger,J.R., Pezzetta,D. and Holt,B.F.,Iii (2016) NUCLEAR FACTOR Y, Subunit C (NF-YC) Transcription Factors Are Positive Regulators of Photomorphogenesis in Arabidopsis thaliana. *PLoS Genet.*, **12**, e1006333.
69. Wang,A.W., Wang,Y.J., Zahm,A.M., Morgan,A.R., Wangenstein,K.J. and Kaestner,K.H. (2020) The Dynamic Chromatin Architecture of the Regenerating Liver. *Cellular and Molecular Gastroenterology and Hepatology*, **9**, 121–143.
70. Boyle,A.P., Song,L., Lee,B.-K., London,D., Keefe,D., Birney,E., Iyer,V.R., Crawford,G.E. and Furey,T.S. (2011) High-resolution genome-wide in vivo footprinting of diverse transcription factors in human cells. *Genome Res.*, **21**,

71. Rauluseviciute,I., Riudavets-Puig,R., Blanc-Mathieu,R., Castro-Mondragon,J.A., Ferenc,K., Kumar,V., Lemma,R.B., Lucas,J., Chèneby,J., Baranasic,D., *et al.* (2024) JASPAR 2024: 20th anniversary of the open-access database of transcription factor binding profiles. *Nucleic Acids Res.*, **52**, D174–D182.
72. Schmitges,F.W., Radovani,E., Najafabadi,H.S., Barazandeh,M., Campitelli,L.F., Yin,Y., Jolma,A., Zhong,G., Guo,H., Kanagalingam,T., *et al.* (2016) Multiparameter functional diversity of human C2H2 zinc finger proteins. *Genome Res.*, **26**, 1742–1752.
73. Stefflova,K., Thybert,D., Wilson,M.D., Streeter,I., Aleksic,J., Karagianni,P., Brazma,A., Adams,D.J., Talianidis,I., Marioni,J.C., *et al.* (2013) Cooperativity and rapid evolution of cobound transcription factors in closely related mammals. *Cell*, **154**, 530–540.
74. Kreibich,E., Kleinendorst,R., Barzaghi,G., Kaspar,S. and Krebs,A.R. (2023) Single-molecule footprinting identifies context-dependent regulation of enhancers by DNA methylation. *Mol. Cell*, **83**, 787–802.e9.
75. Slattey,M., Zhou,T., Yang,L., Dantas Machado,A.C., Gordân,R. and Rohs,R. (09/2014) Absence of a simple code: how transcription factors read the genome. *Trends Biochem. Sci.*, **39**, 381–399.
76. Brand,A., Allen,L., Altman,M., Hlava,M. and Scott,J. (2015) Beyond authorship: attribution, contribution, collaboration, and credit. *Learn. Publ.*, **28**, 151–155.
77. Longo,D.L. and Drazen,J.M. (2016) Data Sharing. *N. Engl. J. Med.*, **374**, 276–277.